EXTRA PROBLEMS · 1ST MIDTERM · PROBLEMS

1st Midterm: Exam-Style Problems

Exam-style problems on Units 1 to 5, worth 10 points each like the Midterm 1 problem, written for this portal and solved step by step. They are practice material, not official exercises; each one was checked with SymPy.

16 exam-style problems on Units 1 to 5, written for this portal. The Midterm 1 papers of 2011 to 2024 are one problem of 10 points with 5 to 7 parts; the Exam simulator draws one of these and times it.

Problem · Unit 1 · 10 points · 60 min · Standard exam level

P01 — Bead on a vertical helical wire: Frenet triad, constant angular acceleration and the reaction

A rigid wire has the shape of a circular helix of radius R wound around the vertical axis Oz of the inertial frame $S_0 = \{O; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$ (z upward, uniform gravity $-g\,\mathbf{k}$). In cylindrical coordinates (ρ, φ, z), with φ measured from the x axis and counter-clockwise seen from above, the wire is

$$\rho = R, \qquad z = b\,\varphi \qquad (b > 0)$$

*so it rises a height $2\pi b$ (the pitch) per turn. A bead P of mass m slides **without friction** along the wire. At $t = 0$ it is released from rest at $\varphi = \varphi_0$.*

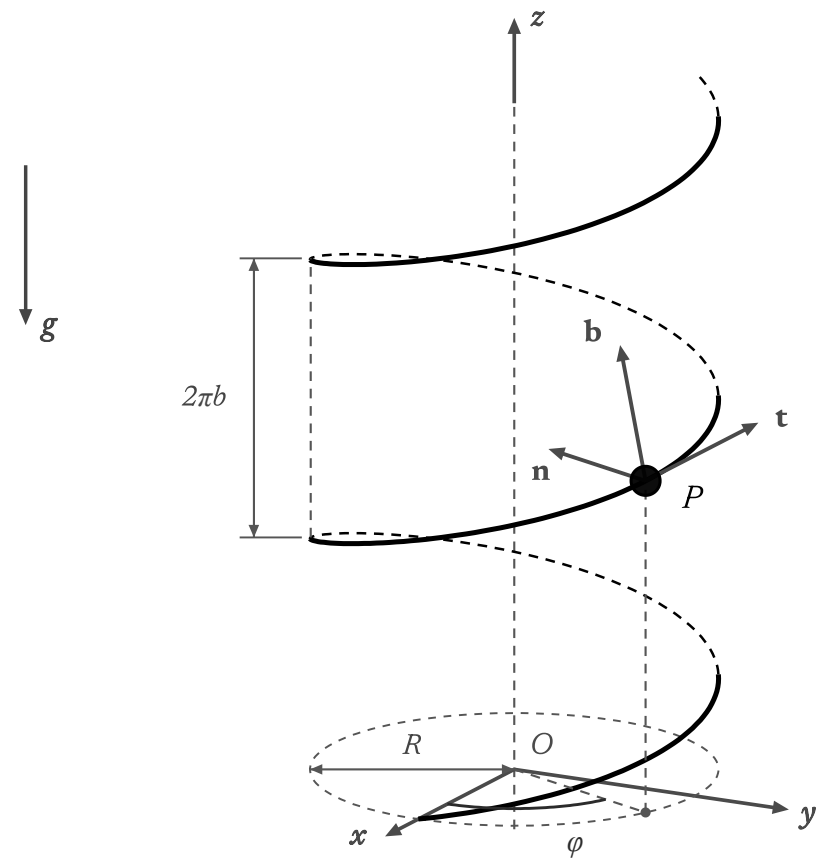


Figure 1. Bead P on the smooth helix $\rho = R$, $z = b\varphi$ (pitch $2\pi b$), z up and gravity $-g\,\mathbf{k}$; φ is the azimuth from the x axis, and $\{\mathbf{t}, \mathbf{n}, \mathbf{b}\}$ is the Frenet triad at P.

Use the cylindrical basis $\{\mathbf{e}_\rho, \mathbf{e}_\varphi, \mathbf{k}\}$ with $\mathbf{e}_\rho = \cos\varphi\,\mathbf{i} + \sin\varphi\,\mathbf{j}$, $\mathbf{e}_\varphi = -\sin\varphi\,\mathbf{i} + \cos\varphi\,\mathbf{j}$, and the intrinsic (Frenet) triad $\{\mathbf{t}, \mathbf{n}, \mathbf{b}\}$ of the helix, with $\mathbf{t}$ oriented towards increasing φ. Write $h = \sqrt{R^2 + b^2}$.

Setup

- Data: m, R, b, g, φ_0. In the numerical part (f): $R = 0.30$ m, $b = 0.40$ m (so $h = 0.50$ m), $g = 9.81$ m s^{-2}.
- Hypotheses: point particle, rigid fixed wire, no friction (the reaction has no component along $\mathbf{t}$), S_0 inertial.
- The bead is threaded on the wire, so the constraint is **bilateral**: the reaction may point either way in the normal plane.
- Notation: the scalar b is the pitch parameter of the helix ($z = b\varphi$); the bold $\mathbf{b}$ is the binormal unit vector. They are unrelated.

Part (a) · 1.5 points

Degrees of freedom and kinematics

How many degrees of freedom has P? Classify the constraints. Write the position, velocity and acceleration of P in the cylindrical basis as functions of φ, $\dot\varphi$, $\ddot\varphi$.

Step 1. Count

The configuration of a free particle needs 3 coordinates (ρ, φ, z). The wire imposes $f_1 = \rho - R = 0$ and $f_2 = z - b\varphi = 0$: two independent holonomic equations (positions only), with no explicit time (scleronomic), and two-sided (bilateral). Hence $3 - 2 = 1$ degree of freedom (Notes §2.7), and φ is a convenient coordinate: ρ and z follow from it.

Step 2. Change of basis

The cylindrical basis turns with φ; its relation to the inertial basis is

$$\left(\mathbf{e}_\rho\ \mathbf{e}_\varphi\ \mathbf{k}\right) = [R]\left(\mathbf{i}\ \mathbf{j}\ \mathbf{k}\right), \qquad [R] = \begin{pmatrix} \cos\varphi & \sin\varphi & 0 \\ -\sin\varphi & \cos\varphi & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Checks: rows 1 and 2 have length $\cos^2\varphi + \sin^2\varphi = 1$ and $-\cos\varphi\sin\varphi + \sin\varphi\cos\varphi = 0$, row 3 is $\mathbf{k}$, so $[R][R]^T = I$; $\det[R] = \cos^2\varphi + \sin^2\varphi = +1$ (right-handed, $\mathbf{e}_\rho \times \mathbf{e}_\varphi = \mathbf{k}$). Differentiating the rows with the chain rule, $\dfrac{d\mathbf{e}_\rho}{d\varphi} = \mathbf{e}_\varphi$ and $\dfrac{d\mathbf{e}_\varphi}{d\varphi} = -\mathbf{e}_\rho$.

Step 3. Position and velocity

$\mathbf{r} = \rho\,\mathbf{e}_\rho + z\,\mathbf{k} = R\,\mathbf{e}_\rho + b\varphi\,\mathbf{k}$. The cylindrical velocity (Notes §2.6.2) is $\mathbf{v} = \dot\rho\,\mathbf{e}_\rho + \rho\dot\varphi\,\mathbf{e}_\varphi + \dot z\,\mathbf{k}$. Here $\dot\rho = 0$ and, by the chain rule, $\dot z = \dfrac{dz}{d\varphi}\dot\varphi = b\dot\varphi$:

$$\mathbf{v} = R\dot\varphi\,\mathbf{e}_\varphi + b\dot\varphi\,\mathbf{k}, \qquad |\mathbf{v}| = \sqrt{R^2\dot\varphi^2 + b^2\dot\varphi^2} = h\,|\dot\varphi|$$

Step 4. Acceleration

The cylindrical acceleration is $\mathbf{a} = (\ddot\rho - \rho\dot\varphi^2)\mathbf{e}_\rho + (\rho\ddot\varphi + 2\dot\rho\dot\varphi)\mathbf{e}_\varphi + \ddot z\,\mathbf{k}$. With $\rho = R$, $\dot\rho = \ddot\rho = 0$ and $\ddot z = \dfrac{d}{dt}(b\dot\varphi) = b\ddot\varphi$:

$$\mathbf{a} = -R\dot\varphi^2\,\mathbf{e}_\rho + R\ddot\varphi\,\mathbf{e}_\varphi + b\ddot\varphi\,\mathbf{k}$$

Check by direct differentiation of $\mathbf{v}$ with $\dot{\mathbf{e}}_\varphi = -\dot\varphi\,\mathbf{e}_\rho$: $\dot{\mathbf{v}} = R\ddot\varphi\,\mathbf{e}_\varphi + R\dot\varphi(-\dot\varphi\,\mathbf{e}_\rho) + b\ddot\varphi\,\mathbf{k}$, the same result.

KEY RESULTS

- 1 degree of freedom, coordinate φ; two holonomic, scleronomic, bilateral constraints $\rho - R = 0$, $z - b\varphi = 0$.
- $\mathbf{v} = R\dot{\varphi}\,\mathbf{e}_\varphi + b\dot{\varphi}\,\mathbf{k}$, $|\mathbf{v}| = h\,|\dot{\varphi}|$.
- $\mathbf{a} = -R\dot{\varphi}^2\,\mathbf{e}_\rho + R\ddot{\varphi}\,\mathbf{e}_\varphi + b\ddot{\varphi}\,\mathbf{k}$.

Part (b) · 2 points

Intrinsic triad, curvature and torsion

Obtain the unit vectors $\mathbf{t}$, $\mathbf{n}$, $\mathbf{b}$ *of the helix in the cylindrical basis, its curvature* κ *and torsion* τ*, and the intrinsic components of the acceleration of* P*.*

Step 1. Arc length and tangent

$ds = |\mathbf{v}|\,dt = h\,d\varphi$ (taking s increasing with φ), so $s = h(\varphi - \varphi_0)$ and $\dot{s} = h\dot{\varphi}$. The unit tangent (Notes §2.4) is

$$\mathbf{t} = \frac{d\mathbf{r}}{ds} = \frac{d\mathbf{r}}{d\varphi}\frac{d\varphi}{ds} = \frac{1}{h}\big(R\,\mathbf{e}_\varphi + b\,\mathbf{k}\big)$$

using $\dfrac{d\mathbf{e}_\rho}{d\varphi} = \mathbf{e}_\varphi$.

Step 2. Curvature and normal

With $\dfrac{d\mathbf{e}_\varphi}{d\varphi} = -\mathbf{e}_\rho$ and $\mathbf{k}$ constant:

$$\frac{d\mathbf{t}}{ds} = \frac{1}{h}\frac{d\mathbf{t}}{d\varphi} = \frac{1}{h}\cdot\frac{R}{h}\big(-\mathbf{e}_\rho\big) = \frac{R}{h^2}\,(-\mathbf{e}_\rho)$$

The modulus is the curvature and the unit vector the principal normal: $\kappa = R/h^2 = R/(R^2+b^2)$, $\mathbf{n} = -\mathbf{e}_\rho$. The normal is **horizontal** and points to the axis, although the helix is inclined.

Step 3. Binormal and torsion

$\mathbf{b} = \mathbf{t}\times\mathbf{n} = \dfrac{1}{h}(R\,\mathbf{e}_\varphi + b\,\mathbf{k})\times(-\mathbf{e}_\rho)$. With $\mathbf{e}_\varphi\times\mathbf{e}_\rho = -\mathbf{k}$ and $\mathbf{k}\times\mathbf{e}_\rho = \mathbf{e}_\varphi$:

$$\mathbf{b} = \frac{1}{h}\big(R\,\mathbf{k} - b\,\mathbf{e}_\varphi\big)$$

Then $\dfrac{d\mathbf{b}}{ds} = \dfrac{1}{h}\cdot\dfrac{-b}{h}\dfrac{d\mathbf{e}_\varphi}{d\varphi} = \dfrac{b}{h^2}\,\mathbf{e}_\rho = -\dfrac{b}{h^2}\,\mathbf{n}$, so $\tau = b/h^2 = b/(R^2+b^2)$. In matrix form the triad is $(\mathbf{t},\mathbf{n},\mathbf{b})^T = [Q]\,(\mathbf{e}_\rho,\mathbf{e}_\varphi,\mathbf{k})^T$ with rows $(0,\ R/h,\ b/h)$, $(-1,\ 0,\ 0)$, $(0,\ -b/h,\ R/h)$: the rows are unit and mutually orthogonal ($R^2/h^2 + b^2/h^2 = 1$, $-bR/h^2 + bR/h^2 = 0$), so $[Q][Q]^T = I$, and expanding along the second row $\det[Q] = -(-1)\left(\frac{R}{h}\frac{R}{h} + \frac{b}{h}\frac{b}{h}\right) = +1$: $\{\mathbf{t},\mathbf{n},\mathbf{b}\}$ is right-handed. The third Frenet formula $\dfrac{d\mathbf{n}}{ds} = -\kappa\,\mathbf{t} + \tau\,\mathbf{b}$ checks: $\dfrac{1}{h}\dfrac{d(-\mathbf{e}_\rho)}{d\varphi} = -\dfrac{\mathbf{e}_\varphi}{h}$ and $-\dfrac{R}{h^3}(R\mathbf{e}_\varphi + b\mathbf{k}) + \dfrac{b}{h^3}(R\mathbf{k} - b\mathbf{e}_\varphi) = -\dfrac{R^2+b^2}{h^3}\mathbf{e}_\varphi = -\dfrac{\mathbf{e}_\varphi}{h}$.

Step 4. Intrinsic acceleration

The intrinsic acceleration (Notes §2.5) is $\mathbf{a} = \ddot{s}\,\mathbf{t} + \kappa\dot{s}^2\,\mathbf{n}$. With $\dot{s} = h\dot{\varphi}$, $\ddot{s} = h\ddot{\varphi}$ and $\kappa\dot{s}^2 = \dfrac{R}{h^2}h^2\dot{\varphi}^2$:

$$\mathbf{a} = h\ddot{\varphi}\,\mathbf{t} + R\dot{\varphi}^2\,\mathbf{n}$$

Cross-check with (a): $\mathbf{a}\cdot\mathbf{t} = \dfrac{1}{h}(R\cdot R\ddot{\varphi} + b\cdot b\ddot{\varphi}) = h\ddot{\varphi}$, $\mathbf{a}\cdot\mathbf{n} = R\dot{\varphi}^2$, $\mathbf{a}\cdot\mathbf{b} = \dfrac{1}{h}(-bR\ddot{\varphi} + Rb\ddot{\varphi}) = 0$.

KEY RESULTS

- $\mathbf{t} = \dfrac{R\,\mathbf{e}_\varphi + b\,\mathbf{k}}{h}$, $\mathbf{n} = -\mathbf{e}_\rho$ (horizontal, towards the axis), $\mathbf{b} = \dfrac{-b\,\mathbf{e}_\varphi + R\,\mathbf{k}}{h}$.
- $\kappa = \dfrac{R}{R^2+b^2}$, $\tau = \dfrac{b}{R^2+b^2}$.
- $\mathbf{a} = h\ddot{\varphi}\,\mathbf{t} + R\dot{\varphi}^2\,\mathbf{n}$, no binormal component.

Part (c) · 1 point

Forces on the bead

Write the weight and the reaction of the wire in the intrinsic basis. How many unknowns does the problem have?

Step 1. Weight

$\mathbf{k}\cdot\mathbf{t} = b/h$, $\mathbf{k}\cdot\mathbf{n} = -\mathbf{k}\cdot\mathbf{e}_\rho = 0$, $\mathbf{k}\cdot\mathbf{b} = R/h$. Hence

$$\mathbf{W} = -mg\,\mathbf{k} = -\frac{mgb}{h}\,\mathbf{t} - \frac{mgR}{h}\,\mathbf{b}$$

The weight has no component along $\mathbf{n}$, because $\mathbf{n}$ is horizontal.

Step 2. Reaction

The wire is smooth: its reaction is orthogonal to the tangent, i.e. it lies in the normal plane spanned by $\mathbf{n}$ and $\mathbf{b}$ (Notes §6.5):

$$\mathbf{N} = N_n\,\mathbf{n} + N_b\,\mathbf{b}$$

The three scalar equations of Newton's law will determine the three unknowns $\varphi(t)$, N_n, N_b.

KEY RESULTS

- $\mathbf{k} = \dfrac{b}{h}\mathbf{t} + \dfrac{R}{h}\mathbf{b}$, so $\mathbf{W} = -mg\dfrac{b}{h}\,\mathbf{t} - mg\dfrac{R}{h}\,\mathbf{b}$.
- $\mathbf{N} = N_n\,\mathbf{n} + N_b\,\mathbf{b}$; unknowns $\varphi(t)$, N_n, N_b.

Part (d) · 2 points

Equation of motion and reaction

Obtain the equation of motion free of reactions and integrate it. Obtain N_n and N_b as functions of $\dot\varphi$. Does the bead move like a pendulum?

Step 1. Projections

With $\mathbf{a}$ of (b) and the forces of (c):

$$\mathbf{t}:\ mh\ddot\varphi = -\frac{mgb}{h}, \qquad \mathbf{n}:\ mR\dot\varphi^2 = N_n, \qquad \mathbf{b}:\ 0 = -\frac{mgR}{h} + N_b$$

Step 2. Equation of motion

The tangential equation contains no reaction:

$$\ddot\varphi = -\frac{gb}{h^2} = -\frac{gb}{R^2+b^2}$$

The right-hand side does not depend on φ: there is **no restoring term** and no equilibrium. The tangential acceleration is $\ddot s = h\ddot\varphi = -g\,\dfrac{b}{h} = -g\sin\alpha$, with α the constant inclination of the helix ($\sin\alpha = b/h$): the bead moves exactly as on an inclined plane wound around a cylinder. It is not a pendulum.

Step 3. Integration

Integrating once between 0 and t (Barrow's rule, $\dot\varphi(0)=0$): $\dot\varphi(t)-\dot\varphi(0)=\displaystyle\int_0^t \ddot\varphi\,dt' = -\frac{gb}{h^2}\,[t']_0^t$, so $\dot\varphi = -\dfrac{gb}{h^2}t$. Integrating again with $\varphi(0)=\varphi_0$:

$$\varphi(t)=\varphi_0+\int_0^t\Big(-\frac{gb}{h^2}t'\Big)dt'=\varphi_0-\frac{gb}{h^2}\Big[\frac{t'^2}{2}\Big]_0^t=\varphi_0-\frac{gb}{2h^2}\,t^2$$

The bead descends (φ and $z=b\varphi$ decrease).

Step 4. Reactions

$$N_n = mR\dot\varphi^2, \qquad N_b=\frac{mgR}{h}$$

N_b supports the part of the weight normal to the wire and is constant; N_n (pointing to the axis) supplies the centripetal force of the horizontal circular motion and grows with the speed.

KEY RESULTS

- $\ddot\varphi=-\dfrac{gb}{R^2+b^2}$: **constant**. $\varphi(t)=\varphi_0-\dfrac{gb}{2h^2}t^2$.
- $N_n=mR\dot\varphi^2,\ N_b=\dfrac{mgR}{h}$ (constant).

Part (e) · 1.5 points

First integral

Show that the mechanical energy is conserved and use it to write $\dot\varphi^2$ and N_n as functions of φ.

Step 1. Power of the forces

$\mathbf{v}=h\dot\varphi\,\mathbf{t}$ and $\mathbf{N}\cdot\mathbf{t}=0$, so $\mathbf{N}\cdot\mathbf{v}=0$: the reaction of a fixed smooth curve does no work (Notes §6.4). The weight is conservative with $V=mgz=mgb\varphi$. Hence (Notes §4.5)

$$E=\tfrac12 m|\mathbf{v}|^2+mgz=\tfrac12 mh^2\dot\varphi^2+mgb\varphi=\text{const}$$

Step 2. Same result from the equation of motion

Multiply $h^2\ddot{\varphi} = -gb$ by $\dot{\varphi}$: $h^2\dot{\varphi}\ddot{\varphi} = \dfrac{d}{dt}\left(\tfrac{1}{2}h^2\dot{\varphi}^2\right)$ (chain rule) and $-gb\dot{\varphi} = \dfrac{d}{dt}(-gb\varphi)$. Integrating from 0 to t with Barrow's rule: $\tfrac{1}{2}h^2\dot{\varphi}^2 - 0 = -gb(\varphi - \varphi_0)$.

Step 3. Speed and reaction along the wire

$$\dot{\varphi}^2 = \frac{2gb(\varphi_0 - \varphi)}{h^2}, \qquad N_n = mR\dot{\varphi}^2 = \frac{2mgRb(\varphi_0 - \varphi)}{h^2}$$

With $\kappa = R/h^2$ and the height fallen $\Delta z = b(\varphi_0 - \varphi)$: $N_n = 2mg\kappa\,\Delta z$, i.e. $N_n = m\kappa v^2$ with $v^2 = 2g\Delta z$.

KEY RESULTS

- $E = \tfrac{1}{2}mh^2\dot{\varphi}^2 + mgb\varphi = mgb\varphi_0$.
- $\dot{\varphi}^2 = \dfrac{2gb(\varphi_0 - \varphi)}{h^2}$, $N_n = \dfrac{2mgRb(\varphi_0 - \varphi)}{h^2} = 2mg\,\kappa\,\Delta z$ with $\Delta z = b(\varphi_0 - \varphi)$.

Part (f) · 2 points

One full turn: time, speed and reaction

Find the time t_1 the bead needs to complete one turn and its speed then. For $R = 0.30$ m, $b = 0.40$ m, $g = 9.81$ m s^{-2} give κ, τ, $\ddot{\varphi}$, t_1, the speed and the reaction (in units of mg) at t_1. Is the acceleration vector of the bead constant?

Step 1. Time for one turn

From $\varphi(t_1) = \varphi_0 - 2\pi$: $\dfrac{gb}{2h^2}t_1^2 = 2\pi \Rightarrow t_1 = \sqrt{\dfrac{4\pi h^2}{gb}} = 2h\sqrt{\dfrac{\pi}{gb}}$.

Step 2. Speed after one turn

From (e) with $\varphi_0 - \varphi = 2\pi$: $\dot{\varphi}_1^2 = \dfrac{4\pi gb}{h^2}$ and $v_1 = h|\dot{\varphi}_1| = \sqrt{4\pi gb}$. Equivalently $v_1 = h|\dot{\varphi}(t_1)| = h\dfrac{gb}{h^2}t_1 = \dfrac{gb}{h}\cdot 2h\sqrt{\dfrac{\pi}{gb}} = \sqrt{4\pi gb}$. Since the fall per turn is $2\pi b$, $v_1 = \sqrt{2g\cdot 2\pi b}$: the speed depends only on the height fallen, not on R.

Step 3. Numbers

$h = \sqrt{0.09 + 0.16} = 0.5$ m; $\kappa = 0.3/0.25 = 1.2$ m^{-1}; $\tau = 0.4/0.25 = 1.6$ m^{-1}; $\ddot{\varphi} = -9.81 \cdot 0.4/0.25 = -15.696$ rad s^{-2}, $\ddot{s} = h\ddot{\varphi} = -7.848$ m s^{-2} ($\sin\alpha = 0.8$, $\alpha = 53.13^\circ$).

$$t_1 = 2(0.5)\sqrt{\frac{\pi}{9.81\cdot 0.4}} = 0.8948\ \text{s}, \qquad v_1 = \sqrt{4\pi\cdot 9.81\cdot 0.4} = 7.022\ \text{m/s}, \qquad |\dot{\varphi}_1| = \frac{v_1}{h} = 14.04\ \text{rad/s}$$

Step 4. Reaction after one turn

$$\frac{N_n}{mg} = \frac{R\dot\varphi_1^2}{g} = \frac{4\pi R b}{h^2} = 4\pi \cdot \frac{0.12}{0.25} = 1.92\pi \approx 6.032, \qquad \frac{N_b}{mg} = \frac{R}{h} = 0.6$$

$|\mathbf{N}| = mg\sqrt{6.032^2 + 0.6^2} \approx 6.062\, mg$. The tangential acceleration is constant, but $\mathbf{a} = h\ddot\varphi\,\mathbf{t} + R\dot\varphi^2\,\mathbf{n}$ has a growing normal part and a direction that turns with the triad: the acceleration vector is not constant.

KEY RESULTS

- $t_1 = 2h\sqrt{\dfrac{\pi}{gb}},\ v_1 = \sqrt{4\pi g b} = \sqrt{2g\,(2\pi b)}$ (independent of R).
- $\kappa = 1.2\ \mathrm{m^{-1}}, \tau = 1.6\ \mathrm{m^{-1}}, \ddot\varphi = -15.70\ \mathrm{rad\ s^{-2}}, \ddot s = -7.848\ \mathrm{m\ s^{-2}}, t_1 = 0.8948\ \mathrm{s}, v_1 = 7.022$ m/s, $|\dot\varphi_1| = 14.04$ rad/s.
- $N_n = 1.92\pi\, mg \approx 6.032\, mg$, $N_b = 0.6\, mg$, $|\mathbf{N}| \approx 6.062\, mg$. The acceleration is not constant: its normal part $R\dot\varphi^2$ grows.

Interpretation and checks

- **Dimensional check:** κ and τ are $\mathrm{m/m^2} = \mathrm{m^{-1}}$; gb/h^2 is $(\mathrm{m\ s^{-2}})(\mathrm{m})/\mathrm{m^2} = \mathrm{s^{-2}}$, an angular acceleration; $t_1 = 2h\sqrt{\pi/(gb)}$ has $\mathrm{m}/\sqrt{\mathrm{m^2\ s^{-2}}} = \mathrm{s}$; $N = mR\dot\varphi^2$ is $\mathrm{kg\ m\ s^{-2}} = \mathrm{N}$.
- **Limits:** $b \to 0$ (horizontal ring): $\ddot\varphi \to 0$, the bead stays at rest and $N_b \to mg$ carries the whole weight. $R \to 0$ (vertical straight wire): $\ddot z = b\ddot\varphi = -g\,b^2/h^2 \to -g$ (free fall), $\kappa \to 0$ and the reaction vanishes. κ is largest for $b \to 0$ and τ is largest for $b = R$.
- **Common mistakes in the exam:** (1) expecting a pendulum-type equation: on a helix the gravity component along the wire is the same everywhere, so $\ddot\varphi$ is constant; (2) taking $\mathbf{n}$ along the "radius of the coil" tilted with the helix: it is exactly $-\mathbf{e}_\rho$, horizontal; (3) forgetting N_b, which exists even at rest; (4) concluding that the acceleration is constant because $\ddot\varphi$ is: the normal acceleration $R\dot\varphi^2$ grows linearly with the height fallen.

Problem · Unit 4 · 10 points · 60 min · High difficulty / trap

P02 — Particle on a smooth paraboloid of revolution: axial angular momentum, circular orbits and radial oscillations

*A particle P of mass m moves on the **inner** face of a smooth, fixed paraboloid of revolution whose axis Oz is vertical (upward; uniform gravity $-g\,\mathbf{k}$). In cylindrical coordinates (r,φ,z) of the inertial frame $S_0=\{O;\mathbf{i},\mathbf{j},\mathbf{k}\}$ the surface is*

$$z=\frac{r^2}{2c}\qquad (c>0)$$

with O at its vertex. The particle rests on the surface from above (the contact can only push P away from the surface, towards the inside of the bowl). The azimuth φ is measured from the x axis, counter-clockwise seen from above.

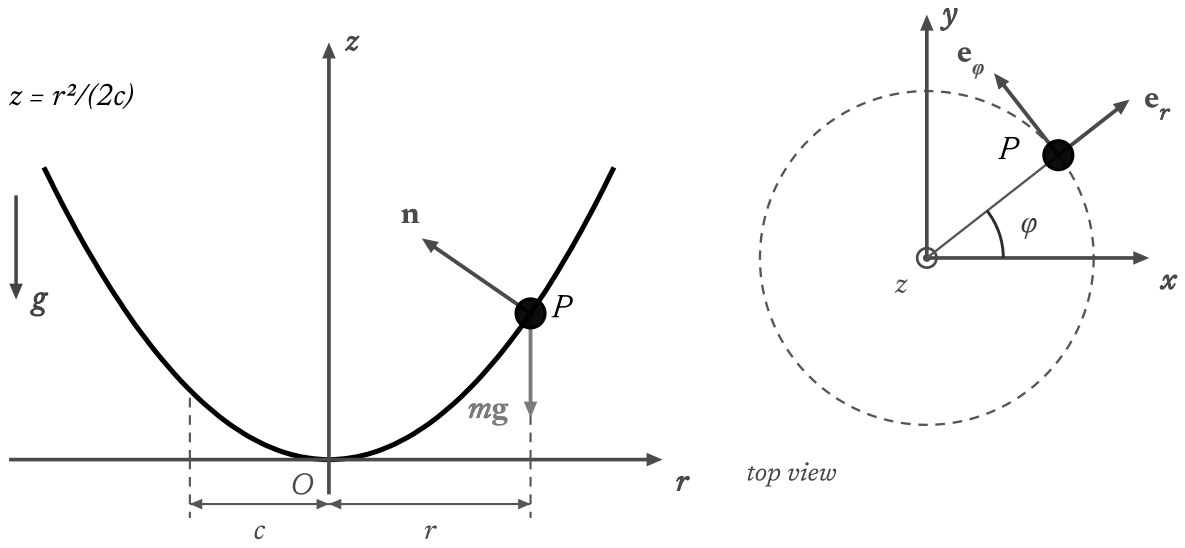


Figure 1. Particle P on the smooth inner surface $z=r^2/(2c)$ (axis Oz vertical, gravity $-g\,\mathbf{k}$). Left: meridian plane through P with the unit normal $\mathbf{n}$; right: top view with the azimuth φ.

Use the cylindrical basis $\{\mathbf{e}_r,\mathbf{e}_\varphi,\mathbf{k}\}$, and write $q=\sqrt{1+r^2/c^2}$ and $\Omega_0=\sqrt{g/c}$.

Setup

- Data: m, c, g. In part (f): initial radius $r_0 = c/2$, $\dot r(0) = 0$, initial velocity horizontal and tangent to the parallel with speed $v_0 = 2r_0\Omega_0 = c\,\Omega_0$; for numbers, $c = 1$ m and $g = 9.81$ m s^{-2}.
- Hypotheses: point particle, no friction, S_0 inertial. The contact is **unilateral**: the model is valid while the normal reaction is non-negative (checked in (f)).

Part (a) · 1.5 points

Degrees of freedom and kinematics

How many degrees of freedom has P while it stays on the surface? Write its velocity and acceleration in the cylindrical basis using the constraint.

Step 1. Count

A free particle needs (r, φ, z). The surface $f = z - r^2/(2c) = 0$ is one holonomic equation without explicit time: $3 - 1 = 2$ degrees of freedom (Notes §2.7), with generalized coordinates r and φ. The constraint is unilateral, because the particle is only pressed against the surface.

Step 2. Derivatives of z

By the chain rule, $\dot z = \dfrac{dz}{dr}\dot r = \dfrac{r\dot r}{c}$. By the product rule, $\ddot z = \dfrac{1}{c}\dfrac{d}{dt}(r\dot r) = \dfrac{\dot r\,\dot r + r\ddot r}{c} = \dfrac{\dot r^2 + r\ddot r}{c}$.

Step 3. Change of basis

The cylindrical basis turns with φ; its relation to the inertial basis is

$$\left(\mathbf{e}_r\ \mathbf{e}_\varphi\ \mathbf{k}\right) = [R]\left(\mathbf{i}\ \mathbf{j}\ \mathbf{k}\right), \qquad [R] = \begin{pmatrix} \cos\varphi & \sin\varphi & 0 \\ -\sin\varphi & \cos\varphi & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Checks: rows 1 and 2 have length $\cos^2\varphi + \sin^2\varphi = 1$ and $-\cos\varphi\sin\varphi + \sin\varphi\cos\varphi = 0$, row 3 is $\mathbf{k}$, so $[R][R]^T = I$; $\det[R] = \cos^2\varphi + \sin^2\varphi = +1$ (right-handed, $\mathbf{e}_r \times \mathbf{e}_\varphi = \mathbf{k}$). By the chain rule, $\dot{\mathbf{e}}_r = \dot\varphi\,\mathbf{e}_\varphi$ and $\dot{\mathbf{e}}_\varphi = -\dot\varphi\,\mathbf{e}_r$.

Step 4. Velocity and acceleration

In cylindrical coordinates (Notes §2.6.2), $\mathbf{v} = \dot r\,\mathbf{e}_r + r\dot\varphi\,\mathbf{e}_\varphi + \dot z\,\mathbf{k}$ and $\mathbf{a} = (\ddot r - r\dot\varphi^2)\mathbf{e}_r + (r\ddot\varphi + 2\dot r\dot\varphi)\mathbf{e}_\varphi + \ddot z\,\mathbf{k}$. With Step 2:

$$\mathbf{v} = \dot r\,\mathbf{e}_r + r\dot\varphi\,\mathbf{e}_\varphi + \frac{r\dot r}{c}\,\mathbf{k}, \qquad |\mathbf{v}|^2 = q^2\dot r^2 + r^2\dot\varphi^2$$

$$\mathbf{a} = (\ddot r - r\dot\varphi^2)\,\mathbf{e}_r + (2\dot r\dot\varphi + r\ddot\varphi)\,\mathbf{e}_\varphi + \frac{\dot r^2 + r\ddot r}{c}\,\mathbf{k}$$

KEY RESULTS

- 2 degrees of freedom, coordinates (r, φ); one holonomic, scleronomic, unilateral constraint.
- $\dot z = r\dot r/c,\ \ddot z = (\dot r^2 + r\ddot r)/c$.
- $\mathbf{a} = (\ddot r - r\dot\varphi^2)\mathbf{e}_r + (2\dot r\dot\varphi + r\ddot\varphi)\mathbf{e}_\varphi + \dfrac{\dot r^2 + r\ddot r}{c}\mathbf{k}$.

Part (b) · 1 point

Forces

Obtain the unit normal of the surface and write the forces on P.

Step 1. Normal vector

$\nabla f = \dfrac{\partial f}{\partial r}\mathbf{e}_r + \dfrac{1}{r}\dfrac{\partial f}{\partial \varphi}\mathbf{e}_\varphi + \dfrac{\partial f}{\partial z}\mathbf{k} = -\dfrac{r}{c}\,\mathbf{e}_r + \mathbf{k}$, of modulus $\sqrt{1 + r^2/c^2} = q$ (Notes §6.3). Its $\mathbf{k}$ component is positive, so it points into the bowl, the side on which the particle lies:

$$\mathbf{n} = \frac{1}{q}\Big(-\frac{r}{c}\,\mathbf{e}_r + \mathbf{k}\Big)$$

Check: $\mathbf{n}\cdot\mathbf{v} = \dfrac{1}{q}\Big(-\dfrac{r\dot r}{c} + \dfrac{r\dot r}{c}\Big) = 0$.

Step 2. Forces

The weight $\mathbf{W} = -mg\,\mathbf{k}$ and the smooth normal reaction $\mathbf{N} = N\,\mathbf{n}$ (Notes §6.4). Contact requires $N \geq 0$. Note that $\mathbf{N}$ lies in the meridian plane (it has no $\mathbf{e}_\varphi$ component) but does **not** pass through O in general.

KEY RESULTS

- $\mathbf{n} = \dfrac{-(r/c)\,\mathbf{e}_r + \mathbf{k}}{q}$ (towards the inside of the bowl).
- $\mathbf{W} = -mg\,\mathbf{k},\ \mathbf{N} = N\mathbf{n}$ with $N \geq 0$ for contact.

Part (c) · 2 points

Equations of motion and first integrals

Project Newton's law, show that the component $H = r^2\dot\varphi$ of the specific angular momentum along the axis is conserved, write the energy and obtain one equation for $r(t)$ free of N. Give N as a function of the state.

Step 1. Projections

With $\mathbf{a}$ of (a), $\mathbf{N}\cdot\mathbf{e}_r = -Nr/(cq)$, $\mathbf{N}\cdot\mathbf{k} = N/q$:

$$\mathbf{e}_r:\ m(\ddot r - r\dot\varphi^2) = -\frac{Nr}{c\,q} \qquad \mathbf{e}_\varphi:\ m(2\dot r\dot\varphi + r\ddot\varphi) = 0 \qquad \mathbf{k}:\ m\frac{\dot r^2 + r\ddot r}{c} = \frac{N}{q} - mg$$

Step 2. Axial angular momentum

By the product rule, $\dfrac{d}{dt}(r^2\dot\varphi) = 2r\dot r\dot\varphi + r^2\ddot\varphi = r(2\dot r\dot\varphi + r\ddot\varphi)$, so the $\mathbf{e}_\varphi$ equation is $\dfrac{m}{r}\dfrac{d}{dt}(r^2\dot\varphi) = 0$: $H = r^2\dot\varphi$ is constant. Physical reason (Notes §4.8): $\mathbf{k}\cdot(\mathbf{r}\times\mathbf{F})$ is the moment about the z axis; $\mathbf{W}$ is parallel to the axis and $\mathbf{N}$ lies in the meridian plane, which contains the axis, so both have zero moment about it. The full vector $\mathbf{H}_O$ is **not** conserved ($\mathbf{r}\times\mathbf{N} \neq \mathbf{0}$ in general): only its axial component is.

Step 3. Equation for r

From $\mathbf{k}$: $\dfrac{N}{q} = m\Big(\dfrac{\dot r^2 + r\ddot r}{c} + g\Big)$. Substituting in $\mathbf{e}_r$ (which contains $\dfrac{r}{c}\cdot\dfrac{N}{q}$) and using $\dot\varphi = H/r^2$:

$$\ddot r - \frac{H^2}{r^3} = -\frac{r}{c}\Big(\frac{\dot r^2 + r\ddot r}{c} + g\Big) \implies \Big(1 + \frac{r^2}{c^2}\Big)\ddot r + \frac{r\dot r^2}{c^2} - \frac{H^2}{r^3} + \frac{gr}{c} = 0$$

Step 4. Energy

$\mathbf{N}\cdot\mathbf{v} = 0$ (fixed smooth surface) and $\mathbf{W}$ derives from $V = mgz = mgr^2/(2c)$, so $E = \tfrac12 m|\mathbf{v}|^2 + V$ is constant (Notes §4.5). With $|\mathbf{v}|^2$ of (a) and $r\dot\varphi = H/r$:

$$\frac{E}{m} = \frac12\Big[\Big(1 + \frac{r^2}{c^2}\Big)\dot r^2 + \frac{H^2}{r^2}\Big] + \frac{gr^2}{2c}$$

Check: $\dfrac{d}{dt}\dfrac{E}{m} = \dot r\Big[\Big(1 + \dfrac{r^2}{c^2}\Big)\ddot r + \dfrac{r\dot r^2}{c^2} - \dfrac{H^2}{r^3} + \dfrac{gr}{c}\Big] = 0$ by Step 3 (the term $\dfrac{r\dot r^3}{c^2}$ comes from the chain rule on $\dfrac{r^2}{c^2}\dot r^2$).

Step 5. Reaction

$N = mq\Big(\dfrac{\dot r^2 + r\ddot r}{c} + g\Big)$, with $\ddot r$ from Step 3.

KEY RESULTS

- $H = r^2\dot\varphi = \text{const}$ (the moment about the axis of $\mathbf{W}$ and $\mathbf{N}$ is zero; $\mathbf{H}_O$ itself is not conserved).
- $\Big(1 + \dfrac{r^2}{c^2}\Big)\ddot r + \dfrac{r\dot r^2}{c^2} - \dfrac{H^2}{r^3} + \dfrac{gr}{c} = 0.$
- $\dfrac{E}{m} = \tfrac{1}{2}\Big[\Big(1 + \dfrac{r^2}{c^2}\Big)\dot r^2 + \dfrac{H^2}{r^2}\Big] + \dfrac{gr^2}{2c};\ N = mq\Big(\dfrac{\dot r^2 + r\ddot r}{c} + g\Big).$

Part (d) · 1.5 points

Horizontal circular orbits

Find the angular rate of a horizontal circular motion at radius r_0, its speed and the normal reaction. Discuss the dependence on r_0.

Step 1. Rate

With $\dot r = \ddot r = 0$ and $H = r_0^2\dot\varphi$: $\dfrac{H^2}{r_0^3} = r_0\dot\varphi^2 = \dfrac{gr_0}{c}$, hence

$$\dot\varphi^2 = \frac{g}{c} \quad\Longrightarrow\quad \Omega_0 = \sqrt{\frac{g}{c}}, \qquad T = 2\pi\sqrt{\frac{c}{g}}, \qquad v = r_0\Omega_0$$

All horizontal circles are described in the same time, whatever their radius (isochronous orbits). This is the trap of the item: on a cone or on a sphere the rate depends on the radius; on the paraboloid the slope $dz/dr = r/c$ grows exactly in proportion to the centripetal acceleration $r\Omega^2$ needed.

Step 2. Reaction

$N = mq_0(0 + g) = mg\sqrt{1 + r_0^2/c^2}$. Its vertical component $N/q_0 = mg$ carries the weight; its horizontal component $Nr_0/(cq_0) = mgr_0/c = mr_0\Omega_0^2$ is the centripetal force.

KEY RESULTS

- $\dot\varphi^2 = g/c$: $\Omega_0 = \sqrt{g/c}$, **the same for every** r_0; period $2\pi\sqrt{c/g}$; speed $r_0\Omega_0$.
- $N = mg\sqrt{1 + r_0^2/c^2}$.

Part (e) · 2 points

Small radial oscillations about a circular orbit

Perturb the circular orbit of radius r_0 keeping $H = r_0^2\Omega_0$. Obtain the frequency ω_r of the small radial oscillations and the ratio ω_r/Ω_0. For which r_0 is the perturbed orbit closed after one revolution? What happens for $r_0 \to 0$ and $r_0 = c$?

Step 1. Linearisation

$G(r,\dot r) = \dfrac{F(r) - r\dot r^2/c^2}{1 + r^2/c^2}$ with $F(r) = \dfrac{H^2}{r^3} - \dfrac{gr}{c}$, and $F(r_0) = 0$ by (d). Put $r = r_0 + \varepsilon$ with ε, $\dot\varepsilon$ small. $\partial G/\partial \dot r = -2r\dot r/(c^2 q^2)$ vanishes at $\dot r = 0$. Since $F(r_0) = 0$, $\left.\dfrac{\partial G}{\partial r}\right|_{r_0} = \dfrac{F'(r_0)}{1 + r_0^2/c^2}$ (the derivative of the denominator multiplies $F(r_0) = 0$). With $F'(r) = -\dfrac{3H^2}{r^4} - \dfrac{g}{c}$ and $H^2 = r_0^4\, g/c$:

$$F'(r_0) = -\frac{3g}{c} - \frac{g}{c} = -\frac{4g}{c} \qquad \Longrightarrow \qquad \ddot\varepsilon + \frac{4g}{c\,(1 + r_0^2/c^2)}\,\varepsilon = 0$$

Step 2. Frequency and ratio

$$\omega_r^2 = \frac{4gc}{c^2 + r_0^2}, \qquad \frac{\omega_r}{\Omega_0} = \sqrt{\frac{4c^2}{c^2 + r_0^2}} = \frac{2c}{\sqrt{c^2 + r_0^2}}$$

The orbit is stable (positive ω_r^2) for every r_0. The trap: the radial frequency is **not** Ω_0, and the factor $1 + r_0^2/c^2$ (the inertia of the vertical motion that comes with $\dot r$) cannot be dropped.

Step 3. Special radii

- $r_0 \to 0$: ratio 2: two radial oscillations per revolution, a centred ellipse, as for an isotropic harmonic bowl ($z \approx r^2/(2c)$ is then the potential of a linear spring of stiffness mg/c).
- $r_0 = c$: ratio $\sqrt{2}$, irrational: the perturbed orbit precesses and never closes.
- $r_0 = \sqrt{3}\,c$: ratio $2c/\sqrt{4c^2} = 1$: one radial oscillation per revolution, a closed orbit.

KEY RESULTS

- $\omega_r^2 = \dfrac{4g}{c\,(1 + r_0^2/c^2)} = \dfrac{4gc}{c^2 + r_0^2}, \quad \dfrac{\omega_r}{\Omega_0} = \dfrac{2c}{\sqrt{c^2 + r_0^2}}.$
- Ratio 1 for $r_0 = \sqrt{3}\,c$ (closed after one revolution); 2 for $r_0 \to 0$; $\sqrt{2}$ for $r_0 = c$ (not closed).

Part (f) · 2 points

Turning points and contact

The particle starts at r_0 with $\dot r = 0$ and a horizontal velocity v_0 tangent to the parallel. Find the turning radii. Then take $r_0 = c/2$, $v_0 = 2r_0\Omega_0$: give the turning radii, $\ddot r$ and N at both, and decide whether the particle ever leaves the surface.

Step 1. Constants

$\mathbf{v}(0) = v_0\mathbf{e}_\varphi$, so $H = r_0 \cdot r_0\dot\varphi(0) = r_0 v_0$ and $E/m = \tfrac12 v_0^2 + \dfrac{g r_0^2}{2c}$.

Step 2. Turning radii

At $\dot r = 0$ the energy of (c) gives $\dfrac{H^2}{r^2} + \dfrac{gr^2}{c} = v_0^2 + \dfrac{gr_0^2}{c}$. Multiply by $u = r^2$:

$$\frac{g}{c}u^2 - \Big(v_0^2 + \frac{gr_0^2}{c}\Big)u + r_0^2 v_0^2 = 0 \iff (u - r_0^2)\Big(\frac{gu}{c} - v_0^2\Big) = 0$$

(expanding the product gives back the three terms). Roots: $r = r_0$ and $r^* = v_0\sqrt{c/g} = v_0/\Omega_0$. If $v_0 > r_0\Omega_0$ the particle climbs to $r^* > r_0$; if $v_0 = r_0\Omega_0$ both coincide (circular orbit of (d)). The time between them is a quadrature: from the energy, $dt = \dfrac{q\,dr}{\sqrt{2E/m - H^2/r^2 - gr^2/c}}$.

Step 3. The numerical case

$r_0 = c/2$, $v_0 = c\Omega_0$: $r^* = c\Omega_0/\Omega_0 = c$. So r oscillates between $c/2$ and c (z between $c/8$ and $c/2$). With $H^2 = r_0^2 v_0^2 = c^3 g/4$, $E/m = \tfrac58 gc$. For $c = 1$ m: $\Omega_0 = v_0 = 3.132$ (rad/s and m/s), $\omega_r(r_0) = \tfrac{4}{\sqrt5}\Omega_0 = 5.603$ rad/s for the small oscillations of (e), which do not apply here because the amplitude is large.

At the turning points $\dot r = 0$, so $q^2\ddot r = \dfrac{H^2}{r^3} - \dfrac{gr}{c}$:

- $r = c/2$: $\dfrac{H^2}{r^3} = \dfrac{c^3 g/4}{c^3/8} = 2g$, $\dfrac{gr}{c} = \dfrac{g}{2}$, $q^2 = \dfrac54$: $\ddot r = \dfrac{3g/2}{5/4} = \dfrac65 g$.
- $r = c$: $\dfrac{H^2}{r^3} = \dfrac{g}{4}$, $\dfrac{gr}{c} = g$, $q^2 = 2$: $\ddot r = -\dfrac{3g}{8}$.

Then $N = mq\Big(\dfrac{r\ddot r}{c} + g\Big)$: at $r = c/2$, $N = m\dfrac{\sqrt5}{2}\Big(\dfrac35 g + g\Big) = \dfrac{4}{\sqrt5}mg \approx 1.789\,mg$; at $r = c$, $N = m\sqrt2\Big(-\dfrac38 g + g\Big) = \dfrac{5\sqrt2}{8}mg \approx 0.884\,mg$.

Step 4. Contact along any motion

From Step 3 of (c), multiplied by r: $r\ddot r\,q^2 = \dfrac{H^2}{r^2} - \dfrac{r^2\dot r^2}{c^2} - \dfrac{gr^2}{c}$. Adding $q^2\dot r^2$ to both sides and using $q^2 - r^2/c^2 = 1$:

$$q^2\left(\dot r^2 + r\ddot r\right) = \dot r^2 + \frac{H^2}{r^2} - \frac{gr^2}{c}$$

From the energy of (c), $q^2\dot r^2 = 2E/m - H^2/r^2 - gr^2/c$. Multiplying the previous line by q^2 and substituting $q^2\dot r^2$:

$$q^4\left(\dot r^2 + r\ddot r\right) = \frac{2E}{m} - \frac{H^2}{r^2} - \frac{gr^2}{c} + q^2\frac{H^2}{r^2} - q^2\frac{gr^2}{c} = \frac{2E}{m} + \frac{H^2}{c^2} - \left(2 + \frac{r^2}{c^2}\right)\frac{gr^2}{c}$$

because $(q^2-1)\dfrac{H^2}{r^2} = \dfrac{H^2}{c^2}$ and $(1+q^2)\dfrac{gr^2}{c} = \left(2+\dfrac{r^2}{c^2}\right)\dfrac{gr^2}{c}$. So

$$\dot r^2 + r\ddot r = \frac{1}{q^4}\left[\frac{2E}{m} + \frac{H^2}{c^2} - \left(2 + \frac{r^2}{c^2}\right)\frac{gr^2}{c}\right]$$

Substituting in $N = mq\left[(\dot r^2 + r\ddot r)/c + g\right] = \dfrac{m}{cq^3}\left[q^4(\dot r^2 + r\ddot r) + gc\,q^4\right]$ and using the identity

$$gc\,q^4 - \left(2 + \frac{r^2}{c^2}\right)\frac{gr^2}{c} = gc\left(1 + \frac{2r^2}{c^2} + \frac{r^4}{c^4}\right) - \frac{2gr^2}{c} - \frac{gr^4}{c^3} = gc$$

we get $N = \dfrac{m}{cq^3}\left(\dfrac{2E}{m} + \dfrac{H^2}{c^2} + gc\right)$, and with $cq^3 = (c^2+r^2)^{3/2}/c^2$:

$$N = \frac{m\left(2c^2E/m + H^2 + gc^3\right)}{(c^2+r^2)^{3/2}}$$

With $V = mgz \geq 0$ the energy is positive, so $N > 0$ on **every** motion: the particle never leaves the inner face. In the numerical case $2c^2E/m + H^2 + gc^3 = \tfrac{5}{4}gc^3 + \tfrac{1}{4}gc^3 + gc^3 = \tfrac{5}{2}gc^3$, so $N = \dfrac{5mgc^3}{2(c^2+r^2)^{3/2}}$, decreasing from $1.789\,mg$ at $r = c/2$ to its minimum $0.884\,mg$ at $r = c$, in agreement with Step 3.

KEY RESULTS

- $(u - r_0^2)\left(\dfrac{gu}{c} - v_0^2\right) = 0$: turning radii r_0 and $r^* = v_0\sqrt{c/g} = v_0/\Omega_0$.
- Case: $r^* = c$; $\ddot r(c/2) = \frac{6}{5}g$ (outwards), $\ddot r(c) = -\frac{3}{8}g$; $N(c/2) = \frac{4}{\sqrt{5}}mg \approx 1.789\,mg$, $N(c) = \frac{5\sqrt{2}}{8}mg \approx 0.884\,mg$.
- On every motion $N = \dfrac{m\,(2Ec^2/m + H^2 + gc^3)}{(c^2 + r^2)^{3/2}} > 0$: contact is never lost. Here $N = \dfrac{5mgc^3}{2(c^2 + r^2)^{3/2}}$.

Interpretation and checks

- **Dimensional check:** c is a length, so g/c and $\omega_r^2 = 4gc/(c^2 + r_0^2)$ are s^{-2}; $H = r^2\dot\varphi$ is $\mathrm{m}^2\ \mathrm{s}^{-1}$, H^2/r^3 and gr/c are $\mathrm{m}\ \mathrm{s}^{-2}$; in N, $(2c^2E/m + H^2 + gc^3)$ is $\mathrm{m}^4\ \mathrm{s}^{-2}$ and $(c^2 + r^2)^{3/2}$ is m^3: N is $\mathrm{kg}\ \mathrm{m}\ \mathrm{s}^{-2}$.
- **Limits:** $c \to \infty$ (flat floor): $\Omega_0 \to 0$, no horizontal circle without a wall. $r_0 \ll c$: the bowl acts as an isotropic linear spring, $\omega_r = 2\Omega_0$ and the orbits are centred ellipses.
- **Common mistakes in the exam:** (1) claiming that the angular momentum about O is conserved because "the reaction is normal": only the axial component is; (2) expecting the orbital rate to depend on r_0 (it does not on a paraboloid); (3) linearising without the factor $1 + r_0^2/c^2$, which gives $\omega_r = 2\Omega_0$ for every radius; (4) solving the quartic for the turning points instead of noticing that r_0 is a root; (5) not checking $N \geq 0$ for a unilateral contact.

Problem · Unit 5 · 10 points · 60 min · High difficulty / trap

P03 — Bead on a parabolic wire spinning about its axis: Coriolis side force, Jacobi integral and the critical rate

*A smooth rigid wire lies in a vertical plane and has the shape of the parabola $y = x^2/(2p)$ ($p > 0$), with its vertex at O and its axis along the upward vertical. The wire is driven by a motor so that it spins about that vertical axis with **constant** angular rate $\Omega > 0$. A bead P of mass m is threaded on the wire.*

Let $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ be inertial ($\mathbf{j}_0$ vertical upward, gravity $-g\,\mathbf{j}_0$) and $S_1 = \{O; \mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$ be attached to the wire, with $\mathbf{j}_1 = \mathbf{j}_0$, the wire in the plane x_1y_1 and $\boldsymbol{\omega}_{10} = \Omega\,\mathbf{j}_0$. At $t = 0$ both frames coincide, so $\mathbf{i}_1$ has turned the angle $\theta = \Omega t$ about $\mathbf{j}_0$. The position of P on the wire is given by its abscissa x along $\mathbf{i}_1$ (positive or negative).

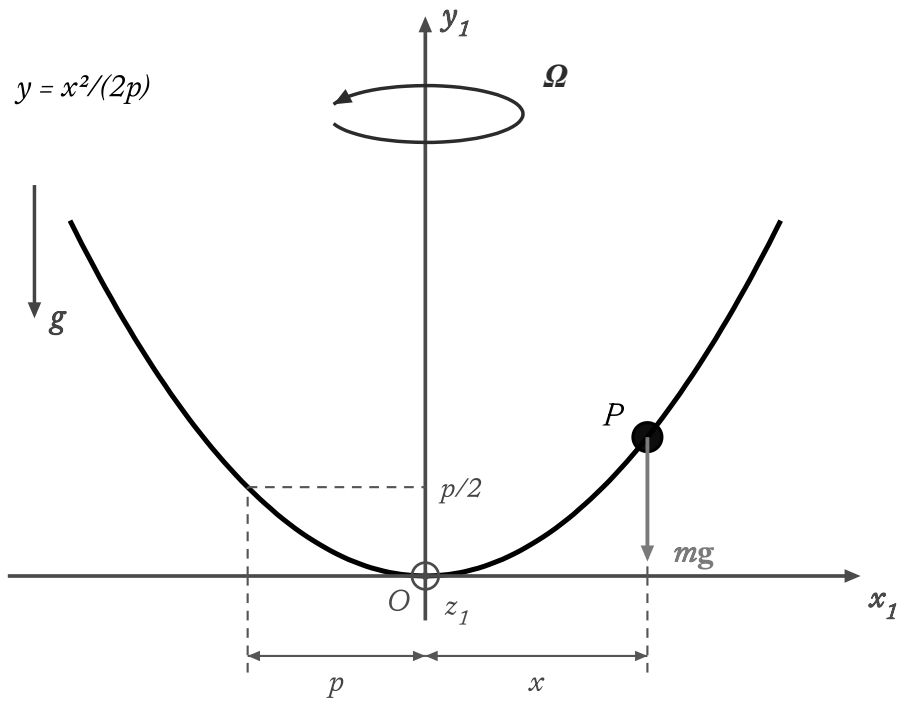


Figure 1. Wire $y = x^2/(2p)$ in the plane x_1y_1 of S_1, which turns with $\boldsymbol{\omega}_{10} = \Omega\,\mathbf{j}$ about the vertical axis; $\mathbf{k}_1$ is normal to the plane of the wire, gravity $-g\,\mathbf{j}$.

Setup

- Data: m, p, g, Ω. In part (g): $\Omega^2 = 2g/p$ and the bead is released **at rest relative to the wire** at $x_0 = p/2$.
- Hypotheses: point particle, no friction, bead threaded on the wire (bilateral constraint), S_0 inertial.
- Write $q = \sqrt{1 + x^2/p^2}$.

Part (a) · 1 point

Degrees of freedom and frames

Count the degrees of freedom and classify the constraints in S_0 and in S_1. Write the change-of-basis matrix $[{}_0R_1]$, check it, and give $\boldsymbol{\omega}_{10}$ and $\boldsymbol{\alpha}_{10}$.

Step 1. Count

In S_1 the wire is fixed: $z_1 = 0$ and $y_1 - x_1^2/(2p) = 0$ are two holonomic equations without time, so $3 - 2 = 1$ degree of freedom, coordinate x (Notes §2.7). Seen from S_0 the same curve moves, so the constraints contain t through $\theta = \Omega t$: rheonomic. This is why we work in S_1.

Step 2. Change of basis

The columns of $[{}_0R_1]$ are the components of $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ in $\mathcal{B}_0$ (Notes §1.5): $\mathbf{i}_1 = \cos\theta\,\mathbf{i}_0 - \sin\theta\,\mathbf{k}_0$, $\mathbf{j}_1 = \mathbf{j}_0$, $\mathbf{k}_1 = \sin\theta\,\mathbf{i}_0 + \cos\theta\,\mathbf{k}_0$:

$$[{}_0R_1] = \begin{pmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{pmatrix}, \qquad \det[{}_0R_1] = \cos^2\theta + \sin^2\theta = 1, \qquad [{}_0R_1][{}_0R_1]^T = I$$

Step 3. Angular velocity

$\dot{\mathbf{i}}_1 = \dot\theta(-\sin\theta\,\mathbf{i}_0 - \cos\theta\,\mathbf{k}_0) = -\Omega\,\mathbf{k}_1$ and $\boldsymbol{\omega} \times \mathbf{i}_1 = \Omega\,\mathbf{j}_1 \times \mathbf{i}_1 = -\Omega\,\mathbf{k}_1$ (Poisson, Notes §1.7), so $\boldsymbol{\omega}_{10} = \dot\theta\,\mathbf{j}_0 = \Omega\,\mathbf{j}_0$, and $\boldsymbol{\alpha}_{10} = \dot{\boldsymbol{\omega}}_{10} = \mathbf{0}$ because Ω is constant.

KEY RESULTS

- 1 degree of freedom, coordinate x. Two holonomic constraints, scleronomic in S_1 ($z_1 = 0$, $y_1 = x_1^2/(2p)$), rheonomic in S_0.
- $[{}_0R_1] = \begin{pmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{pmatrix}$, $\det = 1$, $RR^T = I$; $\boldsymbol{\omega}_{10} = \Omega\,\mathbf{j}_0$, $\boldsymbol{\alpha}_{10} = \mathbf{0}$.

Part (b) · 1.5 points

Relative and absolute kinematics

Write $\mathbf{r}$, $\mathbf{v}_{rel}$, $\mathbf{a}_{rel}$ in $\mathcal{B}_1$, and the absolute acceleration of P with the Coriolis theorem.

Step 1. Relative motion

$\mathbf{r} = x\,\mathbf{i}_1 + \dfrac{x^2}{2p}\mathbf{j}_1$. Differentiating in S_1 (where $\mathbf{i}_1, \mathbf{j}_1$ are constant), with the chain rule $\dfrac{d}{dt}\dfrac{x^2}{2p} = \dfrac{x\dot x}{p}$ and the product rule $\dfrac{d}{dt}\dfrac{x\dot x}{p} = \dfrac{\dot x^2 + x\ddot x}{p}$:

$$\mathbf{v}_{rel} = \dot{x}\,\mathbf{i}_1 + \frac{x\dot{x}}{p}\mathbf{j}_1, \qquad \mathbf{a}_{rel} = \ddot{x}\,\mathbf{i}_1 + \frac{x\ddot{x} + \dot{x}^2}{p}\mathbf{j}_1$$

Step 2. Transport and Coriolis terms

O is fixed and $\boldsymbol{\alpha}_{10} = \mathbf{0}$, so (Notes §3.2) $\mathbf{a} = \mathbf{a}_{rel} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}) + 2\boldsymbol{\omega} \times \mathbf{v}_{rel}$. With $\mathbf{j}_1 \times \mathbf{i}_1 = -\mathbf{k}_1$ and $\mathbf{j}_1 \times \mathbf{j}_1 = \mathbf{0}$:

$$\boldsymbol{\omega} \times \mathbf{r} = -\Omega x\,\mathbf{k}_1, \qquad \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}) = \Omega\,\mathbf{j}_1 \times (-\Omega x\,\mathbf{k}_1) = -\Omega^2 x\,\mathbf{i}_1, \qquad 2\boldsymbol{\omega} \times \mathbf{v}_{rel} = -2\Omega\dot{x}\,\mathbf{k}_1$$

$$\mathbf{a} = \big(\ddot{x} - \Omega^2 x\big)\mathbf{i}_1 + \frac{x\ddot{x} + \dot{x}^2}{p}\mathbf{j}_1 - 2\Omega\dot{x}\,\mathbf{k}_1$$

KEY RESULTS

- $\mathbf{v}_{rel} = \dot{x}\Big(\mathbf{i}_1 + \dfrac{x}{p}\mathbf{j}_1\Big)$, $\mathbf{a}_{rel} = \ddot{x}\Big(\mathbf{i}_1 + \dfrac{x}{p}\mathbf{j}_1\Big) + \dfrac{\dot{x}^2}{p}\mathbf{j}_1$.
- $\mathbf{a}_{tr} = -\Omega^2 x\,\mathbf{i}_1$, $\mathbf{a}_{cor} = -2\Omega\dot{x}\,\mathbf{k}_1$, $\mathbf{a} = \mathbf{a}_{rel} + \mathbf{a}_{tr} + \mathbf{a}_{cor}$.

Part (c) · 1.5 points

Forces and inertia forces

Write the applied forces, the reaction of the wire and the inertia forces of S_1 in $\mathcal{B}_1$. Which of them do work in S_1?

Step 1. Applied force and reaction

$\mathbf{W} = -mg\,\mathbf{j}_1$. The tangent of the wire is along $\boldsymbol{\tau} = \mathbf{i}_1 + \dfrac{x}{p}\mathbf{j}_1$; the normal plane is spanned by the in-plane normal $\mathbf{n} = \dfrac{1}{q}\Big(-\dfrac{x}{p}\mathbf{i}_1 + \mathbf{j}_1\Big)$ and by $\mathbf{k}_1$. A smooth wire gives (Notes §6.5) $\mathbf{N} = N_n\,\mathbf{n} + N_z\,\mathbf{k}_1$, two unknowns.

Step 2. Inertia forces

In S_1 Newton's law reads $m\mathbf{a}_{rel} = \mathbf{W} + \mathbf{N} + \mathbf{F}_{cf} + \mathbf{F}_{cor}$ (Notes §4.9), with

$$\mathbf{F}_{cf} = -m\,\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}) = m\Omega^2 x\,\mathbf{i}_1, \qquad \mathbf{F}_{cor} = -2m\,\boldsymbol{\omega} \times \mathbf{v}_{rel} = 2m\Omega\dot{x}\,\mathbf{k}_1$$

There is no Euler term ($\boldsymbol{\alpha} = \mathbf{0}$) and no translational term (O fixed).

Step 3. Work in S_1

$\mathbf{N} \cdot \mathbf{v}_{rel} = 0$ and $\mathbf{F}_{cor} \cdot \mathbf{v}_{rel} = 0$ (both orthogonal to the tangent). $\mathbf{W}$ has potential $mgy = \dfrac{mgx^2}{2p}$ and $\mathbf{F}_{cf}$ has potential $V_{cf} = -\tfrac{1}{2}m\Omega^2 x^2$ (check: $-\dfrac{dV_{cf}}{dx} = m\Omega^2 x$).

KEY RESULTS

- $\mathbf{W} = -mg\,\mathbf{j}_1$; $\mathbf{N} = N_n\mathbf{n} + N_z\mathbf{k}_1$ with $\mathbf{n} = \dfrac{-(x/p)\mathbf{i}_1 + \mathbf{j}_1}{q}$.
- $\mathbf{F}_{cf} = m\Omega^2 x\,\mathbf{i}_1$ (potential $-\tfrac{1}{2}m\Omega^2x^2$), $\mathbf{F}_{cor} = 2m\Omega\dot{x}\,\mathbf{k}_1$ (no work in S_1).

Part (d) · 2 points

Equation of motion and reactions

Obtain the equation of motion free of reactions and the two components of the reaction.

Step 1. Tangential projection

Dot $m\mathbf{a}_{rel} = \mathbf{W} + \mathbf{N} + \mathbf{F}_{cf} + \mathbf{F}_{cor}$ with $\boldsymbol{\tau} = \mathbf{i}_1 + \dfrac{x}{p}\mathbf{j}_1$. Left: $m\left[\ddot{x} + \dfrac{x}{p}\cdot\dfrac{x\ddot{x} + \dot{x}^2}{p}\right]$. Right: $\mathbf{W}\cdot\boldsymbol{\tau} = -\dfrac{mgx}{p}$, $\mathbf{F}_{cf}\cdot\boldsymbol{\tau} = m\Omega^2 x$, and $\mathbf{N}\cdot\boldsymbol{\tau} = \mathbf{F}_{cor}\cdot\boldsymbol{\tau} = 0$:

$$\left(1 + \frac{x^2}{p^2}\right)\ddot{x} + \frac{x}{p^2}\dot{x}^2 = x\left(\Omega^2 - \frac{g}{p}\right)$$

Step 2. Out-of-plane projection

$\mathbf{a}_{rel}\cdot\mathbf{k}_1 = 0$, so $0 = N_z + 2m\Omega\dot{x}$:

$$N_z = -2m\Omega\dot{x}$$

The wire must cancel the Coriolis force. Seen from S_0: a bead moving outwards must gain the azimuthal speed Ωx of the wire; the drag direction of the rotation at P is $\boldsymbol{\omega} \times \mathbf{r} = -\Omega x\, \mathbf{k}_1$, and for $x > 0, \dot{x} > 0$ the wire indeed pushes along $-\mathbf{k}_1$.

Step 3. In-plane normal projection

Dot with $q\,\mathbf{n} = -\dfrac{x}{p}\mathbf{i}_1 + \mathbf{j}_1$. Left: $m\Big[-\dfrac{x}{p}\ddot{x} + \dfrac{x\ddot{x} + \dot{x}^2}{p}\Big] = \dfrac{m\dot{x}^2}{p}$. Right: $-mg$ (weight) $-\dfrac{m\Omega^2 x^2}{p}$ (centrifugal) $+qN_n$:

$$N_n = \frac{m}{q}\Big(\frac{\dot{x}^2}{p} + \frac{\Omega^2 x^2}{p} + g\Big)$$

a sum of positive terms: the wire always pushes the bead towards the concave side.

Step 4. Cross-check in S_0

As an extension (Lagrange equations are not in the course notes), the Lagrange equation of $L = \frac{1}{2}m|\mathbf{v}|^2 - mgy$ with the absolute velocity $\mathbf{v} = \mathbf{v}_{rel} - \Omega x\,\mathbf{k}_1$, $|\mathbf{v}|^2 = q^2\dot{x}^2 + \Omega^2 x^2$, gives the same equation (checked with SymPy).

KEY RESULTS

- $\Big(1 + \dfrac{x^2}{p^2}\Big)\ddot{x} + \dfrac{x}{p^2}\dot{x}^2 = x\Big(\Omega^2 - \dfrac{g}{p}\Big)$.
- $N_z = -2m\Omega\dot{x};\ N_n = \dfrac{m}{q}\Big(\dfrac{\dot{x}^2}{p} + \dfrac{\Omega^2 x^2}{p} + g\Big) > 0$.

Part (e) · 1 point

Jacobi first integral

Obtain a first integral of the motion. Is it the mechanical energy of the bead in S_0?

Step 1. Exact derivatives

By the product and chain rules, $\dfrac{d}{dt}\Big[\frac{1}{2}\Big(1 + \dfrac{x^2}{p^2}\Big)\dot{x}^2\Big] = \Big(1 + \dfrac{x^2}{p^2}\Big)\dot{x}\ddot{x} + \dfrac{x\dot{x}}{p^2}\dot{x}^2 = \dot{x}\Big[\Big(1 + \dfrac{x^2}{p^2}\Big)\ddot{x} + \dfrac{x}{p^2}\dot{x}^2\Big]$, and $\dot{x}\cdot x\Big(\Omega^2 - \dfrac{g}{p}\Big) = \dfrac{d}{dt}\Big[\frac{1}{2}\Big(\Omega^2 - \dfrac{g}{p}\Big)x^2\Big]$. Integrating from 0 to t (Barrow's rule):

$$J = \frac{1}{2}\Big(1 + \frac{x^2}{p^2}\Big)\dot{x}^2 + \frac{1}{2}\Big(\frac{g}{p} - \Omega^2\Big)x^2 = J(0)$$

Step 2. Meaning

$\frac{1}{2}|\mathbf{v}_{rel}|^2 = \frac{1}{2}q^2\dot{x}^2$ and the potentials of (c): $J = \frac{1}{2}|\mathbf{v}_{rel}|^2 + gy - \frac{1}{2}\Omega^2 x^2$. In S_1 only $\mathbf{W}$ and $\mathbf{F}_{cf}$ work, so this "relative energy" is conserved. In S_0, $\mathbf{N}\cdot\mathbf{v} = N_z\mathbf{k}_1\cdot(-\Omega x\,\mathbf{k}_1) = 2m\Omega^2 x\dot{x} \neq 0$: the motor feeds energy to the bead.

KEY RESULTS

- $J = \frac{1}{2}\left(1+\dfrac{x^2}{p^2}\right)\dot{x}^2 + \frac{1}{2}\left(\dfrac{g}{p}-\Omega^2\right)x^2 = \text{const}$ (per unit mass).
- It is the energy **in** S_1 including the centrifugal potential; the energy in S_0 is not conserved (the motor works through N_z).

Part (f) · 1.5 points

Relative equilibrium and the critical rate

Find the positions of relative equilibrium and their stability for $\Omega^2 < g/p$, $\Omega^2 = g/p$ and $\Omega^2 > g/p$. Give the frequency of small oscillations when it exists. What happens at the critical rate?

Step 1. Equilibria

$\dot{x} = \ddot{x} = 0$ in (d) gives $x(\Omega^2 - g/p) = 0$. If $\Omega^2 \neq g/p$ only $x = 0$ (the vertex). $V''_{eff}(0) = m(g/p - \Omega^2)$ (Notes §4.6): a minimum (stable) for $\Omega^2 < g/p$, a maximum (unstable) for $\Omega^2 > g/p$.

Step 2. Small oscillations

Linearising (d) about $x = 0$ ($q \to 1$, the $x\dot{x}^2$ term is of third order): $\ddot{x} + \left(\dfrac{g}{p} - \Omega^2\right)x = 0$, so $\omega = \sqrt{g/p - \Omega^2}$ for $\Omega^2 < g/p$.

Step 3. The critical rate (the trap)

For $\Omega^2 = g/p$ the linearised equation is $\ddot{x} = 0$, which does not decide anything by itself. The full equation becomes $q^2\ddot{x} + \dfrac{x}{p^2}\dot{x}^2 = 0$, so $\ddot{x} = 0$ whenever $\dot{x} = 0$, **at any** x: every point is a relative equilibrium. J reduces to $\frac{1}{2}q^2\dot{x}^2 = |\mathbf{v}_{rel}|^2/2 = \text{const}$: the bead slides with constant relative speed and never comes back (neutral, not stable). Geometric reason: at relative rest the resultant of gravity and centrifugal force must be normal to the wire, i.e. $dy/dx = \Omega^2 x/g$; the parabola has $dy/dx = x/p$, which matches at every x exactly when $\Omega^2 = g/p$ (it is the shape of the free surface of a liquid spinning at that rate).

KEY RESULTS

- $x = 0$ is an equilibrium for every Ω: stable for $\Omega^2 < g/p$ ($\omega^2 = g/p - \Omega^2$), unstable for $\Omega^2 > g/p$.
- $\Omega^2 = g/p$: **every** point of the wire is an equilibrium (neutral), and a bead with relative speed slides at constant speed along the wire: $q^2\dot{x}^2 = \text{const}$.

Part (g) · 1.5 points

Release at $\Omega^2 = 2g/p$: escape and side force

For $\Omega^2 = 2g/p$ and release from relative rest at $x_0 = p/2$: show that the bead moves outwards, give $\dot{x}^2(x)$ and $N_z(x)$, their values at $x = p$ and their limits for $x \to \infty$.

Step 1. Direction

With $\dot{x} = 0$ in (d): $q_0^2 \ddot{x} = x_0\left(\frac{2g}{p} - \frac{g}{p}\right) = \frac{g x_0}{p} > 0$. The bead starts outwards (the vertex is unstable, part (f)).

Step 2. Speed

$J(x, \dot{x}) = J(x_0, 0)$ with $g/p - \Omega^2 = -g/p$:

$$\frac{1}{2}\left(1 + \frac{x^2}{p^2}\right)\dot{x}^2 - \frac{g}{2p}x^2 = -\frac{g}{2p}x_0^2 \implies \dot{x}^2 = \frac{g}{p}\,\frac{x^2 - x_0^2}{1 + x^2/p^2}$$

$\dot{x}^2 \geq 0$ only for $|x| \geq x_0$: the bead never returns towards the vertex, and x grows monotonically.
Consistency: $\frac{d(\dot{x}^2)}{dx} = 2\ddot{x}$ reproduces (d) (checked with SymPy).

Step 3. Side force

$N_z = -2m\Omega\dot{x}$ with $\Omega = \sqrt{2g/p}$ and $\dot{x} > 0$:

$$N_z = -2m\sqrt{\frac{2g}{p}}\sqrt{\frac{g}{p}\,\frac{x^2 - x_0^2}{1 + x^2/p^2}} = -\frac{2\sqrt{2}\,mg}{p}\sqrt{\frac{x^2 - x_0^2}{1 + x^2/p^2}}$$

Step 4. Values

At $x = p$ ($x_0 = p/2$): $\dot{x}^2 = \frac{g}{p}\cdot\frac{p^2 - p^2/4}{2} = \frac{3}{8}gp$, and $N_z = -2m\sqrt{\frac{2g}{p}}\sqrt{\frac{3gp}{8}} = -2mg\sqrt{\frac{3}{4}} = -\sqrt{3}\,mg$.
The in-plane reaction there: $\frac{\dot{x}^2}{p} + \frac{\Omega^2 x^2}{p} + g = \frac{3}{8}g + 2g + g = \frac{27}{8}g$ and $q = \sqrt{2}$, so $N_n = \frac{27}{8\sqrt{2}}mg \approx 2.386\,mg$. The relative speed is then $|\mathbf{v}_{rel}|^2 = q^2\dot{x}^2 = \frac{3}{4}gp$.

Step 5. Limits

For $x \to \infty$, $\frac{x^2 - x_0^2}{1 + x^2/p^2} \to p^2$, so $\dot{x}^2 \to gp$ and $|N_z| \to 2\sqrt{2}\,mg$. The abscissa grows at a bounded rate, while the speed along the wire $q|\dot{x}|$ grows without bound because the wire becomes steep.

KEY RESULTS

- $\ddot{x}(x_0) = \dfrac{(g/p)\,x_0}{1 + x_0^2/p^2} > 0;\ \dot{x}^2 = \dfrac{g}{p}\,\dfrac{x^2 - x_0^2}{1 + x^2/p^2}$.
- $N_z = -\dfrac{2\sqrt{2}\,mg}{p}\sqrt{\dfrac{x^2 - x_0^2}{1 + x^2/p^2}}$; at $x = p$: $\dot{x}^2 = \tfrac{3}{8}gp$, $N_z = -\sqrt{3}\,mg \approx -1.732\,mg$, $N_n = \dfrac{27}{8\sqrt{2}}mg \approx 2.386\,mg$.
- $x \to \infty$: $\dot{x}^2 \to gp$ and $|N_z| \to 2\sqrt{2}\,mg \approx 2.83\,mg$.

Interpretation and checks

- **Dimensional check:** g/p and Ω^2 are s^{-2}; in (d) every term is $\mathrm{m\ s^{-2}}$ ($x\dot{x}^2/p^2$ is $\mathrm{m\ (m/s)^2/m^2}$); $N_z = 2m\Omega\dot{x}$ is $\mathrm{kg\ s^{-1}\ m\ s^{-1}}$ = N; $\dot{x}^2 \to gp$ is $\mathrm{m^2\ s^{-2}}$.
- **Limits:** $\Omega = 0$: a bead in a fixed parabolic bowl, $\omega = \sqrt{g/p}$ about the vertex and $N_z = 0$; $p \to \infty$ (straight horizontal wire): always $\Omega^2 > g/p$, and $x(t)$ grows as $\cosh \Omega t$, the classic rotating rod.
- **Common mistakes in the exam:** (1) forgetting N_z (the reaction of a wire has two components; the Coriolis force needs one of them); (2) writing the Jacobi integral as the absolute energy (the motor does work in S_0); (3) at $\Omega^2 = g/p$, concluding "stable" or "unstable" from $\ddot{x} = 0$: every point is an equilibrium and a moving bead keeps a constant relative speed; (4) dropping the $(x/p^2)\dot{x}^2$ term, which comes from differentiating the constraint $y = x^2/(2p)$ twice.

Problem · Unit 5 · 10 points · 60 min · Standard exam level

P04 — Radial tube on a turntable with increasing rate: Euler force, three regimes and the motor torque

A horizontal turntable turns about the fixed vertical axis $O\mathbf{k}$ of the inertial frame $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}\}$ ($\mathbf{k}$ vertical upward, gravity $-g\,\mathbf{k}$). A motor imposes the angle $\theta(t)$ between $\mathbf{i}_0$ and a radial line of the table, with

$$\dot{\theta} = \Omega(t) = \Omega_0 + \beta t, \qquad \theta(0) = 0$$

*($\Omega_0 > 0$ and β constants). A smooth, thin, straight tube is fixed on the table along a diameter of the table, from the rim at $x = -L$ through O to the rim at $x = L$. A bead P of mass m moves inside the tube and is joined to O by a spring of stiffness k and **zero natural length**. The frame $S_1 = \{O; \mathbf{i}_1, \mathbf{j}_1, \mathbf{k}\}$ turns with the table, $\mathbf{i}_1$ along the tube and $\mathbf{j}_1 = \mathbf{k} \times \mathbf{i}_1$; x is the signed coordinate of P along $\mathbf{i}_1$ (so $x < 0$ means on the other side of O). The table and the tube have moment of inertia I_T about the axis.*

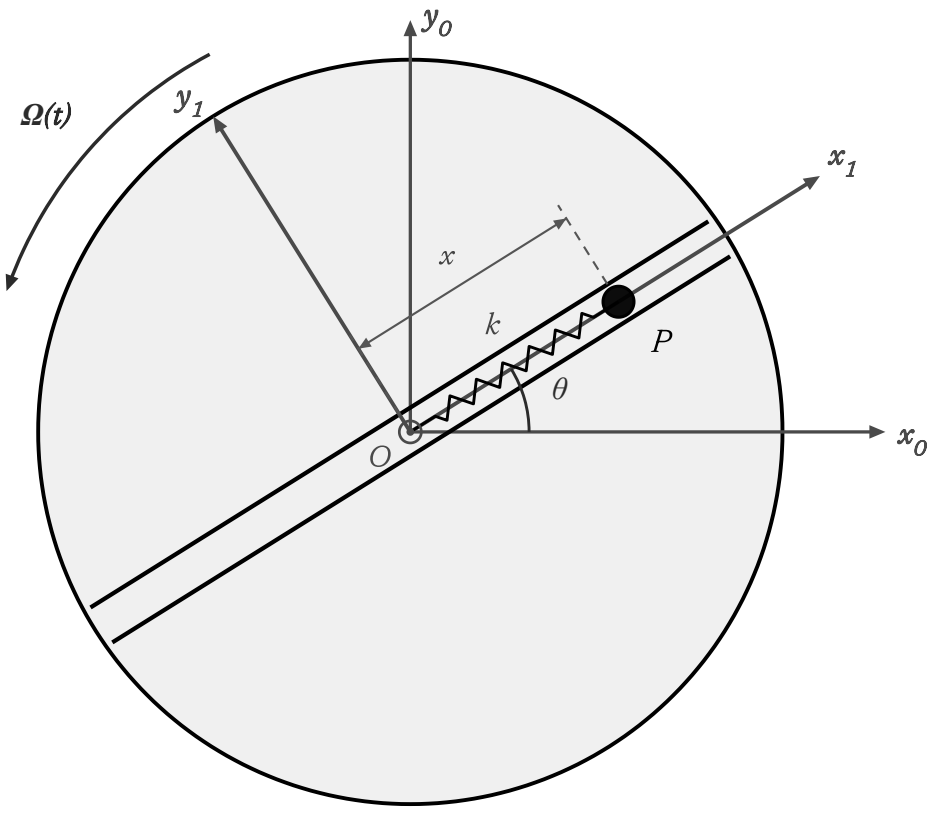


Figure 1. Top view: turntable with rate $\Omega(t) = \dot{\theta} = \Omega_0 + \beta t$ about the vertical axis $O\mathbf{k}$ (out of the page); smooth tube along $\mathbf{i}_1$, bead P at distance x from O, spring of stiffness k and zero natural length.

(The scalar k is the stiffness; the bold $\mathbf{k}$ is the vertical unit vector.)

Setup

- Data: m, k, g, Ω_0, β, I_T, L.
- Numerical parts (e), (f): $m = 0.2$ kg, $k = 20$ N/m (so $\omega_n = \sqrt{k/m} = 10$ rad/s), $\beta = 0$, $x(0) = x_0 = 0.1$ m, $\dot{x}(0) = v_0 = 0$, $L = 0.5$ m, $g = 9.81$ m s^{-2}, and $\Omega_0 = 6$, 10 or 12.5 rad/s.
- Hypotheses: point particle, no friction in the tube, S_0 inertial, the motor delivers whatever torque is needed to keep $\Omega(t)$.

Part (a) · 1 point

Degrees of freedom and frames

Count the degrees of freedom of P. Give $[{}_0R_1]$, $\boldsymbol{\omega}_{10}$ and $\boldsymbol{\alpha}_{10}$.

Step 1. Count

In S_1 the tube is the line $y_1 = 0$, $z_1 = 0$: two holonomic constraints, so $3 - 2 = 1$ degree of freedom, x (Notes §2.7). In S_0 the tube turns with the prescribed $\theta(t)$, so the constraints depend explicitly on time (rheonomic).

Step 2. Frames

$\mathbf{i}_1 = \cos\theta\,\mathbf{i}_0 + \sin\theta\,\mathbf{j}_0$, $\mathbf{j}_1 = -\sin\theta\,\mathbf{i}_0 + \cos\theta\,\mathbf{j}_0$: the columns of $[{}_0R_1]$ above, a simple rotation about $\mathbf{k}$ (Notes §1.5) with $\det = \cos^2\theta + \sin^2\theta = 1$ and $RR^T = I$. Then $\boldsymbol{\omega}_{10} = \dot{\theta}\,\mathbf{k} = \Omega(t)\,\mathbf{k}$ and $\boldsymbol{\alpha}_{10} = \dot{\Omega}\,\mathbf{k} = \beta\,\mathbf{k}$.

KEY RESULTS

- 1 degree of freedom, x; constraints $y_1 = 0$, $z_1 = 0$ (rheonomic in S_0).
- $[{}_0R_1] = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$, $\boldsymbol{\omega}_{10} = \Omega(t)\,\mathbf{k}$, $\boldsymbol{\alpha}_{10} = \beta\,\mathbf{k}$.

Part (b) · 1.5 points

Kinematics

Write the relative velocity and acceleration of P and its absolute acceleration in $\mathcal{B}_1$, (i) with the Coriolis theorem and (ii) by differentiating $\mathbf{r} = x\,\mathbf{i}_1$ directly in S_0.

Step 1. Coriolis theorem

O is fixed, so (Notes §3.2) $\mathbf{a} = \mathbf{a}_{rel} + \boldsymbol{\alpha} \times \mathbf{r} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}) + 2\boldsymbol{\omega} \times \mathbf{v}_{rel}$, with $\mathbf{r} = x\,\mathbf{i}_1$ and $\mathbf{k} \times \mathbf{i}_1 = \mathbf{j}_1$, $\mathbf{k} \times \mathbf{j}_1 = -\mathbf{i}_1$:

$$\boldsymbol{\alpha}\times\mathbf{r}=\beta x\,\mathbf{j}_1,\qquad \boldsymbol{\omega}\times(\boldsymbol{\omega}\times\mathbf{r})=\Omega\,\mathbf{k}\times\Omega x\,\mathbf{j}_1=-\Omega^2 x\,\mathbf{i}_1,\qquad 2\boldsymbol{\omega}\times\mathbf{v}_{rel}=2\Omega\dot{x}\,\mathbf{j}_1$$

$$\mathbf{a}=(\ddot{x}-\Omega^2 x)\,\mathbf{i}_1+(2\Omega\dot{x}+\beta x)\,\mathbf{j}_1$$

Step 2. Direct differentiation

$\mathbf{v}=\dfrac{d}{dt}(x\,\mathbf{i}_1)=\dot{x}\,\mathbf{i}_1+x\Omega\,\mathbf{j}_1$ (product rule). Again, with $\dfrac{d}{dt}(x\Omega)=\dot{x}\Omega+x\dot{\Omega}=\dot{x}\Omega+x\beta$:

$$\mathbf{a}=\ddot{x}\,\mathbf{i}_1+\dot{x}\Omega\,\mathbf{j}_1+(\dot{x}\Omega+x\beta)\,\mathbf{j}_1+x\Omega(-\Omega\,\mathbf{i}_1)$$

which is the same result. The term $\beta x\,\mathbf{j}_1$ is the tangential acceleration of the point of the table under P.

KEY RESULTS

- $\mathbf{v}_{rel}=\dot{x}\,\mathbf{i}_1$, $\mathbf{a}_{rel}=\ddot{x}\,\mathbf{i}_1$.
- $\mathbf{a}=(\ddot{x}-\Omega^2 x)\,\mathbf{i}_1+(2\Omega\dot{x}+\beta x)\,\mathbf{j}_1$.

Part (c) · 1.5 points

Forces and inertia forces

Write all the forces on P in S_1, including the three inertia forces.

Step 1. Applied forces and reaction

The spring has zero natural length, so its force is $-k\,\overrightarrow{OP}=-kx\,\mathbf{i}_1$. The weight is $-mg\,\mathbf{k}$. The tube is smooth: its reaction has no $\mathbf{i}_1$ component, $\mathbf{N}=N_y\,\mathbf{j}_1+N_z\,\mathbf{k}$.

Step 2. Inertia forces of S_1

From the terms of (b) multiplied by $-m$ (Notes §4.9):

$$\mathbf{F}_E=-m\beta x\,\mathbf{j}_1,\qquad \mathbf{F}_{cf}=m\Omega^2 x\,\mathbf{i}_1,\qquad \mathbf{F}_{cor}=-2m\Omega\dot{x}\,\mathbf{j}_1$$

The Euler (azimuthal) force exists only because the rotation accelerates; it points against the sense of rotation for $\beta>0$.

KEY RESULTS

- $\mathbf{F}_k = -kx\,\mathbf{i}_1$, $\mathbf{W} = -mg\,\mathbf{k}$, $\mathbf{N} = N_y\,\mathbf{j}_1 + N_z\,\mathbf{k}$.
- $\mathbf{F}_E = -m\beta x\,\mathbf{j}_1$ (Euler), $\mathbf{F}_{cf} = m\Omega^2 x\,\mathbf{i}_1$, $\mathbf{F}_{cor} = -2m\Omega\dot{x}\,\mathbf{j}_1$.

Part (d) · 1.5 points

Equation of motion and reactions

Obtain the equation of motion free of reactions, and N_y, N_z. Why does the equation contain Ω^2 but not β?

Step 1. Projections

$$\mathbf{i}_1:\ m\ddot{x} = -kx + m\Omega^2 x \qquad \mathbf{j}_1:\ 0 = N_y - m\beta x - 2m\Omega\dot{x} \qquad \mathbf{k}:\ 0 = N_z - mg$$

Step 2. Result

$$\ddot{x} + \left(\frac{k}{m} - \Omega(t)^2\right)x = 0, \qquad N_y = m(\beta x + 2\Omega\dot{x}), \qquad N_z = mg$$

Step 3. Why no β

The Euler force $-m\beta x\,\mathbf{j}_1$ and the Coriolis force are perpendicular to the tube: the wall absorbs them and they appear only in N_y. The angular acceleration still acts on x, but only through the time-dependent coefficient $\Omega(t)^2 = (\Omega_0 + \beta t)^2$. (In S_0 the same equation is the $\mathbf{i}_1$ component of $m\mathbf{a}$ of (b).)

KEY RESULTS

- $\ddot{x} + \left(\frac{k}{m} - \Omega(t)^2\right)x = 0$.
- $N_y = m(\beta x + 2\Omega\dot{x})$, $N_z = mg$.

Part (e) · 2 points

Constant rate: the three regimes

Take $\beta = 0$. Solve the equation of motion with $x(0) = x_0$, $\dot{x}(0) = v_0$ for $\Omega_0^2 < k/m$, $\Omega_0^2 = k/m$ and $\Omega_0^2 > k/m$, write the Jacobi integral, and evaluate each case for the data of the Setup ($\Omega_0 = 6$, 10, 12.5 rad/s).

Step 1. $\Omega_0^2 < k/m$

$\ddot{x} + \omega^2 x = 0$ with $\omega^2 = k/m - \Omega_0^2 > 0$ (Notes Ch. 5): $x = A\cos\omega t + B\sin\omega t$. $x(0) = A = x_0$; $\dot{x} = -A\omega\sin\omega t + B\omega\cos\omega t$ gives $\dot{x}(0) = B\omega = v_0$. Data, $\Omega_0 = 6$: $\omega = \sqrt{100 - 36} = 8$ rad/s, $x = 0.1\cos 8t$ m, period $2\pi/8 = 0.7854$ s. The spring wins over the centrifugal force.

Step 2. $\Omega_0^2 = k/m$ **(the trap)**

$\ddot{x} = 0$: integrating twice with Barrow's rule, $\dot{x} = v_0$ and $x = x_0 + v_0 t$. With $v_0 = 0$ the bead stays where it is, wherever that is: the spring force kx is exactly the centrifugal force $m\Omega_0^2 x$ at every x. With $v_0 \neq 0$ the bead drifts at constant relative speed. It is neither an oscillation of zero frequency nor an exponential escape. Data, $\Omega_0 = 10$: $x \equiv 0.1$ m.

Step 3. $\Omega_0^2 > k/m$

$\ddot{x} - \lambda^2 x = 0$ with $\lambda^2 = \Omega_0^2 - k/m$: $x = A\cosh\lambda t + B\sinh\lambda t$, with $x(0) = A = x_0$ and $\dot{x}(0) = B\lambda = v_0$. Writing it as $\tfrac{1}{2}(x_0 + v_0/\lambda)e^{\lambda t} + \tfrac{1}{2}(x_0 - v_0/\lambda)e^{-\lambda t}$, the growing exponential disappears only for $v_0 = -\lambda x_0$, when $x = x_0 e^{-\lambda t}$. Data, $\Omega_0 = 12.5$: $\lambda = \sqrt{156.25 - 100} = 7.5$ s^{-1}, $x = 0.1\cosh 7.5t$ m.

Step 4. Jacobi integral

Multiplying the equation by $\dot{x}$: $\dfrac{d}{dt}\left[\tfrac{1}{2}\dot{x}^2 + \tfrac{1}{2}\left(\dfrac{k}{m} - \Omega_0^2\right)x^2\right] = 0$, so

$$\tfrac{1}{2}\dot{x}^2 + \tfrac{1}{2}\left(\frac{k}{m} - \Omega_0^2\right)x^2 = \text{const}$$

the energy in S_1 with the spring and centrifugal potentials. $\mathbf{N}$, $\mathbf{F}_{cor}$ and $\mathbf{F}_E$ do no work in S_1 because they are orthogonal to $\mathbf{v}_{rel}$.

KEY RESULTS

- $\Omega_0^2 < \omega_n^2$: $x = x_0\cos\omega t + \dfrac{v_0}{\omega}\sin\omega t$, $\omega = \sqrt{\omega_n^2 - \Omega_0^2}$.
- $\Omega_0^2 = \omega_n^2$: $x = x_0 + v_0 t$ (every position is a neutral relative equilibrium).
- $\Omega_0^2 > \omega_n^2$: $x = x_0\cosh\lambda t + \dfrac{v_0}{\lambda}\sinh\lambda t$, $\lambda = \sqrt{\Omega_0^2 - \omega_n^2}$; it decays only if $v_0 = -\lambda x_0$.
- Data: $\omega = 8$ rad/s, $x = 0.1\cos 8t$ m, period 0.7854 s; $x \equiv 0.1$ m; $\lambda = 7.5$ s^{-1}, $x = 0.1\cosh 7.5t$ m.

Part (f) · 1 point

Reaching the end of the tube

For $\Omega_0 = 12.5$ rad/s, $\beta = 0$ and the data of the Setup, find the time at which the bead reaches the rim ($x = L$), its relative speed then, and the reaction there.

Step 1. Time

$0.1 \cosh 7.5 t_L = 0.5 \Rightarrow \cosh 7.5 t_L = 5 \Rightarrow t_L = \dfrac{\operatorname{arccosh} 5}{7.5} = \dfrac{2.2924}{7.5} = 0.3057$ s.

Step 2. Speed

$\dot x = x_0 \lambda \sinh \lambda t$ with $\sinh \lambda t_L = \sqrt{25 - 1} = \sqrt{24}$: $\dot x = 0.1 \cdot 7.5 \cdot \sqrt{24} = 3.674$ m/s. Check with the Jacobi integral: $\dot x^2 = \lambda^2 (L^2 - x_0^2) = 56.25 \cdot 0.24 = 13.5$ m^2 s^{-2}.

Step 3. Reaction

$N_y = m(0 + 2\Omega_0 \dot x) = 2 \cdot 0.2 \cdot 12.5 \cdot 3.674 = 18.37$ N, about nine times the weight $N_z = mg = 1.962$ N.

KEY RESULTS

- $t_L = \dfrac{\operatorname{arccosh} 5}{7.5} = 0.3057$ s; $\dot x = 0.75\sqrt{24} = 3.674$ m/s.
- $N_y = 2 m \Omega_0 \dot x = 18.37$ N, $N_z = mg = 1.962$ N.

Part (g) · 1.5 points

Motor torque and power balance

With the angular-momentum theorem about the axis, find the torque M the motor must deliver to keep $\Omega(t)$, and check it against N_y. Write the power balance of the bead in S_0. For the case of (f), give M (beyond $I_T \beta$) and the power at $x = L$, and the work of the motor on the bead between x_0 and L.

Step 1. Angular momentum of the bead about the axis

$\mathbf{r} \times m\mathbf{v} = x\,\mathbf{i}_1 \times m(\dot x\,\mathbf{i}_1 + \Omega x\,\mathbf{j}_1) = m x^2 \Omega\,\mathbf{k}$, so $H_z = m x^2 \Omega$. By the product rule:

$$\frac{dH_z}{dt} = m\big(2x\dot x\,\Omega + x^2 \dot\Omega\big) = m\big(2x\dot x\,\Omega + x^2 \beta\big)$$

Step 2. Torque on the bead

Moments about the axis (Notes §4.8): the spring force passes through O; $\mathbf{r}\times\mathbf{W} = x\,\mathbf{i}_1\times(-mg\,\mathbf{k}) = mgx\,\mathbf{j}_1$ and $\mathbf{r}\times N_z\mathbf{k}$ are horizontal; only $\mathbf{r}\times N_y\mathbf{j}_1 = xN_y\,\mathbf{k}$ counts. Hence $\dfrac{dH_z}{dt} = xN_y$, and with N_y of (d): $xN_y = mx(\beta x + 2\Omega\dot{x})$, the same expression as Step 1. The two routes agree.

Step 3. Motor torque

The table receives the motor torque M and the moment $-xN_y$ of the bead's push on the tube: $I_T\beta = M - xN_y$, so

$$M = I_T\beta + mx\big(\beta x + 2\Omega\dot{x}\big)$$

Even at constant rate ($\beta = 0$) the motor must supply $2m\Omega x\dot{x}$ whenever the bead moves radially.

Step 4. Power balance in S_0

$T = \tfrac{1}{2}m(\dot{x}^2 + \Omega^2x^2)$, $V = \tfrac{1}{2}kx^2$. Differentiating, $\dfrac{d}{dt}(T+V) = m\dot{x}\ddot{x} + m\Omega\beta x^2 + m\Omega^2x\dot{x} + kx\dot{x}$, and with $m\ddot{x} = -kx + m\Omega^2x$ from (d): $\dfrac{d}{dt}(T+V) = 2m\Omega^2x\dot{x} + m\Omega\beta x^2 = \Omega\,xN_y$. This is the power of $N_y\mathbf{j}_1$ on the absolute velocity, $N_y\,\Omega x$ (Notes §4.4): in S_0 the reaction of the moving tube works; in S_1 it does not.

Step 5. Numbers at $x = L$ (case (f), $\beta = 0$)

$xN_y = 0.5\cdot 18.37 = 9.186$ N m and the power is $\Omega_0 xN_y = 12.5\cdot 9.186 = 114.8$ W. Since $\beta = 0$, the work of the motor on the bead is $\displaystyle\int M\Omega_0\,dt = \Omega_0\int\frac{dH_z}{dt}dt = \Omega_0\,[H_z]_0^{t_L} = \Omega_0\big(H_z(t_L) - H_z(0)\big) = m\Omega_0^2(L^2 - x_0^2) = 0.2\cdot 156.25\cdot 0.24 = 7.5$ J. Check: T goes from 0.15625 to $\tfrac{1}{2}\cdot 0.2\,(13.5 + 39.0625) = 5.25625$ J and V from 0.1 to 2.5 J, so $\Delta(T+V) = 5.1 + 2.4 = 7.5$ J.

KEY RESULTS

- $H_z = mx^2\Omega$, $\dfrac{dH_z}{dt} = m(2x\dot{x}\Omega + x^2\beta) = xN_y$; $M = I_T\beta + mx(\beta x + 2\Omega\dot{x})$.
- $\dfrac{d}{dt}(T+V) = \Omega\,xN_y = M_{bead}\,\Omega$ (the reaction works in S_0, not in S_1).
- Case (f): $xN_y = 9.186$ N m, power 114.8 W, work $= \Omega_0\Delta H_z = m\Omega_0^2(L^2 - x_0^2) = 7.5$ J $= \Delta(T+V)$.

Interpretation and checks

- **Dimensional check:** k/m and Ω^2 are s^{-2}; $N_y = m(\beta x + 2\Omega\dot{x})$ is kg (s^{-2} m) = N; $H_z = mx^2\Omega$ is kg m^2 s^{-1}; M is N m and $M\Omega$ is W.
- **Limits:** $k = 0$ gives the classic bead on a rotating rod, $x = x_0\cosh\Omega_0 t$; $\Omega_0 = 0$, $\beta = 0$ gives the plain oscillator $\omega = \omega_n$. With $\beta > 0$ the coefficient $k/m - \Omega(t)^2$ eventually becomes negative: a bead that oscillates at first is pushed outwards once $\Omega(t) > \omega_n$.

- **Common mistakes in the exam:** (1) forgetting the Euler force $-m\boldsymbol{\alpha}\times\mathbf{r}$ (it changes N_y and the torque, not the equation of motion); (2) at $\Omega_0^2=k/m$ writing an exponential or a harmonic solution instead of $x=x_0+v_0t$; (3) claiming that the energy $T+V$ of the bead is conserved because the tube is smooth: in S_0 the tube moves and its reaction works; (4) computing the motor torque without the $I_T\beta$ term of the table, or with the wrong sign of the bead's push.

Problem · Unit 5 · 10 points · 60 min · Standard exam level

P05 — Flight pendulum on an accelerating support: inertia forces, equilibrium and a change of regime

A flight-test pendulum hangs from a rail fixed to the ceiling of an aircraft cabin. The support A slides along the rail with a ***constant acceleration*** $\mathbf{a}_A = a_0\,\mathbf{i}$ *with respect to the inertial frame* $S_0 = \{O;\mathbf{i},\mathbf{j},\mathbf{k}\}$ *(x horizontal, forward; y vertical, upward; uniform gravity $-g\,\mathbf{j}$), starting from rest at $t = 0$. The bob P of mass m is joined to A by a massless, inextensible string of length ℓ that stays taut. The angle θ is measured from the downward vertical, positive towards $+x$. At $t = 0$ the bob hangs at rest with respect to the cabin ($\theta = 0$, $\dot\theta = 0$).*

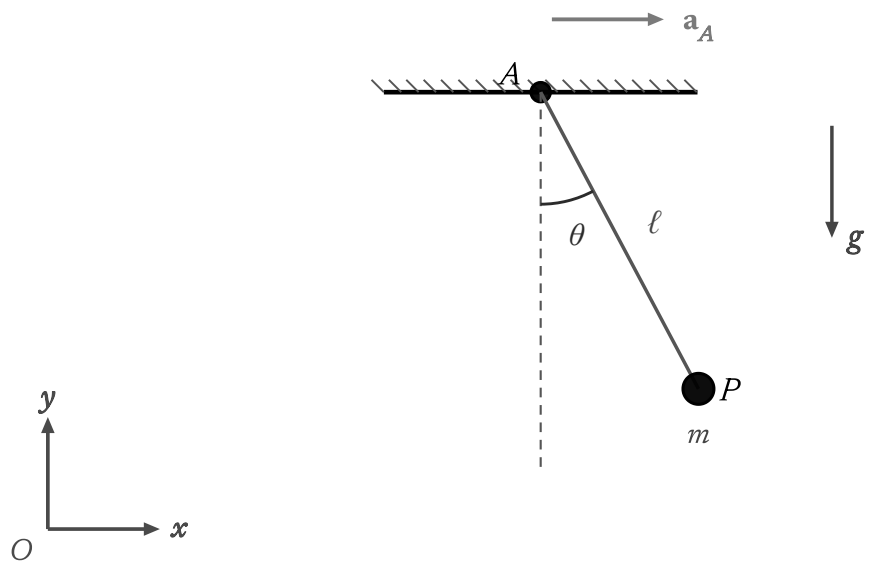


Figure 1. Flight pendulum: support A with acceleration $\mathbf{a}_A = a_0\,\mathbf{i}$, string of length ℓ, bob P of mass m, angle θ from the downward vertical, positive towards $+x$.

Use the reference frame $S_1 = \{A;\mathbf{i},\mathbf{j},\mathbf{k}\}$ that translates with A (its axes stay parallel to those of S_0), and, only to project vectors, the unit vectors $\mathbf{e}_r = \sin\theta\,\mathbf{i} - \cos\theta\,\mathbf{j}$ and $\mathbf{e}_\theta = \cos\theta\,\mathbf{i} + \sin\theta\,\mathbf{j}$.

Setup

- Data: m, ℓ, g, $a_0 > 0$. In the numerical parts (f) and (g): $a_0 = g/\sqrt{3}$.
- Hypotheses: point particle, planar motion, string always taut (checked in (g)), no friction, S_0 inertial.
- S_1 is a **translating** frame: $\boldsymbol{\omega}_{10} = \mathbf{0}$, so the only inertia force is $-m\,\mathbf{a}_A$ (no Coriolis, no centrifugal force).

Part (a) · 1 point

Degrees of freedom and frames

How many effective degrees of freedom has P? State the angular velocity of S_1 with respect to S_0 and the kind of inertia force that appears in S_1.

Step 1. Count

A free particle in the plane has 2 degrees of freedom. The string gives the holonomic constraint $|\mathbf{r}_{AP}| = \ell$, which removes one: θ describes the configuration. The motion of A is given ($\mathbf{a}_A = a_0\mathbf{i}$), so it adds no unknown: the constraint is **rheonomic** in S_0 and scleronomic in S_1.

Step 2. The frame

S_1 keeps its axes parallel to those of S_0, so $\boldsymbol{\omega}_{10} = \mathbf{0}$. In the relative-motion law $\mathbf{F} + \mathbf{F}_{in} = m\mathbf{a}_{rel}$ (Notes §4.9) the Coriolis term $-2m\boldsymbol{\omega} \times \mathbf{v}_{rel}$ and the centrifugal term vanish, and only $\mathbf{F}_{in} = -m\mathbf{a}_A$ remains.

KEY RESULTS

- 1 degree of freedom, coordinate $\theta(t)$.
- $\boldsymbol{\omega}_{10} = \mathbf{0}$; only the translational inertia force $-ma_0\mathbf{i}$.

Part (b) · 1.5 points

Kinematics of P

*Write the position, velocity and acceleration of P **relative to** S_1, and the absolute velocity and acceleration of P in S_0, in the basis $\{\mathbf{e}_r, \mathbf{e}_\theta\}$.*

Change of basis

The projection basis $\{\mathbf{e}_r, \mathbf{e}_\theta\}$ is related to $\{\mathbf{i}, \mathbf{j}\}$ (planar, $\mathbf{k}$ common) by

$$\begin{pmatrix} \mathbf{e}_r \\ \mathbf{e}_\theta \end{pmatrix} = R \begin{pmatrix} \mathbf{i} \\ \mathbf{j} \end{pmatrix}, \qquad R = \begin{pmatrix} \sin\theta & -\cos\theta \\ \cos\theta & \sin\theta \end{pmatrix}$$

$$\det R = \sin^2\theta + \cos^2\theta = 1, \qquad RR^T = \begin{pmatrix} \sin^2\theta + \cos^2\theta & 0 \\ 0 & \cos^2\theta + \sin^2\theta \end{pmatrix} = I$$

so R is orthonormal and right-handed ($\mathbf{e}_r \times \mathbf{e}_\theta = \mathbf{k}$). The inverse is R^T: $\mathbf{i} = \sin\theta\,\mathbf{e}_r + \cos\theta\,\mathbf{e}_\theta$, used below.

Step 1. Derivatives of the unit vectors

From $\mathbf{e}_r = \sin\theta\,\mathbf{i} - \cos\theta\,\mathbf{j}$ and the chain rule $\frac{d}{dt} f(\theta(t)) = \frac{df}{d\theta}\dot\theta$:

$$\dot{\mathbf{e}}_r = \dot\theta\,(\cos\theta\,\mathbf{i} + \sin\theta\,\mathbf{j}) = \dot\theta\,\mathbf{e}_\theta, \qquad \dot{\mathbf{e}}_\theta = \dot\theta\,(-\sin\theta\,\mathbf{i} + \cos\theta\,\mathbf{j}) = -\dot\theta\,\mathbf{e}_r$$

Step 2. Relative motion

$$\mathbf{r}_{rel} = \ell\,\mathbf{e}_r, \qquad \mathbf{v}_{rel} = \ell\dot\theta\,\mathbf{e}_\theta, \qquad \mathbf{a}_{rel} = \ell\ddot\theta\,\mathbf{e}_\theta + \ell\dot\theta\,\dot{\mathbf{e}}_\theta = \ell\ddot\theta\,\mathbf{e}_\theta - \ell\dot\theta^2\,\mathbf{e}_r$$

(ℓ is constant, so the string produces no radial relative velocity.)

Step 3. Absolute motion

The support starts from rest with constant acceleration: $\mathbf{v}_A = a_0 t\,\mathbf{i}$, $\mathbf{r}_A = \frac{1}{2}a_0 t^2\,\mathbf{i}$. Since S_1 does not rotate, $\mathbf{v}_P = \mathbf{v}_A + \mathbf{v}_{rel}$ and $\mathbf{a}_P = \mathbf{a}_A + \mathbf{a}_{rel}$:

$$\mathbf{v}_P = a_0 t\,\mathbf{i} + \ell\dot\theta\,\mathbf{e}_\theta, \qquad \mathbf{a}_P = a_0\,\mathbf{i} + \ell\ddot\theta\,\mathbf{e}_\theta - \ell\dot\theta^2\,\mathbf{e}_r$$

Projecting $\mathbf{i} = \sin\theta\,\mathbf{e}_r + \cos\theta\,\mathbf{e}_\theta$ (the dot products $\mathbf{i}\cdot\mathbf{e}_r = \sin\theta$, $\mathbf{i}\cdot\mathbf{e}_\theta = \cos\theta$):

$$\mathbf{a}_P = \big(a_0\sin\theta - \ell\dot\theta^2\big)\mathbf{e}_r + \big(a_0\cos\theta + \ell\ddot\theta\big)\mathbf{e}_\theta$$

KEY RESULTS

- $\mathbf{a}_{rel} = \ell\ddot\theta\,\mathbf{e}_\theta - \ell\dot\theta^2\mathbf{e}_r$.
- $\mathbf{a}_P = a_0\mathbf{i} + \mathbf{a}_{rel}$ with $\mathbf{i} = \sin\theta\,\mathbf{e}_r + \cos\theta\,\mathbf{e}_\theta$.

Part (c) · 1.5 points

Forces on P in S_1

Write every force acting on P in S_1 in the basis $\{\mathbf{e}_r, \mathbf{e}_\theta\}$, including the inertia force. Which of them can do work in S_1?

Step 1. Weight

$\mathbf{W} = -mg\,\mathbf{j}$ with $\mathbf{W}\cdot\mathbf{e}_r = -mg(\mathbf{j}\cdot\mathbf{e}_r) = mg\cos\theta$ and $\mathbf{W}\cdot\mathbf{e}_\theta = -mg\sin\theta$:

$$\mathbf{W} = mg\cos\theta\,\mathbf{e}_r - mg\sin\theta\,\mathbf{e}_\theta$$

Step 2. Tension and inertia force

The string is massless and taut, so its force on P points to A: $\mathbf{T} = -T\,\mathbf{e}_r$ with $T \geq 0$ unknown. The inertia force of the translating frame is

$$\mathbf{F}_{in} = -m\,\mathbf{a}_A = -ma_0\,\mathbf{i} = -ma_0\sin\theta\,\mathbf{e}_r - ma_0\cos\theta\,\mathbf{e}_\theta$$

Step 3. Work in S_1

The power of a force is $\mathbf{F}\cdot\mathbf{v}_{rel}$ with $\mathbf{v}_{rel} = \ell\dot\theta\,\mathbf{e}_\theta$: only the $\mathbf{e}_\theta$ components work. $\mathbf{T}\cdot\mathbf{v}_{rel} = 0$. $\mathbf{W}$ and $\mathbf{F}_{in}$ are uniform forces, hence conservative, with potentials $V_g = -mg\ell\cos\theta$ and $V_{in} = ma_0\ell\sin\theta$ (check: $-\partial V/\partial\theta/\ell$ gives their $\mathbf{e}_\theta$ components).

KEY RESULTS

- $\mathbf{W} = mg(\cos\theta\,\mathbf{e}_r - \sin\theta\,\mathbf{e}_\theta)$, $\mathbf{T} = -T\mathbf{e}_r$, $\mathbf{F}_{in} = -ma_0(\sin\theta\,\mathbf{e}_r + \cos\theta\,\mathbf{e}_\theta)$.
- $\mathbf{T}$ does no work (it is orthogonal to $\mathbf{v}_{rel}$); $\mathbf{W}$ and $\mathbf{F}_{in}$ are conservative.

Part (d) · 2 points

Equation of motion and tension

Obtain the equation $\ddot\theta = f(\theta)$ that does not contain T, and the expression of T as a function of θ and $\dot\theta$.

Step 1. Newton in S_1

$$m\big(\ell\ddot\theta\,\mathbf{e}_\theta - \ell\dot\theta^2\mathbf{e}_r\big) = \mathbf{W} + \mathbf{T} + \mathbf{F}_{in}$$

Step 2. Tangential projection

$$m\ell\ddot\theta = -mg\sin\theta - ma_0\cos\theta \implies \ell\ddot\theta = -g\sin\theta - a_0\cos\theta$$

This equation has one unknown, $\theta(t)$, because the tension is orthogonal to $\mathbf{e}_\theta$.

Step 3. Radial projection

$$-m\ell\dot\theta^2 = mg\cos\theta - T - ma_0\sin\theta \implies T = m\big(\ell\dot\theta^2 + g\cos\theta - a_0\sin\theta\big)$$

Step 4. Cross-check with the absolute frame

Newton in S_0 along $\mathbf{e}_r$: $m\,\mathbf{a}_P\cdot\mathbf{e}_r = m(a_0\sin\theta - \ell\dot\theta^2) = \mathbf{W}\cdot\mathbf{e}_r - T = mg\cos\theta - T$, which gives the same T.

KEY RESULTS

- $\ell\ddot\theta = -g\sin\theta - a_0\cos\theta$.
- $T = m\big(\ell\dot\theta^2 + g\cos\theta - a_0\sin\theta\big)$.

Part (e) · 1 point

Equilibrium and small oscillations

Find the equilibrium angle θ_e and the frequency of the small oscillations about it, as functions of a_0, g, ℓ. What does the pendulum indicate?

Step 1. Equilibrium

Setting $\ddot\theta = 0$: $g\sin\theta + a_0\cos\theta = 0 \Rightarrow \tan\theta_e = -a_0/g$. Two solutions differ by π; the stable one hangs below A: $\theta_e = -\arctan(a_0/g)$.

Step 2. Rewrite the equation

With $\cos\alpha = g/g_{eff}$, $\sin\alpha = a_0/g_{eff}$ ($\alpha = \arctan(a_0/g)$) and the identity $\sin(\theta+\alpha) = \sin\theta\cos\alpha + \cos\theta\sin\alpha$:

$$\ell\ddot\theta = -g_{eff}\big(\sin\theta\cos\alpha + \cos\theta\sin\alpha\big) = -g_{eff}\sin(\theta+\alpha)$$

so $\theta_e = -\alpha$. Define $\varphi = \theta - \theta_e = \theta + \alpha$: $\ell\ddot\varphi = -g_{eff}\sin\varphi$, an ordinary pendulum in a gravity g_{eff}.

Step 3. Linearisation

For $|\varphi| \ll 1$, $\sin\varphi \approx \varphi$ and $\ddot\varphi + \dfrac{g_{eff}}{\ell}\varphi = 0$:

$$\omega = \sqrt{\frac{g_{eff}}{\ell}} = \frac{\sqrt[4]{g^2 + a_0^2}}{\sqrt{\ell}}, \qquad \tau = 2\pi\sqrt{\frac{\ell}{g_{eff}}}$$

The pendulum aligns with the **apparent gravity** $\mathbf{g} - \mathbf{a}_A$ (the specific force) and oscillates faster than at rest.

KEY RESULTS

- $\theta_e = -\arctan(a_0/g)$ (the bob lags behind), $g_{eff} = \sqrt{g^2 + a_0^2}$.
- $\omega^2 = g_{eff}/\ell$, stable equilibrium.

Part (f) · 1.5 points

First integral and the swing from $\theta = 0$

Write a first integral of the motion in S_1. For $a_0 = g/\sqrt{3}$, with the initial conditions of the statement, find the extreme angles of the swing and $\ell\dot\theta^2$ when the bob passes θ_e.

Step 1. Energy in S_1

In S_1 the work is done by $\mathbf{W}$ and $\mathbf{F}_{in}$ (conservative, part (c)); $\mathbf{T}$ does none. Hence $E = T_{kin} + V_g + V_{in}$ is constant:

$$\frac{E}{m} = \tfrac{1}{2}\ell^2\dot\theta^2 - g\ell\cos\theta + a_0\ell\sin\theta$$

Check: $\dfrac{d}{dt}\dfrac{E}{m} = \ell\dot\theta\big(\ell\ddot\theta + g\sin\theta + a_0\cos\theta\big) = 0$ by part (d). With $-g\cos\theta + a_0\sin\theta = -g_{eff}\cos(\theta + \alpha)$:

$$\frac{E}{m} = \tfrac{1}{2}\ell^2\dot\theta^2 - g_{eff}\,\ell\cos\varphi$$

Step 2. Numbers

For $a_0 = g/\sqrt{3}$: $\tan\alpha = 1/\sqrt{3}$, so $\alpha = 30^\circ$, $\theta_e = -30^\circ$ and $g_{eff} = g\sqrt{1 + 1/3} = 2g/\sqrt{3}$. At $t = 0$: $\theta = 0 \Rightarrow \varphi_0 = 30^\circ$, $\dot\theta = 0$.

Step 3. Extremes and speed

Equating E at φ_0 and at a generic φ:

$$\ell^2\dot\theta^2 = 2g_{eff}\,\ell\,\big(\cos\varphi - \cos 30^\circ\big)$$

It vanishes at $\varphi = \pm 30^\circ$, i.e. $\theta = 0$ and $\theta = -60^\circ$. At $\varphi = 0$ the factor is $\cos 0 - \cos 30^\circ = 1 - \cos 30^\circ$:

$$\ell\dot\theta^2\Big|_{\theta_e} = 2g_{eff}\,(1-\cos 30^\circ) = \frac{4g}{\sqrt3}\Big(1-\frac{\sqrt3}{2}\Big) = \Big(\frac{4}{\sqrt3}-2\Big)g \approx 0.309\,g$$

KEY RESULTS

- $\tfrac12\ell^2\dot\theta^2 - g\ell\cos\theta + a_0\ell\sin\theta = \text{const}$, i.e. $\tfrac12\ell^2\dot\theta^2 - g_{eff}\ell\cos\varphi = \text{const}$.
- $\theta_e = -30^\circ$, extremes $\theta = 0$ and $\theta = -60^\circ$; $\ell\dot\theta^2(\theta_e) = 2g_{eff}(1-\cos 30^\circ) = (4/\sqrt3 - 2)\,g \approx 0.309\,g$.

Part (g) · 1.5 points

Tension and a change of regime

For $a_0 = g/\sqrt3$: (i) compute T at θ_e and show that the string never goes slack. (ii) At the instant the bob passes θ_e for the first time (moving backwards, $\dot\theta < 0$) the support stops accelerating ($a_0 \to 0$, A keeps its velocity). Find the maximum angle of the following motion and the tension when the bob passes the vertical.

Step 1. Tension along the swing

From (d), $T = m(\ell\dot\theta^2 + g\cos\theta - a_0\sin\theta)$ and $g\cos\theta - a_0\sin\theta = g_{eff}\cos\varphi$. With $\ell\dot\theta^2 = 2g_{eff}(\cos\varphi - \cos 30^\circ)$ of (f):

$$T(\varphi) = mg_{eff}\big(3\cos\varphi - 2\cos 30^\circ\big)$$

At $\varphi = 0$: $T = m\frac{2g}{\sqrt3}(3-\sqrt3) = mg(2\sqrt3-2) \approx 1.464\,mg$. T decreases with $|\varphi|$; its minimum is at $\varphi = \pm 30^\circ$: $T = mg_{eff}\cos 30^\circ = mg > 0$. The string is taut during the whole swing.

Step 2. Initial conditions of phase 2

At the switch: $\theta_1 = -30^\circ$, $\ell\dot\theta_1^2 = (4/\sqrt3 - 2)g$. Afterwards $\ell\ddot\theta = -g\sin\theta$ (ordinary pendulum), with the energy per unit mass $\tfrac12\ell^2\dot\theta^2 - g\ell\cos\theta = \text{const}$. The bob has kinetic energy at θ_1, which is **not** the equilibrium of the new motion.

Step 3. Maximum angle

Setting $\dot\theta = 0$ at θ_m:

$$-g\ell\cos\theta_m = \tfrac12\ell^2\dot\theta_1^2 - g\ell\cos 30^\circ \implies \cos\theta_m = \frac{\sqrt3}{2} - \frac12\Big(\frac{4}{\sqrt3}-2\Big) = 1 - \frac{1}{2\sqrt3} = 1 - \frac{\sqrt3}{6}$$

so $\theta_m = \arccos(0.7113) = 44.66^\circ$: the bob, which was at -30° moving backwards, first reaches -44.66° (beyond the old equilibrium), then swings through the vertical to $+44.66^\circ$: the new motion is symmetric, $\theta \in [-44.66^\circ, +44.66^\circ]$, about the vertical.

Step 4. Tension at the vertical

The energy $\frac{1}{2}\ell^2\dot\theta^2 - g\ell\cos\theta$ is the same at $\theta_1 = -30^\circ$ and at $\theta = 0$, so $\ell\dot\theta^2(0) = \ell\dot\theta_1^2 + 2g(\cos 0 - \cos 30^\circ)$ with $\cos 0 - \cos 30^\circ = 1 - \sqrt{3}/2$:

$$\ell\dot\theta^2(0) = \Big(\frac{4}{\sqrt{3}} - 2\Big)g + 2g\Big(1 - \frac{\sqrt{3}}{2}\Big) = \Big(\frac{4}{\sqrt{3}} - \sqrt{3}\Big)g = \frac{g}{\sqrt{3}}$$

and with $a_0 = 0$: $T(0) = m\big(\ell\dot\theta^2 + g\big) = mg\big(1 + \frac{1}{\sqrt{3}}\big) \approx 1.577\,mg$.

KEY RESULTS

- $T(\varphi) = mg_{eff}(3\cos\varphi - 2\cos 30^\circ)$; $T(\theta_e) = mg(2\sqrt{3} - 2) \approx 1.46\,mg$; $T_{min} = mg$ at the turning points.
- Phase 2: $\cos\theta_{max} = 1 - \sqrt{3}/6$, $\theta_{max} \approx 44.66^\circ$ (the swing is $\pm 44.66^\circ$ about the vertical; the first turning point is **behind** the vertical, at -44.66°); $T(0) = mg(1 + 1/\sqrt{3}) \approx 1.58\,mg$.

Interpretation and checks

- **Dimensional check:** a_0, g and $\ell\dot\theta^2$ are accelerations (m s^{-2}); $\omega^2 = g_{eff}/\ell$ is s^{-2}; T is $m\times$ acceleration = N.
- **Limits:** $a_0 \to 0$ gives $\theta_e = 0$, $g_{eff} = g$ and the ordinary pendulum; $a_0 \gg g$ gives $\theta_e \to -90^\circ$ (the string lies horizontal, behind the support) and $\omega^2 \approx a_0/\ell$.
- **Common mistakes in the exam:** (1) writing the inertia force with the wrong sign (it is $-m\mathbf{a}_A$, opposite to the support's acceleration); (2) forgetting that $\mathbf{T}$ is orthogonal to $\mathbf{v}_{rel}$ and adding its work; (3) taking the old equilibrium as the centre of the new motion after the support stops: the bob keeps its speed and swings **past** the vertical; (4) using g instead of g_{eff} in the period.

Problem · Unit 4 · 10 points · 60 min · Standard exam level

P06 — Precessing orbit under an attractive inverse-cube perturbation: Binet equation, circular orbits and apsidal advance

A particle P of mass m moves about a fixed centre O (inertial frame $S_0 = \{O;\mathbf{i},\mathbf{j},\mathbf{k}\}$) under the single force

$$\mathbf{F} = -\left(\frac{m\mu}{r^2} + \frac{m\varepsilon}{r^3}\right)\mathbf{e}_r$$

where $r = |\mathbf{OP}|$, $\mathbf{e}_r = \mathbf{OP}/r$, $\mu > 0$ is the gravitational parameter of the centre and $\varepsilon > 0$ (units $m^4\,s^{-2}$) measures a small attractive perturbation (a model perturbation added to the Newtonian attraction). Once the motion is shown to be planar, use polar coordinates (r,θ) in the plane of the orbit, with θ measured counter-clockwise from a fixed reference line through O, the unit vectors $\mathbf{e}_r$, $\mathbf{e}_\theta$ ($\mathbf{e}_\theta$ points towards increasing θ) and $h = r^2\dot\theta$.

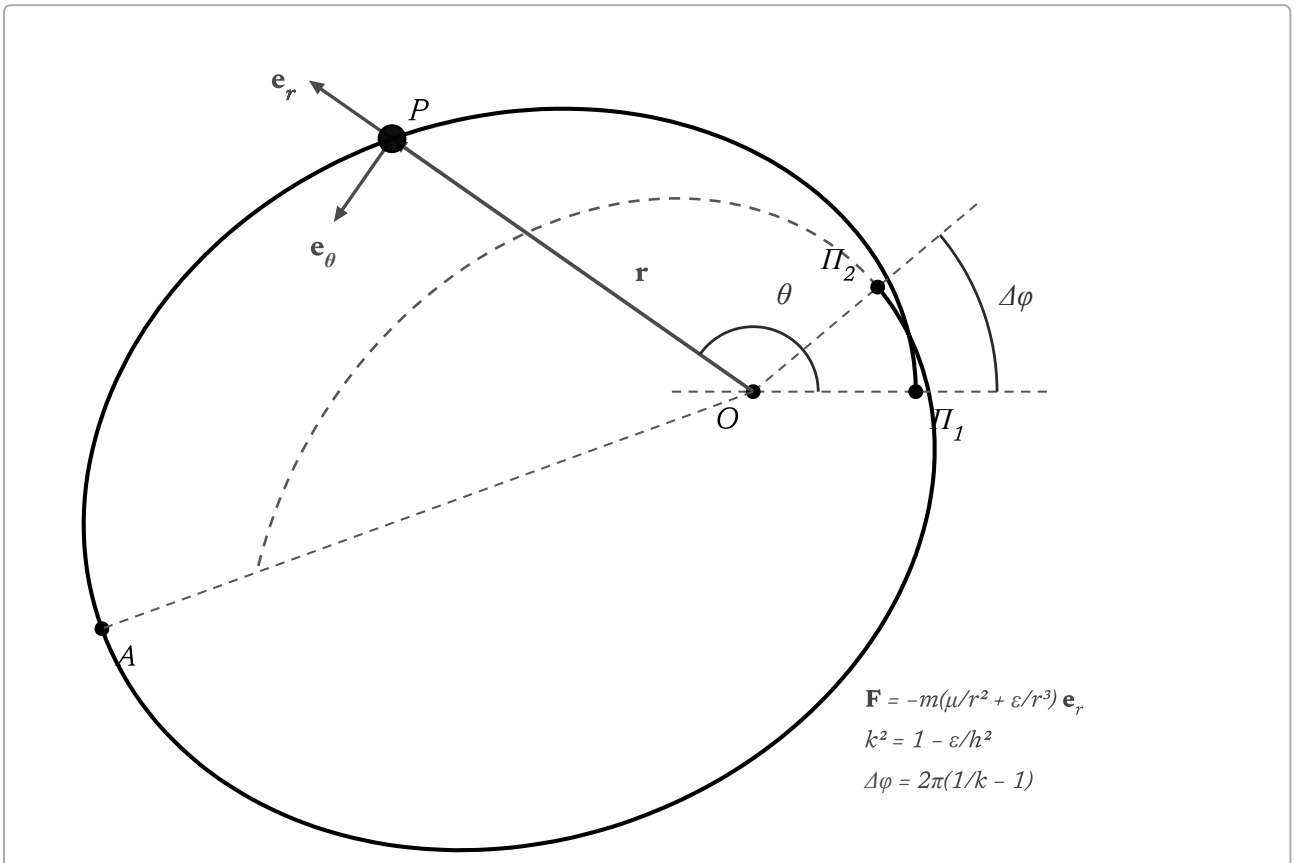


Figure 1. Orbit of P about O for $\varepsilon=0.19\,h^2$, $e = 0.62$: one radial cycle from the perigee Π_1 through the apogee A to the next perigee Π_2 (solid) and the start of the next cycle (dashed); r, θ, the polar unit vectors at P and the apsidal advance $\Delta\phi$.

A perigee is a point of minimum r and an apogee a point of maximum r. In the numerical part the reference line $\theta = 0$ passes through a perigee.

Setup

- Data: m, μ, ε and the initial conditions. Numerical case (part (f)): $\mu = 4.0 \times 10^{14}$ m^3 s^{-2}; at $t = 0$ the particle is at the perigee $r_p = 8.0 \times 10^{6}$ m with the speed $v_p = 1.0 \times 10^{4}$ m/s perpendicular to $\mathbf{OP}$; $\varepsilon = 0.19\, h^2$.
- Hypotheses: point particle, O fixed and S_0 inertial, no other force.
- Frames: S_0 (inertial, Cartesian) and the polar basis $\{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{k}\}$ attached to P, with $\mathbf{e}_r = \cos\theta\,\mathbf{i} + \sin\theta\,\mathbf{j}$ and $\mathbf{e}_\theta = -\sin\theta\,\mathbf{i} + \cos\theta\,\mathbf{j}$ ($\mathbf{i}, \mathbf{j}$ in the plane of the orbit, $\mathbf{k}$ along $\mathbf{H}_O$, see (a)).
- Change of basis:

$$\left(\mathbf{e}_r\ \mathbf{e}_\theta\ \mathbf{k}\right) = [R]\left(\mathbf{i}\,\mathbf{j}\,\mathbf{k}\right), \qquad [R] = \begin{pmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

with rows of unit length ($\cos^2\theta + \sin^2\theta = 1$) and mutually orthogonal ($-\cos\theta\sin\theta + \sin\theta\cos\theta = 0$), so $[R][R]^T = I$, and $\det[R] = \cos^2\theta + \sin^2\theta = +1$: the polar basis is right-handed, $\mathbf{e}_r \times \mathbf{e}_\theta = \mathbf{k}$.

Part (a) · 1.5 points

Planar motion and conservation of angular momentum

Show that the angular momentum $\mathbf{H}_O$ of P about O is constant, that the motion takes place in a fixed plane through O, and that $h = r^2\dot\theta$ is a constant of the motion. How many degrees of freedom remain?

Step 1. Torque of a central force

$\mathbf{F}$ is parallel to $\mathbf{r} = r\,\mathbf{e}_r$, so

$$\mathbf{M}_O = \mathbf{r} \times \mathbf{F} = -\left(\frac{m\mu}{r^2} + \frac{m\varepsilon}{r^3}\right) r\,(\mathbf{e}_r \times \mathbf{e}_r) = \mathbf{0}$$

By the angular momentum theorem about a fixed point (Notes §4.8), $\dot{\mathbf{H}}_O = \mathbf{M}_O = \mathbf{0}$, so $\mathbf{H}_O = m\,\mathbf{r} \times \mathbf{v}$ is a constant vector.

Step 2. Plane of the motion

For every t, $\mathbf{r}\cdot\mathbf{H}_O = m\,\mathbf{r}\cdot(\mathbf{r}\times\mathbf{v}) = 0$ (a triple product with a repeated vector). The position is always orthogonal to the fixed vector $\mathbf{H}_O$: P moves in the fixed plane through O orthogonal to $\mathbf{H}_O$ (if $\mathbf{H}_O = \mathbf{0}$ the motion is radial, along a fixed line). Choose $\mathbf{k}$ along $\mathbf{H}_O$ (Notes §4.8.1).

Step 3. The constant h

In the plane, $\mathbf{r} = r\,\mathbf{e}_r$ and, with $\dot{\mathbf{e}}_r = \dot\theta\,\mathbf{e}_\theta$ (chain rule on $\mathbf{e}_r(\theta(t))$),

$$\mathbf{v} = \dot r\,\mathbf{e}_r + r\dot\theta\,\mathbf{e}_\theta, \qquad \mathbf{H}_O = m\,r\,\mathbf{e}_r\times\left(\dot r\,\mathbf{e}_r + r\dot\theta\,\mathbf{e}_\theta\right) = mr^2\dot\theta\,\mathbf{k}$$

So $h = |\mathbf{H}_O|/m = r^2\dot\theta$ is constant (twice the areal velocity). The particle has 3 degrees of freedom in space; the plane is fixed by the initial conditions, and in it 2 remain, r and θ, linked by the first integral h.

KEY RESULTS

- $\mathbf{M}_O = \mathbf{r}\times\mathbf{F} = \mathbf{0}$, so $\mathbf{H}_O = m\,\mathbf{r}\times\mathbf{v}$ is constant.
- $\mathbf{r}\cdot\mathbf{H}_O = 0$: P stays in the plane through O orthogonal to $\mathbf{H}_O$; in it $\mathbf{H}_O = mr^2\dot\theta\,\mathbf{k}$, so $h = r^2\dot\theta =$ const.
- 2 degrees of freedom, (r,θ), with one first integral (h).

Part (b) · 1.5 points

Equations of motion and energy

Write the equations of motion in polar components. Show that $\mathbf{F}$ is conservative, find its potential energy, and write the energy integral as a one-dimensional problem in r with an effective potential $U_{eff}(r)$.

Step 1. Acceleration in polar components

Differentiating $\mathbf{v} = \dot r\,\mathbf{e}_r + r\dot\theta\,\mathbf{e}_\theta$ with the product rule and $\dot{\mathbf{e}}_r = \dot\theta\,\mathbf{e}_\theta$, $\dot{\mathbf{e}}_\theta = -\dot\theta\,\mathbf{e}_r$:

$$\mathbf{a} = \ddot r\,\mathbf{e}_r + \dot r\dot\theta\,\mathbf{e}_\theta + (\dot r\dot\theta + r\ddot\theta)\,\mathbf{e}_\theta - r\dot\theta^2\,\mathbf{e}_r = (\ddot r - r\dot\theta^2)\,\mathbf{e}_r + (2\dot r\dot\theta + r\ddot\theta)\,\mathbf{e}_\theta$$

and $2\dot r\dot\theta + r\ddot\theta = \frac{1}{r}\frac{d}{dt}(r^2\dot\theta)$ (product rule: $\frac{d}{dt}(r^2\dot\theta) = 2r\dot r\dot\theta + r^2\ddot\theta$).

Step 2. Newton's second law

$m\mathbf{a} = \mathbf{F}$ projected on $\mathbf{e}_r$ and $\mathbf{e}_\theta$ (Notes §8.3):

$$\ddot r - r\dot\theta^2 = -\frac{\mu}{r^2} - \frac{\varepsilon}{r^3}, \qquad \frac{1}{r}\frac{d}{dt}(r^2\dot\theta) = 0$$

The second equation is the conservation of h of part (a).

Step 3. Potential energy

$\mathbf{F} = f(r)\,\mathbf{e}_r$ with $f(r) = -m\mu/r^2 - m\varepsilon/r^3$ depends only on r, so $\mathbf{F} = -\nabla V$ with $V'(r) = -f(r)$ (Notes §4.4). Choosing $V \to 0$ as $r \to \infty$, $V(r) = -\displaystyle\int_\infty^r f(r')\,dr'$. An antiderivative of $-f(r') = \dfrac{m\mu}{r'^2} + \dfrac{m\varepsilon}{r'^3}$ is $G(r') = -\dfrac{m\mu}{r'} - \dfrac{m\varepsilon}{2r'^2}$, and Barrow's rule with $G(r') \to 0$ as $r' \to \infty$ gives

$$V(r) = \int_\infty^r \Big(\frac{m\mu}{r'^2} + \frac{m\varepsilon}{r'^3}\Big)dr' = \Big[-\frac{m\mu}{r'} - \frac{m\varepsilon}{2r'^2}\Big]_\infty^r = \Big(-\frac{m\mu}{r} - \frac{m\varepsilon}{2r^2}\Big) - 0 = -\frac{m\mu}{r} - \frac{m\varepsilon}{2r^2}$$

Check: $-V'(r) = -\frac{m\mu}{r^2} - \frac{m\varepsilon}{r^3} = f(r)$.

Step 4. Energy integral and effective potential

The only force is conservative, so $E = \frac{1}{2}m|\mathbf{v}|^2 + V$ is constant (Notes §4.5), with $|\mathbf{v}|^2 = \dot r^2 + r^2\dot\theta^2 = \dot r^2 + h^2/r^2$:

$$\frac{E}{m} = \tfrac{1}{2}\dot r^2 + \frac{h^2}{2r^2} - \frac{\mu}{r} - \frac{\varepsilon}{2r^2} = \tfrac{1}{2}\dot r^2 + U_{eff}(r), \qquad U_{eff}(r) = \frac{h^2 - \varepsilon}{2r^2} - \frac{\mu}{r}$$

The perturbation enters exactly as a reduction of the centrifugal barrier, $h^2 \to h^2 - \varepsilon$.

KEY RESULTS

- $\ddot r - r\dot\theta^2 = -\dfrac{\mu}{r^2} - \dfrac{\varepsilon}{r^3}$ and $\dfrac{d}{dt}(r^2\dot\theta) = 0$.
- $V = -\dfrac{m\mu}{r} - \dfrac{m\varepsilon}{2r^2}$.
- $\dfrac{E}{m} = \tfrac{1}{2}\dot r^2 + U_{eff}(r)$, $U_{eff} = \dfrac{h^2 - \varepsilon}{2r^2} - \dfrac{\mu}{r}$.

Part (c) · 1.5 points

Circular orbit and its stability

For a given h, find the radius r_c and the speed of the circular orbit. Under what condition does it exist? Is it stable against small radial perturbations? Compare the frequency of the small radial oscillations with the angular rate on the circle.

Step 1. Equilibrium of the radial problem

$$U_{eff}'(r) = -\frac{h^2 - \varepsilon}{r^3} + \frac{\mu}{r^2} = 0 \implies r_c = \frac{h^2 - \varepsilon}{\mu}$$

A positive radius requires $h^2 > \varepsilon$. If $h^2 \le \varepsilon$ the barrier term is non-positive, U_{eff} decreases monotonically towards $r \to 0$, and the particle with $\dot r < 0$ falls into the centre: no circular orbit exists.

Step 2. Speed

On the circle $\dot r = 0$, so $v_c = r_c\dot\theta = h/r_c$. Equivalently, the radial equation with $\ddot r = 0$ gives $r_c\dot\theta^2 = \mu/r_c^2 + \varepsilon/r_c^3$, i.e. $v_c^2 = \mu/r_c + \varepsilon/r_c^2$; with $h^2 = \mu r_c + \varepsilon$ both forms agree.

Step 3. Stability

$$U_{eff}''(r) = \frac{3(h^2 - \varepsilon)}{r^4} - \frac{2\mu}{r^3} \implies U_{eff}''(r_c) = \frac{3\mu r_c}{r_c^4} - \frac{2\mu}{r_c^3} = \frac{\mu}{r_c^3} > 0$$

r_c is a minimum of U_{eff}: the circular orbit is stable. Linearising $\ddot r = -U_{eff}'(r)$ about r_c with $\rho = r - r_c$ gives $\ddot\rho + U_{eff}''(r_c)\rho = 0$, so the radial oscillations have $\omega_r^2 = \mu/r_c^3$.

Step 4. Radial frequency versus angular rate

On the circle $\dot\theta_c = h/r_c^2$, so $\dot\theta_c^2 = h^2/r_c^4$ and

$$\frac{\omega_r^2}{\dot\theta_c^2} = \frac{\mu r_c}{h^2} = \frac{h^2 - \varepsilon}{h^2} = 1 - \frac{\varepsilon}{h^2}$$

For $\varepsilon = 0$ (Kepler) the ratio is 1: r returns to its minimum exactly once per turn and the slightly perturbed orbit closes (an ellipse). For $\varepsilon > 0$ the radial oscillation is slower than the rotation, so the minimum of r is reached **after** more than one turn: the perigee advances. Part (e) makes this exact for any amplitude.

KEY RESULTS

- $r_c = \dfrac{h^2 - \varepsilon}{\mu}$, it exists only if $h^2 > \varepsilon$; $v_c^2 = \dfrac{h^2}{r_c^2} = \dfrac{\mu}{r_c} + \dfrac{\varepsilon}{r_c^2}$.
- $U''_{eff}(r_c) = \mu / r_c^3 > 0$: stable. $\omega_r^2 = \mu / r_c^3,\ \omega_r / \dot\theta_c = \sqrt{1 - \varepsilon / h^2} < 1$.

Part (d) · 2 points

Binet equation and the orbit

With $u = 1/r$ and θ as the independent variable, derive the Binet equation of this force field, solve it with the perigee at $\theta = 0$, and give the perigee and apogee radii.

Step 1. First derivative

$r = 1/u$ with $u = u(\theta(t))$. Chain rule:

$$\dot r = \frac{d}{dt}\Big(\frac{1}{u}\Big) = -\frac{1}{u^2}\frac{du}{d\theta}\dot\theta = -\frac{1}{u^2}u'\,hu^2 = -h\,u'$$

using $\dot\theta = h/r^2 = hu^2$.

Step 2. Second derivative

h is constant, so again by the chain rule

$$\ddot r = \frac{d}{dt}(-h\,u') = -h\frac{du'}{d\theta}\dot\theta = -h\,u''\,hu^2 = -h^2u^2u''$$

Step 3. Radial equation in u

Substitute $\ddot r$, $r\dot\theta^2 = \frac{1}{u}\,h^2u^4 = h^2u^3$, $\mu/r^2 = \mu u^2$ and $\varepsilon/r^3 = \varepsilon u^3$ in the radial equation of (b):

$$-h^2u^2u'' - h^2u^3 = -\mu u^2 - \varepsilon u^3$$

Dividing by $-h^2u^2$ (Notes §8.3, Eq. 8.12 with the extra term):

$$u'' + \Big(1 - \frac{\varepsilon}{h^2}\Big)u = \frac{\mu}{h^2}$$

Step 4. Solution

With the scalar $k^2 = 1 - \varepsilon/h^2 > 0$ (not the unit vector $\mathbf{k}$; the condition $h^2 > \varepsilon$ of (c)), this is a forced harmonic oscillator in θ with "frequency" k. A particular solution is the constant $u_0 = \mu/(h^2k^2)$; the homogeneous solution is $C\cos(k\theta - \theta_0)$. Choosing $\theta = 0$ at a perigee (maximum of u) gives $\theta_0 = 0$ and $C \geq 0$; write $C = u_0 e$:

$$u = \frac{\mu}{h^2k^2}\big(1 + e\cos k\theta\big), \qquad r = \frac{r_c}{1 + e\cos k\theta}, \qquad r_c = \frac{h^2k^2}{\mu} = \frac{h^2 - \varepsilon}{\mu}$$

Check: $u'' = -k^2u_0 e\cos k\theta$ and $u'' + k^2u = k^2u_0 = \mu/h^2$.

Step 5. Apsides

r is minimum where $\cos k\theta = 1$ ($\theta = 0$) and maximum where $\cos k\theta = -1$ ($k\theta = \pi$):

$$r_p = \frac{r_c}{1 + e}, \qquad r_a = \frac{r_c}{1 - e} \quad (0 \leq e < 1)$$

The shape looks like a conic with semi-parameter r_c, but the angle that appears is $k\theta$, not θ: the apogee is reached at $\theta = \pi/k > \pi$.

KEY RESULTS

- $u'' + \left(1 - \dfrac{\varepsilon}{h^2}\right)u = \dfrac{\mu}{h^2}$.
- $u = \dfrac{\mu}{h^2k^2}\big(1 + e\cos k\theta\big)$, i.e. $r = \dfrac{r_c}{1 + e\cos k\theta}$ with $r_c = \dfrac{h^2 - \varepsilon}{\mu} = \dfrac{h^2k^2}{\mu}$.
- $r_p = \dfrac{r_c}{1 + e}$ at $\theta = 0$; $r_a = \dfrac{r_c}{1 - e}$ at $k\theta = \pi$ (bounded orbit if $0 \leq e < 1$).

Part (e) · 1.5 points

Apsidal precession

Find the angle swept between two consecutive perigees and the advance $\Delta\phi$ of the perigee per radial cycle; give its first-order form for $\varepsilon \ll h^2$. Is the orbit closed? Is the time between two perigees the time the particle needs to turn 360°?

Step 1. Angle between perigees

Perigees are at $\cos k\theta = 1$, i.e. $k\theta = 0, 2\pi, 4\pi, \dots$ Two consecutive ones are separated by $\theta = 2\pi/k$, and since $k < 1$ this is more than one turn. The advance per radial cycle is

$$\Delta\phi = \frac{2\pi}{k} - 2\pi = 2\pi\Big(\frac{1}{\sqrt{1-\varepsilon/h^2}} - 1\Big)$$

positive: the perigee advances in the sense of the motion.

Step 2. Small perturbation

With $x = \varepsilon/h^2 \ll 1$ and the binomial series $(1-x)^{-1/2} = 1 + \tfrac{1}{2}x + O(x^2)$:

$$\Delta\phi = 2\pi\Big(\tfrac{1}{2}x + O(x^2)\Big) \approx \frac{\pi\varepsilon}{h^2}$$

Step 3. Closed or not

The orbit closes if after q radial cycles the particle has made an integer number p of turns: $q\,(2\pi/k) = 2\pi p$, i.e. $k = q/p$ rational. Otherwise the perigee directions never repeat and the trajectory densely fills the annulus between r_p and r_a.

Step 4. Time between perigees (the trap)

Substituting $\dot\theta = h/r^2$ in the radial equation of (b):

$$\ddot r = \frac{h^2}{r^3} - \frac{\mu}{r^2} - \frac{\varepsilon}{r^3} = \frac{\tilde h^2}{r^3} - \frac{\mu}{r^2}, \qquad \tilde h^2 = h^2 - \varepsilon$$

This is the radial equation of a Kepler orbit with angular momentum $\tilde h$ and the same r_p, r_a, so $r(t)$ is periodic with the Kepler period of semi-major axis $a = (r_p + r_a)/2$ (Notes §8.4, Eq. 8.18):

$$T_r = 2\pi\sqrt{\frac{a^3}{\mu}}$$

During T_r the angle advances by $2\pi/k$, not 2π: $\dot\theta = h/r^2 = (h/\tilde h)\,(\tilde h/r^2)$ and $h/\tilde h = 1/k$. The time to turn 360° and the time between perigees are different quantities; equating them is the classic error. The energy also matches: at r_p, $\frac{E}{m} = U_{eff}(r_p) = -\frac{\mu}{2a}$.

KEY RESULTS

- Angle between perigees $2\pi/k$; $\Delta\phi = 2\pi\left(\dfrac{1}{\sqrt{1-\varepsilon/h^2}} - 1\right) \approx \dfrac{\pi\varepsilon}{h^2}$.
- Closed only if k is rational; for irrational k the orbit fills the annulus $r_p \le r \le r_a$ (a rosette).
- Time between perigees: the radial period $T_r = 2\pi\sqrt{a^3/\mu}$, $2a = r_p + r_a$; in that time θ advances $2\pi/k$, **not** 2π.

Part (f) · 2 points

Numerical case

With the data of the Setup ($\mu = 4.0 \times 10^{14}\ m^3\ s^{-2}$, $r_p = 8.0 \times 10^6\ m$, $v_p = 1.0 \times 10^4\ m/s$ perpendicular to **OP**, *$\varepsilon = 0.19\,h^2$), compute h, ε, k, r_c, e, r_a, E/m, the apsidal advance (exact and first-order), and the time between two perigees. Comment on the value of v_p.*

Step 1. Constants of the motion

$h = r_p v_p = 8.0 \times 10^6 \times 1.0 \times 10^4 = 8.0 \times 10^{10}\ \mathrm{m^2/s}$, $h^2 = 6.4 \times 10^{21}\ \mathrm{m^4\ s^{-2}}$, $\varepsilon = 0.19\,h^2 = 1.216 \times 10^{21}\ \mathrm{m^4\ s^{-2}}$ and $k = \sqrt{1 - 0.19} = \sqrt{0.81} = 0.9$.

Step 2. Orbit

$$r_c = \frac{h^2 - \varepsilon}{\mu} = \frac{0.81 \times 6.4 \times 10^{21}}{4.0 \times 10^{14}} = 1.296 \times 10^7\ \mathrm{m}$$

At $\theta = 0$, $r = r_p$: $r_p = r_c/(1+e) \Rightarrow e = r_c/r_p - 1 = 1.62 - 1 = 0.62$. Then $r_a = r_c/(1-e) = 1.296 \times 10^7/0.38 = 3.411 \times 10^7$ m and $a = (r_p + r_a)/2 = 2.105 \times 10^7$ m.

Step 3. Energy

$$\frac{E}{m} = \frac{v_p^2}{2} - \frac{\mu}{r_p} - \frac{\varepsilon}{2r_p^2} = 5.0 \times 10^7 - 5.0 \times 10^7 - \frac{1.216 \times 10^{21}}{1.28 \times 10^{14}} = -9.50 \times 10^6\ \mathrm{J/kg}$$

in agreement with $-\mu/(2a) = -4.0 \times 10^{14}/(4.211 \times 10^7) = -9.50 \times 10^6$ J/kg. Note that $\tfrac{1}{2}v_p^2 = \mu/r_p$: v_p is exactly the **escape speed** of the unperturbed problem. Without the perturbation the orbit would be a parabola; the extra attraction makes it bound.

Step 4. Precession and times

$$\Delta\phi = 2\pi\left(\frac{1}{0.9} - 1\right) = \frac{2\pi}{9} = 40^\circ, \qquad \frac{\pi\varepsilon}{h^2} = 0.19\,\pi = 0.597\ \text{rad} = 34.2^\circ$$

The first-order formula underestimates by 15 %: $\varepsilon/h^2 = 0.19$ is not small. The apogee is at $\theta = \pi/k = 200^\circ$ and the next perigee at $\theta = 400^\circ$. Since $k = 9/10$, the orbit closes after 9 radial cycles and 10 turns.

$$T_r = 2\pi\sqrt{\frac{(2.105\times10^7)^3}{4.0\times10^{14}}} = 3.035\times10^4\ \text{s} \approx 8.43\ \text{h}$$

A numerical integration of the equations of (b) from the perigee confirms that r is minimum again after 3.035×10^4 s with $\theta = 400.0^\circ$.

KEY RESULTS

- $h = 8.0\times10^{10}\ \text{m}^2/\text{s}$, $\varepsilon = 1.216\times10^{21}\ \text{m}^4\,\text{s}^{-2}$, $k = 0.9$.
- $r_c = 1.296\times10^7$ m, $e = 0.62$, $r_a = 3.411\times10^7$ m, $a = 2.105\times10^7$ m, $E/m = -9.50\times10^6$ J/kg.
- $\Delta\phi = 40^\circ$ exactly ($2\pi/9$); first-order estimate 34.2°; apogee at $\theta = 200^\circ$; $T_r = 3.035\times10^4$ s.

Interpretation and checks

- **Dimensional check:** μ/r^2 and ε/r^3 are accelerations ($\text{m}^3\,\text{s}^{-2}/\text{m}^2$ and $\text{m}^4\,\text{s}^{-2}/\text{m}^3$); ε/h^2 is dimensionless ($\text{m}^4\,\text{s}^{-2}$ over $\text{m}^4\,\text{s}^{-2}$), so k and $\Delta\phi$ are pure numbers; $r_c = (h^2-\varepsilon)/\mu$ has $\text{m}^4\,\text{s}^{-2}/(\text{m}^3\,\text{s}^{-2}) = \text{m}$; $\sqrt{a^3/\mu}$ has s.
- **Limits:** $\varepsilon \to 0$ gives $k = 1$, $\Delta\phi = 0$, the conic $r = (h^2/\mu)/(1+e\cos\theta)$ of Notes Eq. 8.14 and T_r equal to the orbital period. $\varepsilon \to h^2$ gives $k \to 0$: the barrier disappears, $\Delta\phi \to \infty$ and $r_c \to 0$ (the particle spirals in).
- **Common mistakes in the exam:** (1) dropping the εu^3 term when changing to u, or writing it on the wrong side, which changes the sign of the precession; (2) treating the orbit as a closed ellipse with θ in place of $k\theta$ (apogee at 180°); (3) taking the time between perigees as the time for one turn, or dividing $\Delta\phi$ by the period of a Kepler orbit with h instead of $\tilde{h}$; (4) using the first-order $\pi\varepsilon/h^2$ when ε/h^2 is not small.

Problem · Unit 4 · 10 points · 60 min · Standard exam level

P07 — Insertion orbit and circularisation impulse: conic from the energy, apsides, apogee burn and time of flight

A probe P of mass m moves in the gravitational field of a planet of centre O, radius R and gravitational parameter μ: the only force is $\mathbf{F} = -\dfrac{m\mu}{r^2}\mathbf{e}_r$, with $r = |\mathbf{OP}|$, $\mathbf{e}_r = \mathbf{OP}/r$; O is fixed and the frame S_0 attached to it is inertial. At $t = 0$ the probe is at the distance r_0 from O and its velocity $\mathbf{v}_0$ has the magnitude

$$v_0 = \sqrt{\mu / r_0}$$

*(the local circular speed) and makes the **flight-path angle** $\gamma = 30^\circ$ with the local horizontal, pointing away from the planet. Use polar coordinates (r, θ) in the orbital plane, with θ (the true anomaly) measured from the perigee Π of the conic in the sense of the motion (Π is a virtual point: with the data below it lies inside the planet, so the probe never reaches it; the trajectory arc starts at the surface $r = R$), the unit vectors $\mathbf{e}_r$, $\mathbf{e}_\theta$ and $h = r^2\dot\theta > 0$. The local horizontal at P is the direction of $\mathbf{e}_\theta$, so $\mathbf{v}_0 = v_0(\sin\gamma\,\mathbf{e}_r + \cos\gamma\,\mathbf{e}_\theta)$.*

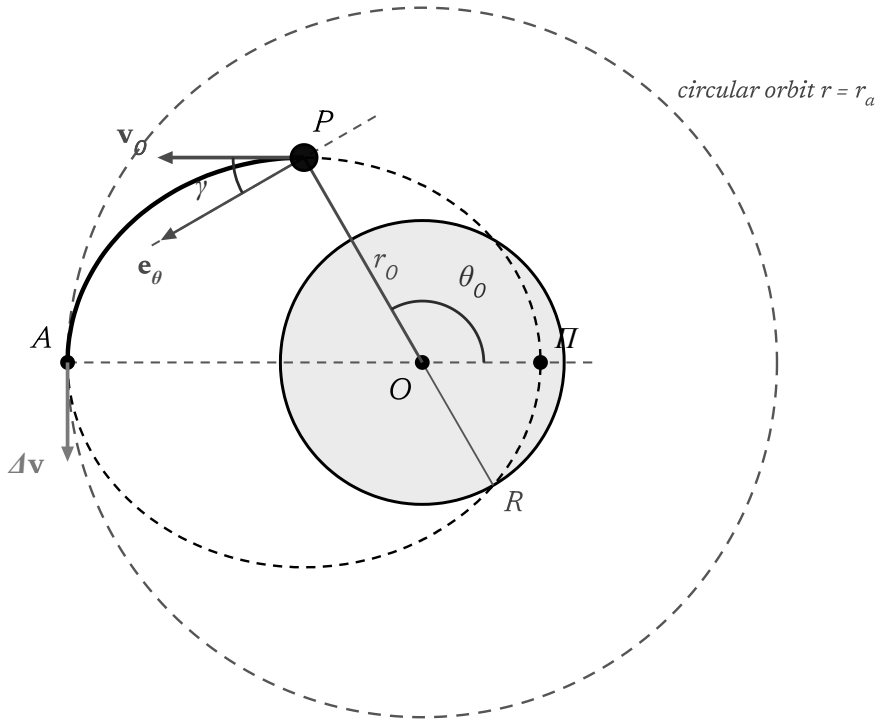


Figure 1. Insertion orbit: focus O (planet of radius R), probe P at r_0 with velocity $\mathbf{v}_0$ and flight-path angle γ above the local horizontal $\mathbf{e}_\theta$; true anomaly θ_0 from the perigee Π; apogee A, impulse $\Delta\mathbf{v}$ and the circular target orbit $r = r_a$ (dashed).

*When the probe reaches the apogee A, its engine fires a short **tangential** impulse $\Delta\mathbf{v}$ that puts it on the circular orbit of radius $r_a = |\mathbf{OA}|$.*

Setup

- Data: μ, r_0, $v_0 = \sqrt{\mu/r_0}$, $\gamma = 30^\circ$. Numerical part (f): $\mu = 4.0 \times 10^{14}$ m^3 s^{-2}, $r_0 = 1.0 \times 10^7$ m, $R = 6.0 \times 10^6$ m.
- Hypotheses: point particle; planet's field spherically symmetric; no drag; the impulse is instantaneous (the position does not change during the burn) and the mass of propellant is irrelevant for the kinematics.
- Frames: $S_0 = \{O; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$ with $\mathbf{i}$ towards Π, $\mathbf{j}$ in the orbital plane and $\mathbf{k}$ along $\mathbf{H}_O$; polar basis $\mathbf{e}_r = \cos\theta\,\mathbf{i} + \sin\theta\,\mathbf{j}$, $\mathbf{e}_\theta = -\sin\theta\,\mathbf{i} + \cos\theta\,\mathbf{j}$. Change of basis:

$$\left(\mathbf{e}_r\ \mathbf{e}_\theta\ \mathbf{k}\right) = [R]\left(\mathbf{i}\,\mathbf{j}\,\mathbf{k}\right), \qquad [R] = \begin{pmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

with $[R][R]^T = I$ (rows of length $\cos^2\theta + \sin^2\theta = 1$, rows 1 and 2 orthogonal: $-\cos\theta\sin\theta + \sin\theta\cos\theta = 0$) and $\det[R] = \cos^2\theta + \sin^2\theta = +1$, so $\mathbf{e}_r \times \mathbf{e}_\theta = \mathbf{k}$.

- Results from Notes §8.4 used: orbit $r = p/(1 + e\cos\theta)$ with $p = h^2/\mu$ (Eq. 8.14), $p = a(1 - e^2)$ (Eq. 8.16), $r_p = a(1 - e)$, $r_a = a(1 + e)$, period $\tau = 2\pi\sqrt{a^3/\mu}$ (Eq. 8.18); vis-viva $v^2 = \mu(2/r - 1/a)$ (Notes §8.5).

Part (a) · 1.5 points

Conserved quantities and type of conic

Explain why h and the energy per unit mass E/m are constant. Compute them from the initial conditions and decide the type of conic and its semi-major axis a.

Step 1. Why they are conserved

$\mathbf{F}$ is central, so its torque about O is $\mathbf{r} \times \mathbf{F} = \mathbf{0}$ and $\mathbf{H}_O$ is constant (Notes §4.8); the motion is planar and $h = r^2\dot\theta$ is constant (Notes §4.8.1). $\mathbf{F} = -\nabla V$ with $V = -m\mu/r$, and it is the only force, so the mechanical energy $E = \frac{1}{2}mv^2 - m\mu/r$ is constant (Notes §4.5).

Step 2. Angular momentum

With $\mathbf{r} = r_0\,\mathbf{e}_r$ and $\mathbf{v}_0 = v_0(\sin\gamma\,\mathbf{e}_r + \cos\gamma\,\mathbf{e}_\theta)$:

$$\mathbf{r} \times \mathbf{v}_0 = r_0 v_0 \cos\gamma\,(\mathbf{e}_r \times \mathbf{e}_\theta) = r_0 v_0 \cos\gamma\,\mathbf{k} \implies h = r_0\sqrt{\frac{\mu}{r_0}}\,\frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{2}\sqrt{\mu r_0}$$

Step 3. Energy and conic

$$\frac{E}{m} = \frac{1}{2}\,\frac{\mu}{r_0} - \frac{\mu}{r_0} = -\frac{\mu}{2r_0} < 0$$

A negative energy means a bounded orbit: an **ellipse**. The vis-viva equation (Notes §8.5) gives $E/m = -\mu/(2a)$, so $a = r_0$. Note the trap: the probe has exactly the circular speed of its radius, yet the orbit is not a circle, because the velocity is not horizontal; the speed fixes a only, the direction fixes the shape.

KEY RESULTS

- $h = r_0 v_0 \cos\gamma = \frac{\sqrt{3}}{2}\sqrt{\mu r_0},\ h^2 = \frac{3}{4}\mu r_0$.
- $E/m = \frac{1}{2}v_0^2 - \mu/r_0 = -\dfrac{\mu}{2r_0} < 0$: ellipse, $a = -\dfrac{\mu}{2E/m} = r_0$.

Part (b) · 2 points

Shape of the orbit and apsides

Find the semi-latus rectum p, the eccentricity e and the perigee and apogee radii r_p, r_a. Verify the apsides independently from the energy equation.

Step 1. Semi-latus rectum

$$p = \frac{h^2}{\mu} = \frac{\frac{3}{4}\mu r_0}{\mu} = \frac{3}{4}r_0$$

Step 2. Eccentricity

From $p = a(1-e^2)$ (Notes Eq. 8.16): $e^2 = 1 - p/a = 1 - \frac{3}{4} = \frac{1}{4}$, so $e = \frac{1}{2}$. Substituting $a = -\mu/(2E/m)$ and $p = h^2/\mu$ gives the general formula $e^2 = 1 + 2(E/m)h^2/\mu^2 = 1 + 2\big(-\frac{\mu}{2r_0}\big)\frac{3\mu r_0}{4\mu^2} = 1 - \frac{3}{4}$, the same value.

Step 3. Apsides

$$r_p = a(1-e) = \frac{r_0}{2}, \qquad r_a = a(1+e) = \frac{3r_0}{2}$$

and indeed $r_p + r_a = 2a$.

Step 4. Independent check

With $v^2 = \dot r^2 + h^2/r^2$, the energy at an apsis ($\dot r = 0$) reads $\dfrac{h^2}{2r^2} - \dfrac{\mu}{r} = \dfrac{E}{m}$. Multiplying by $2r^2/\mu$ with $h^2 = \tfrac{3}{4}\mu r_0$ and $E/m = -\mu/(2r_0)$:

$$\frac{3}{4}r_0 - 2r = -\frac{r^2}{r_0} \implies r^2 - 2r_0 r + \frac{3}{4}r_0^2 = 0 \implies r = r_0 \pm \sqrt{r_0^2 - \tfrac{3}{4}r_0^2} = r_0 \pm \frac{r_0}{2}$$

the same $r_p = r_0/2$ and $r_a = 3r_0/2$.

KEY RESULTS

- $p = \tfrac{3}{4}r_0,\ e = \sqrt{1 - p/a} = \tfrac{1}{2}$ (equivalently $e^2 = 1 + 2(E/m)h^2/\mu^2$).
- $r_p = a(1-e) = \tfrac{1}{2}r_0,\ r_a = a(1+e) = \tfrac{3}{2}r_0$.

Part (c) · 1.5 points

Position on the orbit, apsidal speeds and period

Find the true anomaly θ_0 of the initial point (justify the sign), the speeds v_p, v_a at the apsides of the conic (v_p is the speed the conic would have at its virtual perigee) and the period τ of the full ellipse.

Step 1. True anomaly

$$r_0 = \frac{p}{1 + e\cos\theta_0} \implies 1 + \tfrac{1}{2}\cos\theta_0 = \frac{p}{r_0} = \frac{3}{4} \implies \cos\theta_0 = -\frac{1}{2}$$

so $\theta_0 = \pm 120^\circ$.

Step 2. Sign from the radial velocity

Differentiating $r = p\,(1 + e\cos\theta)^{-1}$ with the chain rule and $\dot\theta = h/r^2$:

$$\dot r = \frac{p\,e\sin\theta\,\dot\theta}{(1+e\cos\theta)^2} = \frac{r^2}{p}\,e\sin\theta\,\frac{h}{r^2} = \frac{h}{p}e\sin\theta = \frac{\mu}{h}e\sin\theta$$

The velocity points away from the planet, $\dot r = v_0\sin\gamma > 0$, so $\sin\theta_0 > 0$ and $\theta_0 = +120^\circ$. Check: $\frac{\mu}{h}\,\frac{1}{2}\,\frac{\sqrt{3}}{2} = \frac{\mu}{\sqrt{3}\sqrt{\mu r_0}}\,\frac{\sqrt{3}}{2} = \frac{1}{2}\sqrt{\mu/r_0} = v_0\sin 30^\circ$.

Step 3. Apsidal speeds

At the apsides the velocity is perpendicular to $\mathbf{r}$ ($\dot r = 0$), so $v = h/r$:

$$v_p = \frac{h}{r_p} = \frac{\frac{\sqrt{3}}{2}\sqrt{\mu r_0}}{r_0/2} = \sqrt{3}\sqrt{\frac{\mu}{r_0}}, \qquad v_a = \frac{h}{r_a} = \frac{\frac{\sqrt{3}}{2}\sqrt{\mu r_0}}{3r_0/2} = \frac{1}{\sqrt{3}}\sqrt{\frac{\mu}{r_0}}$$

Vis-viva check at the apogee: $\mu\left(\frac{2}{r_a} - \frac{1}{a}\right) = \frac{\mu}{r_0}\left(\frac{4}{3} - 1\right) = \frac{\mu}{3r_0} = v_a^2$.

Step 4. Period

Kepler's third law (Notes Eq. 8.18) with $a = r_0$: $\tau = 2\pi\sqrt{a^3/\mu} = 2\pi\sqrt{r_0^3/\mu}$.

KEY RESULTS

- $\cos\theta_0 = -\frac{1}{2}$ and $\dot r > 0$, so $\theta_0 = +120^\circ$ (outbound, between perigee and apogee).
- $v_p = h/r_p = \sqrt{3}\sqrt{\mu/r_0}$, $v_a = h/r_a = \sqrt{\mu/r_0}/\sqrt{3}$; $\tau = 2\pi\sqrt{r_0^3/\mu}$.

Part (d) · 2 points

Circularisation impulse at the apogee

Compute the impulse $\Delta\mathbf{v}$ (magnitude and direction) that makes the orbit circular at r_a, the change of energy per unit mass and the ratio of the new period to τ. Why is the apogee the right place for a purely tangential burn?

Step 1. Why the apogee

A circular orbit has $\dot r = 0$ everywhere, so its velocity is perpendicular to $\mathbf{r}$. At the apogee $\dot r = 0$ already: the velocity $v_a\,\mathbf{e}_\theta$ is horizontal and only its magnitude has to change. At any other point the burn would also have to cancel $\dot r$, so it could not be purely tangential.

Step 2. Impulse

The circular speed at r_a follows from $\mu/r_a^2 = v^2/r_a$ (or from vis-viva with $a = r_a$):

$$v_{circ} = \sqrt{\frac{\mu}{r_a}} = \sqrt{\frac{2\mu}{3r_0}}, \qquad \Delta\mathbf{v} = (v_{circ} - v_a)\,\mathbf{e}_\theta = \left(\sqrt{\tfrac{2}{3}} - \tfrac{1}{\sqrt{3}}\right)\sqrt{\frac{\mu}{r_0}}\,\mathbf{e}_\theta = \frac{\sqrt{2}-1}{\sqrt{3}}\sqrt{\frac{\mu}{r_0}}\,\mathbf{e}_\theta$$

positive: the engine pushes **forward**, in the direction of motion.

Step 3. Energy

The position does not change during the burn, so only the kinetic energy changes:

$$\frac{\Delta E}{m} = \frac{1}{2}\left(v_{circ}^2 - v_a^2\right) = \frac{1}{2}\left(\frac{2}{3} - \frac{1}{3}\right)\frac{\mu}{r_0} = \frac{\mu}{6r_0}$$

Check with $E/m = -\mu/(2a)$: $-\dfrac{\mu}{2r_a} + \dfrac{\mu}{2a} = -\dfrac{\mu}{3r_0} + \dfrac{\mu}{2r_0} = \dfrac{\mu}{6r_0}$.

Step 4. New period

$\tau_{circ} = 2\pi\sqrt{r_a^3/\mu}$, so $\tau_{circ}/\tau = (r_a/a)^{3/2} = (3/2)^{3/2} \approx 1.837$.

KEY RESULTS

- $v_{circ} = \sqrt{\mu/r_a} = \sqrt{2/3}\,\sqrt{\mu/r_0}$; $\Delta v = v_{circ} - v_a = \dfrac{\sqrt{2}-1}{\sqrt{3}}\sqrt{\dfrac{\mu}{r_0}} \approx 0.239\sqrt{\mu/r_0}$, along $+\mathbf{e}_\theta$ (prograde).
- $\Delta E/m = \frac{1}{2}(v_{circ}^2 - v_a^2) = \dfrac{\mu}{6r_0} > 0$; $\tau_{circ}/\tau = (3/2)^{3/2} \approx 1.837$.

Part (e) · 1.5 points

Time of flight from r_0 to the apogee

*Compute the time the probe needs to go from the initial point to the apogee. Show that it is **not** $\tau \cdot (60^\circ/360^\circ)$. (Extension: use the eccentric anomaly $\mathcal{E}$, defined by $r = a(1 - e\cos\mathcal{E})$.)*

Step 1. Eccentric anomaly (extension)

Define $\mathcal{E}$ by $r = a(1 - e\cos\mathcal{E})$; it runs from 0 at the perigee to π at the apogee. Equating with $r = p/(1 + e\cos\theta)$ and $p = a(1 - e^2)$ and solving for $\cos\mathcal{E}$:

$$\cos\mathcal{E} = \frac{1}{e}\left(1 - \frac{1 - e^2}{1 + e\cos\theta}\right) = \frac{e + \cos\theta}{1 + e\cos\theta}$$

At $\theta_0 = 120^\circ$: $\cos\mathcal{E}_0 = (\frac{1}{2} - \frac{1}{2})/(1 - \frac{1}{4}) = 0$, so $\mathcal{E}_0 = \pi/2$ (outbound branch); check $r = a(1 - 0) = r_0$. At the apogee $\mathcal{E} = \pi$.

Step 2. Kepler's equation

The energy equation $\dot r^2 + h^2/r^2 = \mu(2/r - 1/a)$ with $h^2 = \mu a(1-e^2)$ gives

$$\dot r^2 = \frac{\mu}{a\,r^2}\left[2ar - r^2 - a^2(1-e^2)\right]$$

Substituting $r = a(1 - e\cos\mathcal{E})$: $2ar - r^2 - a^2(1-e^2) = a^2\left[2 - 2e\cos\mathcal{E} - 1 + 2e\cos\mathcal{E} - e^2\cos^2\mathcal{E} - 1 + e^2\right] = a^2e^2\sin^2\mathcal{E}$. Hence $\dot r = \pm\sqrt{\mu/a}\;a\,e\sin\mathcal{E}/r$. With $\mathcal{E}$ increasing along the motion, $\sin\mathcal{E} > 0$ on the outbound branch ($0 < \mathcal{E} < \pi, \dot r > 0$) and $\sin\mathcal{E} < 0$ on the inbound branch ($\pi < \mathcal{E} < 2\pi, \dot r < 0$), so the sign is $+$ on **both** branches: $\dot r = \sqrt{\mu/a}\;a\,e\sin\mathcal{E}/r$ for every $\mathcal{E}$. On the other hand, by the chain rule $\dot r = a\,e\sin\mathcal{E}\;\dot{\mathcal{E}}$, also valid on both branches. Equating (and, at the apses where $\sin\mathcal{E} = 0$, by continuity):

$$\dot{\mathcal{E}} = \frac{\sqrt{\mu/a}}{r} = \frac{n}{1 - e\cos\mathcal{E}} \quad \text{for every } \mathcal{E} \implies (1 - e\cos\mathcal{E})\,d\mathcal{E} = n\,dt, \qquad n = \sqrt{\frac{\mu}{a^3}}$$

Step 3. Integration with Barrow's rule

The antiderivative of $1 - e\cos\mathcal{E}$ is $\mathcal{E} - e\sin\mathcal{E}$, so

$$n\,t_{0\to A} = \left[\mathcal{E} - e\sin\mathcal{E}\right]_{\pi/2}^{\pi} = (\pi - 0) - \left(\frac{\pi}{2} - \frac{1}{2}\right) = \frac{\pi}{2} + \frac{1}{2}$$

and with $1/n = \sqrt{r_0^3/\mu}$ (because $a = r_0$): $t_{0\to A} = \left(\frac{\pi}{2} + \frac{1}{2}\right)\sqrt{r_0^3/\mu} \approx 2.071\sqrt{r_0^3/\mu}$. As a fraction of the period, $(\pi/2 + 1/2)/(2\pi) \approx 0.330$.

Step 4. The trap

The angle swept is 60°, one sixth of a turn, but the time is a third of the period: near the apogee the probe is slow ($\dot\theta = h/r^2$ is small), so equal angles do **not** take equal times. A direct numerical quadrature of $dt = r^2\,d\theta/h$ from 120° to 180° gives the same $\pi/2 + 1/2$.

KEY RESULTS

- $\cos\mathcal{E} = \dfrac{e + \cos\theta}{1 + e\cos\theta}$: $\mathcal{E}_0 = 90^\circ$ at $\theta_0 = 120^\circ$, $\mathcal{E} = 180^\circ$ at the apogee.
- $t_{0\to A} = \dfrac{1}{n}\left[\mathcal{E} - e\sin\mathcal{E}\right]_{\pi/2}^{\pi} = \left(\dfrac{\pi}{2} + \dfrac{1}{2}\right)\sqrt{\dfrac{r_0^3}{\mu}} \approx 0.330\,\tau$, twice the naive $\tau/6 \approx 0.167\,\tau$.

Part (f) · 1.5 points

Numerical values and a failed burn

With $\mu = 4.0 \times 10^{14}\ m^3\ s^{-2}$, $r_0 = 1.0 \times 10^7\ m$: compute v_0, h, v_p, v_a, τ, Δv, $\Delta E/m$ and $t_{0\to A}$. If the engine had failed at the apogee, would the probe have survived, given the planet radius $R = 6.0 \times 10^6\ m$? If not, at which true anomaly and how long after the apogee would it hit the surface?

Step 1. Numbers

$\sqrt{\mu/r_0} = \sqrt{4.0 \times 10^7} = 6325$ m/s $= v_0$ and $\sqrt{r_0^3/\mu} = \sqrt{2.5 \times 10^6} = 1581$ s. Then $h = \frac{\sqrt{3}}{2} r_0 v_0 = 5.477 \times 10^{10}$ m^2/s, $v_p = \sqrt{3} \times 6325 = 1.095 \times 10^4$ m/s, $v_a = 6325/\sqrt{3} = 3651$ m/s, $\tau = 2\pi \times 1581 = 9935$ s, $v_{circ} = 5164$ m/s, $\Delta v = 5164 - 3651 = 1512$ m/s, $\Delta E/m = \mu/(6r_0) = 6.67 \times 10^6$ J/kg, $\tau_{circ} = 1.825 \times 10^4$ s and $t_{0\to A} = 2.071 \times 1581 = 3274$ s, against the naive $\tau/6 = 1656$ s.

Step 2. Failed burn

$r_p = r_0/2 = 5.0 \times 10^6$ m is smaller than $R = 6.0 \times 10^6$ m: after the apogee the probe would descend into the planet. Impact: $7.5 \times 10^6/(1 + \frac{1}{2}\cos\theta) = 6.0 \times 10^6 \Rightarrow \cos\theta = \frac{1}{2}$, and on the inbound branch ($\dot r < 0, \sin\theta < 0$) $\theta = 300^\circ$. There $\cos\mathcal{E} = (\frac{1}{2} + \frac{1}{2})/(1 + \frac{1}{4}) = \frac{4}{5}$ with $\mathcal{E}$ in $(\pi, 2\pi)$: $\mathcal{E}_i = 2\pi - \arccos\frac{4}{5} = 5.640$ rad, $\sin\mathcal{E}_i = -\frac{3}{5}$.

Step 3. Time to impact

$$n\,t_{A\to i} = \left[\mathcal{E} - e\sin\mathcal{E}\right]_{\pi}^{\mathcal{E}_i} = \left(5.640 + \frac{1}{2}\cdot\frac{3}{5}\right) - \pi = 2.798 \implies t_{A\to i} = 2.798 \times 1581 = 4424\ \text{s}$$

The circularisation burn is therefore not optional: it also raises the perigee from $0.5\,r_0$ to $1.5\,r_0$.

KEY RESULTS

- $v_0 = 6325$ m/s, $h = 5.477 \times 10^{10}$ m^2/s, $v_p = 1.095 \times 10^4$ m/s, $v_a = 3651$ m/s, $\tau = 9935$ s, $\Delta v = 1512$ m/s, $\Delta E/m = 6.67 \times 10^6$ J/kg, $t_{0\to A} = 3274$ s (naive: 1656 s).
- $r_p = 5.0 \times 10^6$ m $< R$: impact at $\theta = 300^\circ$ ($\cos\theta = \frac{1}{2}$, $\cos\mathcal{E} = \frac{4}{5}$), 4424 s after the apogee.

Interpretation and checks

- **Dimensional check:** $h = r_0 v_0$ in m^2/s; h^2/μ in m^4 s^{-2}/(m^3 s^{-2}) = m; E/m and μ/r in m^2 s^{-2} = J/kg; $\sqrt{a^3/\mu}$ in s; Δv in m/s.
- **Limits:** $\gamma \to 0$ with $v_0 = \sqrt{\mu/r_0}$ gives $h^2 = \mu r_0$, $e = 0$: the probe is already on the circle of radius r_0. $\gamma \to 90^\circ$ gives $h = 0$: a radial (degenerate) orbit, straight up and back down. For any γ, $a = r_0$, so the period does not depend on γ.
- **Common mistakes in the exam:** (1) concluding that the orbit is circular because $v_0 = \sqrt{\mu/r_0}$; (2) using $h = r_0 v_0$ without $\cos\gamma$; (3) taking $\theta_0 = -120^\circ$ (the inbound point) without checking the sign of $\dot r$; (4) computing Δv as $v_{circ}(r_0) - v_0$ or applying it at r_0; (5) converting angles into times with

$\tau\,\Delta\theta/2\pi$, which is wrong for any $e \neq 0$.

Problem · Unit 2 · 10 points · 60 min · Standard exam level

P08 — Seismic accelerometer on a vibrating base: inertia force, frequency response and the two operating regimes

*A vibration sensor is a rigid case whose floor is bolted to a structure that moves **vertically** with the prescribed law $z(t) = Z\cos\nu t$ ($Z > 0$, $\nu > 0$), where z is the height of the case floor above a fixed datum, measured along the upward vertical axis y of the inertial frame S_0. Inside the case, a proof mass m (a particle that can only move along the vertical) is joined to the case floor by a linear spring of stiffness k and natural length L_0 and by a linear viscous damper of constant c (force $-c$ times the velocity of the mass relative to the case). Uniform gravity $-g\,\mathbf{j}$ acts.*

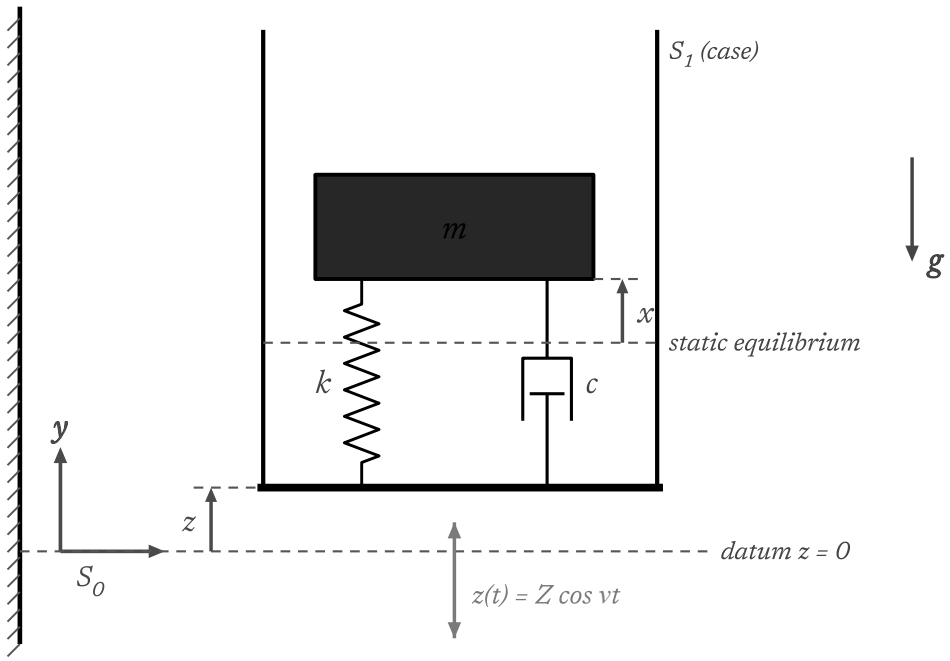


Figure 1. Seismic accelerometer: the case floor moves as $z(t) = Z\cos\nu t$ with respect to the fixed datum of S_0; the proof mass m is held by the spring k and the damper c; x is measured in the case frame S_1 from the static equilibrium level (both z and x positive upwards).

*Let S_1 be the frame attached to the case. The relative coordinate x is the height of the mass in S_1 measured **from its static equilibrium level** (the level where it rests when $z \equiv 0$), positive upwards. Write $\omega_0 = \sqrt{k/m}$ and $\zeta = c/(2\sqrt{km})$ (the Notes' damping parameter is $\gamma = c/(2m) = \zeta\omega_0$), and assume $0 < \zeta < 1$.*

Setup

- Data: m, k, c, L_0, g, Z, ν. Numerical part (f): $m = 0.050$ kg, $k = 500$ N/m, $c = 2.0$ N s/m, $Z = 2.0$ mm, $\nu = 50$ rad/s, $g = 9.81$ m/s^2.
- Hypotheses: point mass with vertical motion only (lateral guides are smooth), massless spring and damper, linear laws, S_0 inertial.
- Frames: $S_0 = \{O; \mathbf{i}, \mathbf{j}\}$ fixed; S_1 attached to the case, a **translating** frame: $\boldsymbol{\omega}_{10} = \mathbf{0}$, origin acceleration $\ddot{z}\,\mathbf{j}$.
- Notation: x is the **vertical** relative coordinate of the mass (along $\mathbf{j}$), not a coordinate along the horizontal axis $\mathbf{i}$ of S_0.

Part (a) · 1.5 points

Degrees of freedom, frames and kinematics

How many degrees of freedom does the mass have? Write the absolute position, velocity and acceleration of the mass in terms of z and x, and identify the relative and the transport (dragging) acceleration.

Step 1. Degrees of freedom

The mass can only move along the vertical: 1 coordinate. The case is not a body of the problem but a prescribed motion, so it adds no unknown; the constraint "the mass moves on the vertical line of the case" is rheonomic in S_0 and fixed in S_1. The coordinate is x.

Step 2. Absolute kinematics

Let ξ be the height of the mass above the case floor, so the absolute height is $y_P = z + \xi$, and $\xi = \xi_e + x$ with ξ_e constant (found in (b)). Differentiating twice (ξ_e is constant):

$$y_P = z + \xi_e + x, \qquad \dot{y}_P = \dot{z} + \dot{x}, \qquad \ddot{y}_P = \ddot{z} + \ddot{x}$$

Step 3. Relative-motion reading

S_1 translates, so $\mathbf{a}_P = \mathbf{a}_{O_1} + \mathbf{a}_{rel}$ (Notes §3.2 with $\boldsymbol{\omega} = \mathbf{0}$): transport acceleration $\ddot{z}\,\mathbf{j}$ with, by the chain rule on $\cos(\nu t)$, $\dot{z} = -\nu Z \sin \nu t$ and $\ddot{z} = -\nu^2 Z \cos \nu t$; relative acceleration $\ddot{x}\,\mathbf{j}$; the Coriolis term $2\boldsymbol{\omega} \times \mathbf{v}_{rel}$ vanishes.

KEY RESULTS

- 1 degree of freedom, $x(t)$; $z(t)$ is prescribed (rheonomic constraint).
- $y_P = z + \xi_e + x$, $\mathbf{a}_P = (\ddot{z} + \ddot{x})\,\mathbf{j}$; $\mathbf{a}_{rel} = \ddot{x}\,\mathbf{j}$, transport $\ddot{z}\,\mathbf{j} = -\nu^2 Z \cos \nu t\,\mathbf{j}$, no Coriolis term.

Part (b) · 1.5 points

Forces in the case frame and static equilibrium

List the forces on the mass in S_1, including the inertia force, in terms of ξ and $\dot\xi$. Find the static equilibrium height ξ_e for $z \equiv 0$ and the static deflection of the spring.

Step 1. Applied forces

- Spring: elongation $\xi - L_0$; the force on the mass is $\mathbf{F}_s = -k(\xi - L_0)\,\mathbf{j}$.
- Damper: opposes the velocity relative to the case, $\mathbf{F}_d = -c\dot\xi\,\mathbf{j} = -c\dot x\,\mathbf{j}$.
- Weight: $\mathbf{W} = -mg\,\mathbf{j}$.
- Guides: a smooth lateral reaction $\mathbf{N}$, horizontal; it has no vertical component and does no work.

Step 2. Inertia force

In the translating frame S_1 the relative-motion law is $m\mathbf{a}_{rel} = \sum \mathbf{F} - m\mathbf{a}_{O_1}$ (Notes §4.9), so

$$\mathbf{F}_{in} = -m\ddot z\,\mathbf{j} = m\nu^2 Z\cos\nu t\,\mathbf{j}$$

When the case accelerates upwards ($\ddot z > 0$) the mass feels pushed down, as in a lift.

Step 3. Static equilibrium

With $z \equiv 0$ there is no inertia force and at rest $\dot\xi = 0$:

$$-k(\xi_e - L_0) - mg = 0 \;\Longrightarrow\; \xi_e = L_0 - \frac{mg}{k}$$

The spring is compressed by $\delta_{st} = mg/k$ and carries the weight.

KEY RESULTS

- $\mathbf{F}_s = -k(\xi - L_0)\,\mathbf{j}$, $\mathbf{F}_d = -c\dot\xi\,\mathbf{j}$, $\mathbf{W} = -mg\,\mathbf{j}$, $\mathbf{F}_{in} = -m\ddot z\,\mathbf{j} = m\nu^2 Z\cos\nu t\,\mathbf{j}$; lateral reaction $\mathbf{N} \perp \mathbf{j}$.
- $\xi_e = L_0 - mg/k$; static deflection $\delta_{st} = mg/k$ (spring compressed).

Part (c) · 1.5 points

Equation of motion

Project the relative-motion law on **j** *and obtain the equation of motion for x. Explain why g does not appear, and write the equation in the normalised form with ω_0 and ζ.*

Step 1. Projection

$$m\ddot{\xi} = -k(\xi - L_0) - c\dot{\xi} - mg - m\ddot{z}$$

Step 2. Change of variable

With $\xi = \xi_e + x, \dot{\xi} = \dot{x}, \ddot{\xi} = \ddot{x}$ and $-k(\xi_e + x - L_0) = -kx - k(\xi_e - L_0) = -kx + mg$ (from (b)):

$$m\ddot{x} = -kx + mg - c\dot{x} - mg - m\ddot{z} \implies m\ddot{x} + c\dot{x} + kx = -m\ddot{z} = m\nu^2 Z\cos\nu t$$

The weight is a constant force; measuring x from the static equilibrium absorbs it into the spring preload, so it drops out of the dynamics. This is true only because the spring is linear: the same constant offset works at every x.

Step 3. Normal form

Dividing by m, with $k/m = \omega_0^2$ and $c/m = 2\zeta\omega_0$:

$$\ddot{x} + 2\zeta\omega_0\dot{x} + \omega_0^2 x = \nu^2 Z\cos\nu t$$

This is Notes Eq. 5.32 with the force amplitude $F_1 = m\nu^2 Z$: the base motion acts as a harmonic force whose amplitude grows with ν^2. Extension (Lagrange equations are not in the course notes): cross-check by Lagrange in S_0: $T = \frac{1}{2}m(\dot{z} + \dot{\xi})^2$, $V = mg(z + \xi) + \frac{1}{2}k(\xi - L_0)^2$ and the damper's generalised force $-c\dot{\xi}$ give the same equation.

KEY RESULTS

- $m\ddot{x} + c\dot{x} + kx = m\nu^2 Z\cos\nu t$.
- $\ddot{x} + 2\zeta\omega_0\dot{x} + \omega_0^2 x = \nu^2 Z\cos\nu t$; g is cancelled by the static preload of the spring.

Part (d) · 2 points

Steady-state response

Using the complex representation, find the amplitude X and the phase φ of the steady-state solution $x_p = X\cos(\nu t + \varphi)$, and write X/Z as a function of $r = \nu/\omega_0$ and ζ. What is the complete solution?

Step 1. Complex equation

The real forcing is $\nu^2 Z\cos\nu t = \mathrm{Re}\big(\nu^2 Z e^{i\nu t}\big)$ (Notes §5.2 and §5.4). With $\tilde{x} = Ae^{i\nu t}$: $\dot{\tilde{x}} = i\nu\tilde{x}$, $\ddot{\tilde{x}} = -\nu^2\tilde{x}$ (chain rule on the exponential), and

$$\big(-\nu^2 + 2i\zeta\omega_0\nu + \omega_0^2\big)A = \nu^2 Z \implies A = \frac{\nu^2 Z}{\omega_0^2 - \nu^2 + 2i\zeta\omega_0\nu}$$

Step 2. Amplitude and phase

Writing $A = Xe^{i\varphi}$, $|A|$ is the modulus of the numerator over that of the denominator:

$$X = \frac{\nu^2 Z}{\sqrt{(\omega_0^2 - \nu^2)^2 + 4\zeta^2\omega_0^2\nu^2}}, \qquad \varphi = -\arg\left(\omega_0^2 - \nu^2 + 2i\zeta\omega_0\nu\right)$$

so $\tan\varphi = -2\zeta\omega_0\nu/(\omega_0^2 - \nu^2)$; the imaginary part of the denominator is positive, hence $-\pi < \varphi < 0$ (the mass lags behind the forcing, Notes Eq. 5.42). The real part is $x_p = X\cos(\nu t + \varphi)$.

Step 3. Dimensionless form

Dividing numerator and denominator by ω_0^2 with $r = \nu/\omega_0$:

$$\frac{X}{Z} = \frac{r^2}{\sqrt{(1 - r^2)^2 + (2\zeta r)^2}}, \qquad \tan\varphi = -\frac{2\zeta r}{1 - r^2}$$

Step 4. Complete solution

Add the general solution of the homogeneous equation (Notes Eq. 5.26, underdamped): $x = X\cos(\nu t + \varphi) + a\,e^{-\zeta\omega_0 t}\cos(\omega_0\sqrt{1-\zeta^2}\,t + \beta)$, with a, β fixed by the initial conditions. The second term is the transient.

KEY RESULTS

- $A = \dfrac{\nu^2 Z}{\omega_0^2 - \nu^2 + 2i\zeta\omega_0\nu}$; $X = |A| = \dfrac{\nu^2 Z}{\sqrt{(\omega_0^2-\nu^2)^2 + 4\zeta^2\omega_0^2\nu^2}}$.
- $\tan\varphi = -\dfrac{2\zeta\omega_0\nu}{\omega_0^2-\nu^2}$ with $-\pi < \varphi < 0$; $\dfrac{X}{Z} = \dfrac{r^2}{\sqrt{(1-r^2)^2 + (2\zeta r)^2}}$.
- $x = X\cos(\nu t + \varphi) + a\,e^{-\zeta\omega_0 t}\cos(\omega_0\sqrt{1-\zeta^2}\,t + \beta)$.

Part (e) · 2 points

Operating regimes and the resonant peak

Discuss the limits $\nu \ll \omega_0$ and $\nu \gg \omega_0$: what does x measure in each case? Then find the forcing frequency at which X is maximum and the maximum value of X/Z. Is it at $\nu = \omega_0$, or at $\omega_0\sqrt{1-2\zeta^2}$ as in Notes §5.4.1?

Step 1. Low frequency: accelerometer

For $r \to 0$ the denominator tends to 1 and $\varphi \to 0$: $X \approx r^2 Z = \nu^2 Z/\omega_0^2$ and

$$x \approx \frac{\nu^2 Z\cos\nu t}{\omega_0^2} = -\frac{\ddot{z}}{\omega_0^2}$$

The relative displacement is proportional to the **acceleration** of the base: the spring force kx supplies the force $-m\ddot{z}$ the mass needs to follow the case.

Step 2. High frequency: vibrometer

For $r \to \infty$: $X/Z \to r^2/r^2 = 1$ and $\tan\varphi = 2\zeta r/(r^2-1) \to 0^+$ with φ in the third quadrant, so $\varphi \to -\pi$, so $x \approx Z\cos(\nu t - \pi) = -z$. The absolute height $y_P = z + \xi_e + x \approx \xi_e$ is constant: the mass cannot follow the fast motion and stays still in S_0, so x measures the **displacement** of the base.

Step 3. Maximum of the response

$(X/Z)^2 = f(s) = s^2/D$ with $D = (1-s)^2 + 4\zeta^2 s$ and $D' = -2(1-s) + 4\zeta^2$. By the quotient rule, $f'(s) = 0$ requires $2sD - s^2 D' = 0$, i.e. $2D = sD'$:

$$2(1-2s+s^2) + 8\zeta^2 s = -2s + 2s^2 + 4\zeta^2 s \implies 2 - 2s + 4\zeta^2 s = 0 \implies s = \frac{1}{1-2\zeta^2}$$

which is admissible ($s > 0$) for $\zeta < 1/\sqrt{2}$. Then $1 - s = -2\zeta^2/(1 - 2\zeta^2)$ and

$$D = \frac{4\zeta^4 + 4\zeta^2(1 - 2\zeta^2)}{(1 - 2\zeta^2)^2} = \frac{4\zeta^2(1 - \zeta^2)}{(1 - 2\zeta^2)^2}, \qquad f = \frac{s^2}{D} = \frac{1}{4\zeta^2(1 - \zeta^2)}$$

so $\nu_{peak} = \omega_0/\sqrt{1 - 2\zeta^2}$ and $(X/Z)_{max} = 1/\left(2\zeta\sqrt{1 - \zeta^2}\right)$.

Step 4. The trap

The Notes' resonant frequency $\sqrt{\omega_0^2 - 2\gamma^2} = \omega_0\sqrt{1 - 2\zeta^2}$ (Eq. 5.46) is for a force of **constant** amplitude F_1. Here the forcing $m\nu^2 Z$ grows with ν^2, which shifts the maximum of the relative amplitude **above** ω_0. Neither $\nu = \omega_0$ (where $X/Z = 1/(2\zeta)$) nor $\omega_0\sqrt{1 - 2\zeta^2}$ is the peak.

KEY RESULTS

- $\nu \ll \omega_0$: $x \approx \nu^2 Z\cos\nu t/\omega_0^2 = -\ddot{z}/\omega_0^2$ (accelerometer). $\nu \gg \omega_0$: $X \to Z$, $\varphi \to -\pi$, $x \approx -z$ (the mass stays still in S_0: vibrometer).
- Peak at $r^2 = \dfrac{1}{1 - 2\zeta^2}$, i.e. $\nu_{peak} = \dfrac{\omega_0}{\sqrt{1 - 2\zeta^2}} > \omega_0$ (for $\zeta < 1/\sqrt{2}$); $\left(\dfrac{X}{Z}\right)_{max} = \dfrac{1}{2\zeta\sqrt{1 - \zeta^2}}$.

Part (f) · 1.5 points

Numerical case

With the data of the Setup, compute ω_0, ζ, the static deflection, X, φ, the energy dissipated per cycle, the decay of the transient, and the error of an accelerometer that reads $\omega_0^2 X$ as the base acceleration amplitude. Locate the resonant peak.

Step 1. Parameters

$\omega_0 = \sqrt{500/0.050} = 100$ rad/s, $\zeta = 2.0/(2\sqrt{500 \times 0.050}) = 2.0/10 = 0.2$, $r = 50/100 = 0.5$, $\delta_{st} = mg/k = 0.050 \times 9.81/500 = 9.81 \times 10^{-4}$ m $= 0.981$ mm.

Step 2. Amplitude and phase

$$\frac{X}{Z} = \frac{0.25}{\sqrt{0.75^2 + 0.2^2}} = \frac{0.25}{\sqrt{0.6025}} = 0.3221 \implies X = 0.644 \text{ mm}, \qquad \varphi = -\arctan\frac{0.2}{0.75} = -14.93^\circ$$

Step 3. Energy per cycle

With $\dot{x} = -\nu X\sin(\nu t+\varphi)$ and $\int_0^{2\pi/\nu}\sin^2(\nu t+\varphi)\,dt = \pi/\nu$ (the antiderivative of $\sin^2 u$ is $\frac{u}{2} - \frac{\sin 2u}{4}$, evaluated between $u=\varphi$ and $u=\varphi+2\pi$):

$$W = c\nu^2 X^2\frac{\pi}{\nu} = \pi c\nu X^2 = \pi\times 2.0\times 50\times(6.442\times10^{-4})^2 = 1.30\times10^{-4}\ \text{J}$$

In steady state this equals the work done per cycle by the inertia force $m\nu^2 Z\cos\nu t$ on the relative motion.

Step 4. Transient

The homogeneous part decays as $e^{-\zeta\omega_0 t} = e^{-20t}$: relaxation time $1/(\zeta\omega_0) = 0.05$ s, and $e^{-20t} = 0.01$ at $t = \ln 100/20 = 0.230$ s. It oscillates at $\omega_d = \omega_0\sqrt{1-\zeta^2} = 97.98$ rad/s.

Step 5. Accelerometer error and peak

The true acceleration amplitude is $\nu^2 Z = 2500\times 0.002 = 5.0$ m/s^2; the instrument reads $\omega_0^2 X = 10^4\times 6.442\times10^{-4} = 6.44$ m/s^2, i.e. $+28.8$ %: at $r=0.5$ with $\zeta=0.2$ the sensor is too close to resonance. The peak is at $\nu_{peak} = 100/\sqrt{0.92} = 104.3$ rad/s with $X/Z = 1/(0.4\sqrt{0.96}) = 2.552$, $X = 5.10$ mm.

KEY RESULTS

- $\omega_0 = 100$ rad/s, $\zeta = 0.2$, $r = 0.5$, $\delta_{st} = 0.981$ mm.
- $X = 0.644$ mm, $\varphi = -14.93°$; $W = \pi c\nu X^2 = 1.30\times10^{-4}$ J per cycle.
- Transient $\propto e^{-20t}$ (relaxation time 0.05 s, below 1 % after 0.230 s), $\omega_d = 97.98$ rad/s.
- $\omega_0^2 X = 6.44$ m/s^2 against $\nu^2 Z = 5.0$ m/s^2: $+28.8$ % error; peak at $\nu = 104.3$ rad/s with $X = 2.552\,Z = 5.10$ mm.

Interpretation and checks

- **Dimensional check:** $\omega_0^2 = k/m$ in (N/m)/kg = s^{-2}; $\zeta = c/(2\sqrt{km})$ in (N s/m)/(kg s^{-1}) = 1; $\nu^2 Z$ in m s^{-2}, the same as $\omega_0^2 x$; $W = \pi c\nu X^2$ in (N s/m)(s^{-1})(m^2) = J.
- **Limits:** $\zeta\to 0$ gives $X/Z = r^2/|1-r^2|$, infinite at $r=1$; $Z\to 0$ leaves the free damped oscillator of Notes §5.3. With $\zeta\approx 0.7$ the factor $1/\sqrt{(1-r^2)^2+(2\zeta r)^2}$ stays close to 1 over a wide band (at $r=0.5$ it is 0.975, a -2.5 % error), which is why real accelerometers are built with that damping.
- **Common mistakes in the exam:** (1) writing the inertia force as $+m\ddot{z}$, or forgetting it and writing the spring force with the absolute coordinate; (2) keeping mg in the equation after measuring x from the static equilibrium (or dropping it when x is measured from the natural length); (3) taking the peak at $\nu=\omega_0$ or at $\omega_0\sqrt{1-2\zeta^2}$; (4) choosing φ in the wrong quadrant with the arctangent above resonance.

Problem · Unit 2 · 10 points · 60 min · High difficulty / trap

P09 — Bead on a vertical hoop held by a spring: reaction, equilibria and a pitchfork bifurcation

A smooth circular hoop of radius a and centre C is fixed in a vertical plane. A bead P of mass m is threaded on it (it can neither leave the hoop nor the plane). A linear spring of stiffness k and natural length l_0 joins P to the fixed point A, which lies on the vertical through C at the height $d = 2a$ above C. The natural length is $l_0 = d - a = a$, so the spring is unstretched when the bead is at the top T of the hoop. Gravity $-g\,\mathbf{j}$ acts. The position of the bead is given by the angle θ between $\mathbf{CT}$ (upward vertical) and $\mathbf{CP}$, positive clockwise in the figure (towards $+x$).

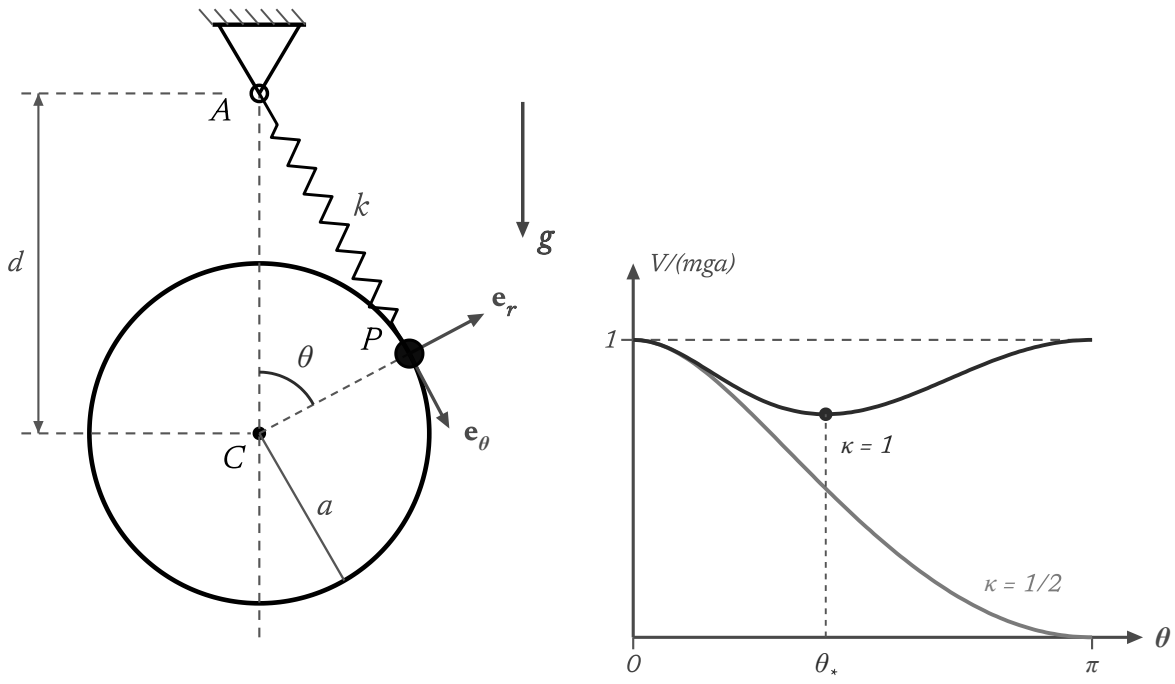


Figure 1. Bead P on the smooth hoop (centre C, radius a), spring of stiffness k from the fixed point A at height $d = 2a$ above C, angle θ from the top; right: $V(\theta)/(mga)$ for $\kappa = ka/(mg) = 1/2$ and $\kappa = 1$.

Use the inertial frame $S_0 = \{C; \mathbf{i}, \mathbf{j}\}$ ($\mathbf{i}$ horizontal, $\mathbf{j}$ upward) and the unit vectors $\mathbf{e}_r = \sin\theta\,\mathbf{i} + \cos\theta\,\mathbf{j}$ (along $\mathbf{CP}$) and $\mathbf{e}_\theta = \cos\theta\,\mathbf{i} - \sin\theta\,\mathbf{j}$ (tangent to the hoop, towards increasing θ). The results are expressed with the dimensionless stiffness

$$\kappa = \frac{ka}{mg}$$

Setup

- Data: m, a, k, g; $d = 2a$, $l_0 = a$. Numerical values in (e) and (f): $a = 0.40$ m, $g = 9.81$ m/s^2.
- Hypotheses: point particle; smooth, rigid, fixed hoop (bilateral constraint); massless linear spring that can be stretched or compressed; S_0 inertial.
- Spring length: $l = |\mathbf{AP}|$ with $\mathbf{AP} = a\,\mathbf{e}_r - d\,\mathbf{j}$.

Part (a) · 1.5 points

Degrees of freedom, geometry and potential energy

How many degrees of freedom has the bead? Find the spring length $l(\theta)$ and the total potential energy $V(\theta)$; check that the applied forces are conservative.

Step 1. Degrees of freedom

A free particle has 3 coordinates. The hoop is a curve: two holonomic, scleronomic constraints (the plane and the circle $|\mathbf{CP}| = a$), so 1 degree of freedom remains, described by θ (Notes §2.7).

Step 2. Spring length

$$l^2 = \mathbf{AP}\cdot\mathbf{AP} = (a\,\mathbf{e}_r - d\,\mathbf{j})\cdot(a\,\mathbf{e}_r - d\,\mathbf{j}) = a^2 - 2ad\,(\mathbf{e}_r\cdot\mathbf{j}) + d^2 = a^2 + d^2 - 2ad\cos\theta$$

With $d = 2a$: $l = a\sqrt{5 - 4\cos\theta}$, from $l(0) = a$ at the top to $l(\pi) = 3a$ at the bottom.

Step 3. Potentials

Weight: uniform, $V_g = mg\,y_P = mga\cos\theta$. Spring: $V_s = \tfrac{1}{2}k(l - l_0)^2$, a function of the position only (Notes §4.4). Both are conservative, the reaction of the smooth fixed hoop does no work (part (b)), so

$$V(\theta) = mga\cos\theta + \tfrac{1}{2}k\big(l(\theta) - l_0\big)^2, \qquad \frac{V}{mga} = \cos\theta + \frac{\kappa}{2}\Big(\sqrt{5 - 4\cos\theta} - 1\Big)^2$$

KEY RESULTS

- 1 degree of freedom, θ (the hoop removes 2 of the 3 spatial coordinates).
- $l^2 = a^2 + d^2 - 2ad\cos\theta = a^2(5 - 4\cos\theta)$; $l(0) = a = l_0$, $l(\pi) = 3a$.
- $V(\theta) = mga\cos\theta + \tfrac{1}{2}k\big(l(\theta) - l_0\big)^2$, i.e. $\dfrac{V}{mga} = \cos\theta + \dfrac{\kappa}{2}\Big(\sqrt{5 - 4\cos\theta} - 1\Big)^2$.

Part (b) · 2 points

Equation of motion and reaction of the hoop

Write the kinematics of P in $\{\mathbf{e}_r, \mathbf{e}_\theta\}$, the forces, the equation of motion free of the reaction, and the reaction $\mathbf{N}$ as a function of θ and $\dot\theta$. Check the equation of motion against $V(\theta)$.

Step 1. Change of basis

The pair $\{\mathbf{e}_r, \mathbf{e}_\theta\}$ turns with θ. Since θ grows clockwise (from $\mathbf{j}$ towards $\mathbf{i}$), the right-handed order in the plane is $\{\mathbf{e}_\theta, \mathbf{e}_r\}$: with $\mathbf{k} = \mathbf{i} \times \mathbf{j}$, $\mathbf{e}_\theta \times \mathbf{e}_r = (\cos\theta\cos\theta + \sin\theta\sin\theta)\,\mathbf{k} = \mathbf{k}$ (the order $\{\mathbf{e}_r, \mathbf{e}_\theta\}$ has determinant -1). Then

$$\begin{pmatrix}\mathbf{e}_\theta \\ \mathbf{e}_r\end{pmatrix} = [R]\begin{pmatrix}\mathbf{i} \\ \mathbf{j}\end{pmatrix}, \qquad [R] = \begin{pmatrix}\cos\theta & -\sin\theta \\ \sin\theta & \cos\theta\end{pmatrix}$$

with $[R][R]^T = \begin{pmatrix}\cos^2\theta + \sin^2\theta & \cos\theta\sin\theta - \sin\theta\cos\theta \\ \sin\theta\cos\theta - \cos\theta\sin\theta & \sin^2\theta + \cos^2\theta\end{pmatrix} = I$ and $\det[R] = \cos^2\theta + \sin^2\theta = +1$.

Step 2. Kinematics

$\mathbf{r}_{CP} = a\,\mathbf{e}_r$. By the chain rule, $\dot{\mathbf{e}}_r = \dot\theta(\cos\theta\,\mathbf{i} - \sin\theta\,\mathbf{j}) = \dot\theta\,\mathbf{e}_\theta$ and $\dot{\mathbf{e}}_\theta = \dot\theta(-\sin\theta\,\mathbf{i} - \cos\theta\,\mathbf{j}) = -\dot\theta\,\mathbf{e}_r$. With a constant:

$$\mathbf{v}_P = a\dot\theta\,\mathbf{e}_\theta, \qquad \mathbf{a}_P = a\ddot\theta\,\mathbf{e}_\theta + a\dot\theta\,\dot{\mathbf{e}}_\theta = a\ddot\theta\,\mathbf{e}_\theta - a\dot\theta^2\,\mathbf{e}_r$$

Step 3. Forces

- Weight: $\mathbf{W} = -mg\,\mathbf{j} = -mg\cos\theta\,\mathbf{e}_r + mg\sin\theta\,\mathbf{e}_\theta$ (using $\mathbf{j}\cdot\mathbf{e}_r = \cos\theta, \mathbf{j}\cdot\mathbf{e}_\theta = -\sin\theta$).
- Spring: $\mathbf{F}_s = -k\,\dfrac{l - l_0}{l}\,\mathbf{AP} = -k\Big(1 - \dfrac{l_0}{l}\Big)\big[(a - d\cos\theta)\,\mathbf{e}_r + d\sin\theta\,\mathbf{e}_\theta\big]$, since $\mathbf{AP}\cdot\mathbf{e}_r = a - d\cos\theta$ and $\mathbf{AP}\cdot\mathbf{e}_\theta = -d\,(\mathbf{j}\cdot\mathbf{e}_\theta) = d\sin\theta$.
- Reaction: the hoop is smooth, so $\mathbf{N}$ is orthogonal to the tangent $\mathbf{e}_\theta$ (Notes §6.5); its component normal to the plane balances nothing (all other forces are in the plane), so $\mathbf{N} = N\,\mathbf{e}_r$.

Step 4. Equation of motion (tangential projection)

$$ma\ddot\theta = mg\sin\theta - k\Big(1 - \frac{l_0}{l}\Big)d\sin\theta = \sin\theta\Big[mg - kd\Big(1 - \frac{l_0}{l}\Big)\Big]$$

N does not appear: this is the reaction-free equation.

Step 5. Reaction (radial projection)

$$-ma\dot\theta^2 = -mg\cos\theta - k\Big(1-\frac{l_0}{l}\Big)(a-d\cos\theta) + N$$

$$N = mg\cos\theta + k\Big(1-\frac{l_0}{l}\Big)(a-d\cos\theta) - ma\dot\theta^2$$

Step 6. Check with V

By the chain rule, $\dfrac{dV}{d\theta} = -mga\sin\theta + k(l-l_0)\dfrac{dl}{d\theta}$ and, differentiating $l^2 = a^2 + d^2 - 2ad\cos\theta$ implicitly, $2l\,l' = 2ad\sin\theta$, so $l' = ad\sin\theta/l$. Hence

$$V'(\theta) = \sin\theta\Big[-mga + kad\Big(1-\frac{l_0}{l}\Big)\Big] = -a\cdot ma\ddot\theta$$

which is the energy form $ma^2\ddot\theta = -V'(\theta)$ of Step 4 (the Lagrange equation, an extension beyond the course notes, gives the same).

KEY RESULTS

- $\mathbf{a}_P = a\ddot\theta\,\mathbf{e}_\theta - a\dot\theta^2\,\mathbf{e}_r$; $\mathbf{N} = N\,\mathbf{e}_r$ (smooth hoop, planar forces).
- $ma\ddot\theta = \sin\theta\Big[mg - kd\Big(1-\dfrac{l_0}{l}\Big)\Big]$, equal to $-V'(\theta)/a$.
- $N = mg\cos\theta + k\Big(1-\dfrac{l_0}{l}\Big)(a-d\cos\theta) - ma\dot\theta^2$ (positive: outwards).

Part (c) · 2 points

Equilibrium positions

Find all the equilibrium positions as functions of κ. Show that besides $\theta = 0$ and $\theta = \pi$ there is a symmetric pair $\theta = \pm\theta^*$ only above a critical value κ_c, and give $\cos\theta^*(\kappa)$.

Step 1. Factorisation

From (b), with $d = 2a$, $l_0 = a$ and $k = \kappa mg/a$:

$$\frac{V'(\theta)}{mga} = \sin\theta\Big[-1 + 2\kappa\Big(1 - \frac{a}{l}\Big)\Big]$$

Equilibria are the zeros of V' (Notes §4.6).

Step 2. $\sin\theta = 0$

$\theta = 0$ and $\theta = \pi$ are equilibria for every κ: by symmetry, the spring force and the weight are both along the vertical there, so they have no tangential component.

Step 3. The symmetric pair

$-1 + 2\kappa(1 - a/l) = 0 \Rightarrow 1 - a/l = 1/(2\kappa) \Rightarrow l^* = \dfrac{a}{1 - 1/(2\kappa)} = \dfrac{2\kappa a}{2\kappa - 1}$, which requires $\kappa > \frac{1}{2}$ for $l^* > 0$. Substituting into $l^2 = a^2(5 - 4\cos\theta)$:

$$\cos\theta^* = \frac{1}{4}\Big(5 - \frac{l^{*2}}{a^2}\Big) = \frac{1}{4}\Big(5 - \frac{4\kappa^2}{(2\kappa - 1)^2}\Big)$$

Step 4. Existence condition

A point of the hoop has $a \le l \le 3a$; a new equilibrium needs $a < l^* < 3a$. $l^* > a$ holds for every $\kappa > \frac{1}{2}$. $l^* < 3a \iff 2\kappa < 3(2\kappa - 1) \iff \kappa > \frac{3}{4}$. So the pair $\pm\theta^*$ exists only for $\kappa > \kappa_c = \frac{3}{4}$; at $\kappa = \frac{3}{4}$, $\cos\theta^* = \frac{1}{4}(5 - 9) = -1$: the pair is born **at the bottom**. As $\kappa \to \infty$, $l^* \to a$ and $\theta^* \to 0$.

KEY RESULTS

- $\theta = 0$ (top) and $\theta = \pi$ (bottom) for every κ.
- $G = 0 \iff 1 - \dfrac{a}{l} = \dfrac{1}{2\kappa} \iff l^* = \dfrac{2\kappa a}{2\kappa - 1}$; $\cos\theta^* = \dfrac{1}{4}\Big(5 - \dfrac{4\kappa^2}{(2\kappa - 1)^2}\Big)$.
- The pair exists only for $\kappa > \kappa_c = \frac{3}{4}$ ($\theta^* \to \pi$ as $\kappa \to \frac{3}{4}^+$, $\theta^* \to 0$ as $\kappa \to \infty$).

Part (d) · 1.5 points

Stability and the bifurcation

Classify the stability of each equilibrium in the two regimes $\kappa < \frac{3}{4}$ and $\kappa > \frac{3}{4}$. Describe what happens at $\kappa = \frac{3}{4}$. Is "the bottom of the hoop is always stable" true?

Step 1. Second derivative

From $V' = \sin\theta\, G(\theta)$ with $G = -mga + kad(1 - l_0/l)$, the product rule gives $V'' = \cos\theta\, G + \sin\theta\, G'$, and by the chain rule $G' = kad\,\dfrac{l_0}{l^2}\,l' = kad\,\dfrac{l_0}{l^2}\,\dfrac{ad\sin\theta}{l}$.

Step 2. Top and bottom

At $\theta = 0$: $l = l_0$, so $G(0) = -mga$ and $V''(0) = -mga < 0$: unstable for every κ (the spring is slack there and cannot hold the bead up).

At $\theta = \pi$: $l = 3a$, $G(\pi) = -mga + k\,a\,2a\,\frac{2}{3} = -mga + \frac{4}{3}ka^2$ and $V''(\pi) = \cos\pi\, G(\pi) = mga - \frac{4}{3}ka^2 = mga\big(1 - \frac{4}{3}\kappa\big)$.

Step 3. The pair

At θ^*, $G = 0$, so $V''(\theta^*) = \sin\theta^* G'(\theta^*) = \dfrac{ka^2 d^2 l_0 \sin^2\theta^*}{l^{*3}} > 0$: a minimum, stable.

Step 4. The two regimes (the trap)

- $\kappa < \frac{3}{4}$ (soft spring): two equilibria; top unstable, **bottom stable**.
- $\kappa > \frac{3}{4}$ (stiff spring): four equilibria; top unstable, **bottom unstable**, $\pm\theta^*$ stable. The stretched spring pulls the bead up harder than gravity pulls it down, and the bead settles on one side.

The bottom is not always stable. At $\kappa = \frac{3}{4}$, $V''(\pi) = 0$ and $V'''(\pi) = 0$ (symmetry), and $V''''(\pi) = \frac{1}{3}mga > 0$: the bottom is still a (flat) minimum. As κ crosses $\frac{3}{4}$ the minimum at the bottom splits into two minima that move away symmetrically: a **supercritical pitchfork bifurcation** (see the energy curves of the figure).

KEY RESULTS

- Top: $V''(0) = -mga < 0$, always unstable.
- Bottom: $V''(\pi) = mga\big(1 - \frac{4}{3}\kappa\big)$: stable for $\kappa < \frac{3}{4}$, **unstable** for $\kappa > \frac{3}{4}$.
- Pair: $V''(\theta^*) = \dfrac{ka^2 d^2 l_0 \sin^2\theta^*}{l^{*3}} > 0$: stable whenever it exists. Supercritical pitchfork at $\kappa = \frac{3}{4}$ ($V''''(\pi) = \frac{1}{3}mga > 0$ there).

Part (e) · 1.5 points

Small oscillations

Compute the angular frequency of small oscillations (i) about the bottom for $\kappa = \frac{1}{2}$ and (ii) about θ^* for $\kappa = 1$ (give θ^*). Numerical values for $a = 0.40$ m.

Step 1. Linearisation

With $\theta = \theta_e + \eta$, $V'(\theta) = V'(\theta_e) + V''(\theta_e)\eta + O(\eta^2)$ and $V'(\theta_e) = 0$, so $ma^2\ddot{\eta} + V''(\theta_e)\eta = 0$ and $\omega^2 = V''(\theta_e)/(ma^2)$ when $V''(\theta_e) > 0$.

Step 2. Bottom, $\kappa = \frac{1}{2}$

$V''(\pi) = mga(1 - \frac{4}{3}\cdot\frac{1}{2}) = \frac{1}{3}mga$, so $\omega^2 = \dfrac{g}{3a} = \dfrac{9.81}{1.2} = 8.175\ \mathrm{s}^{-2}$ and $\omega = 2.859$ rad/s.

Step 3. Side equilibrium, $\kappa = 1$

$l^* = 2a/(2-1) = 2a$, $\cos\theta^* = \frac{1}{4}(5-4) = \frac{1}{4}$, $\theta^* = \arccos\frac{1}{4} = 75.52^\circ$, $\sin^2\theta^* = \frac{15}{16}$. With $k = mg/a$, $d = 2a$, $l_0 = a$:

$$V''(\theta^*) = \frac{k\,a^2\,(4a^2)\,a\cdot\frac{15}{16}}{8a^3} = \frac{15}{32}ka^2 = \frac{15}{32}mga, \qquad \omega^2 = \frac{15}{32}\frac{g}{a}$$

Numerically $\omega^2 = \frac{15}{32}\times\frac{9.81}{0.40} = 11.50\ \mathrm{s}^{-2}$, $\omega = 3.391$ rad/s.

KEY RESULTS

- (i) $\omega^2 = \dfrac{g}{3a}$, $\omega = 2.859$ rad/s.
- (ii) $l^* = 2a$, $\cos\theta^* = \frac{1}{4}$, $\theta^* = 75.52^\circ$; $\omega^2 = \dfrac{15}{32}\dfrac{g}{a}$, $\omega = 3.391$ rad/s.

Part (f) · 1.5 points

Energy integral and the bead at the bottom

Write the energy first integral. For $\kappa = \frac{1}{2}$ the bead leaves the top with a vanishingly small speed. Find its speed at the bottom and the reaction of the hoop there. Would a bead released exactly at rest at the top move?

Step 1. First integral

The reaction is normal to $\mathbf{v}_P = a\dot\theta\,\mathbf{e}_\theta$, so its power $\mathbf{N}\cdot\mathbf{v}_P = 0$ (Notes §6.4 to §6.5); the other forces derive from V. Hence (Notes §4.5)

$$\tfrac{1}{2}ma^2\dot\theta^2 + mga\cos\theta + \tfrac{1}{2}k\big(l(\theta) - l_0\big)^2 = E$$

Check: $\dfrac{dE}{dt} = \dot\theta\big(ma^2\ddot\theta + V'(\theta)\big) = 0$ by (b).

Step 2. Top versus bottom

$V(0) = mga + 0$ and $V(\pi) = -mga + \frac{1}{2}k(3a-a)^2 = -mga + 2ka^2$, so $V(\pi) - V(0) = 2ka^2 - 2mga = 2mga(\kappa - 1)$.

Step 3. Speed at the bottom

$\theta = 0$ is an equilibrium: released **exactly** at rest there, the bead stays. With an infinitesimal push, $E = V(0)$. For $\kappa = \frac{1}{2} < \frac{3}{4}$ the only equilibria are 0 and π, so V decreases monotonically on $(0,\pi)$ and the bead reaches the bottom with

$$\tfrac{1}{2}mv^2 = V(0) - V(\pi) = 2mga(1-\kappa) \implies v^2 = 4ga(1-\kappa) = 2ga, \qquad v = \sqrt{2 \times 9.81 \times 0.40} = 2.801 \text{ m/s}$$

With no spring ($\kappa = 0$) it would be $v^2 = 4ga$: the spring stores part of the energy.

Step 4. Reaction at the bottom

At $\theta = \pi$: $l = 3a$, $1 - l_0/l = \tfrac{2}{3}$, $a - d\cos\pi = 3a$, $a\dot\theta^2 = v^2/a = 2g$ and $k = mg/(2a)$:

$$N = mg\cos\pi + \frac{mg}{2a}\cdot\frac{2}{3}\cdot 3a - m\cdot 2g = -mg + mg - 2mg = -2mg$$

$N < 0$ means the reaction points along $-\mathbf{e}_r$, towards C; at the bottom $-\mathbf{e}_r = \mathbf{j}$, so the hoop pushes the bead **up** with $2mg$. Check on the vertical: $N_{up} - mg + T_{spring} = mv^2/a$ with $T_{spring} = k(3a - a) = mg$ gives $N_{up} = 2mg - mg + mg = 2mg$.

KEY RESULTS

- $\tfrac{1}{2}ma^2\dot\theta^2 + mga\cos\theta + \tfrac{1}{2}k(l-l_0)^2 = E$; $V(\pi) - V(0) = 2mga(\kappa - 1)$.
- $\kappa = \tfrac{1}{2}$: $v^2_{bottom} = 4ga(1-\kappa) = 2ga$, $v = 2.801$ m/s (against $4ga$ without spring).
- $N(\pi) = -2mg$: the hoop pushes the bead **towards** C, i.e. upwards with $2mg$. At rest exactly at the top the bead stays (equilibrium); it needs a disturbance.

Interpretation and checks

- **Dimensional check:** $\kappa = ka/(mg)$ is (N/m)(m)/N = 1; V and mga in J; $\omega^2 = g/a$ in s^{-2}; N in N; $v^2 = ga$ in $\text{m}^2\ \text{s}^{-2}$.
- **Limits:** $\kappa \to 0$ recovers the bead on a hoop under gravity alone (top unstable, bottom stable, $\omega^2 = g/a$ at the bottom: our $V''(\pi)$ gives mga). $\kappa \to \infty$: the spring dominates and the stable positions approach the top, where the spring is unstretched. For $\kappa = 1$, $V(\pi) = V(0)$: a bead leaving the top with a vanishing speed approaches the (now unstable) bottom only asymptotically, never reaching it in finite time.
- **Common mistakes in the exam:** (1) assuming the bottom is always the stable position; (2) missing the side equilibria by cancelling $\sin\theta$ without keeping the second factor; (3) using $l - l_0$ with the wrong natural length (the spring is unstretched at the **top**, not at the bottom); (4) forgetting that the reaction of a bead on a hoop can point either way (bilateral constraint), and assuming $N \geq 0$; (5) "releasing from rest at the top" without noticing that it is an equilibrium.

Problem · Unit 5 · 10 points · 60 min · Standard exam level

P10 — Free fall from a tower on the rotating Earth: eastward deflection, the order of the north-south effect and an upward shot

*A particle P of mass m is released **from rest relative to the ground** from the top of a tower of height h standing at the point A of the Earth's surface at latitude λ (northern hemisphere). The Earth rotates with constant angular velocity of modulus Ω about its polar axis, whose unit vector $\mathbf{k}_E$ points to the north pole; the centre of the Earth is taken as fixed in an inertial frame S_0.*

*The local frame $S_1 = \{A; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$ is attached to the ground at the foot A of the tower and rotates with the Earth: x points **east**, y points **north**, z points **up** along the plumb line. In S_1 the particle feels the effective gravity $-g\,\mathbf{k}$, which already contains the centrifugal force of the point A (gravitation plus $-m\boldsymbol{\Omega}\times(\boldsymbol{\Omega}\times\mathbf{r}_{CA})$, C the centre of the Earth). Air drag is neglected and h is small compared with the radius of the Earth.*

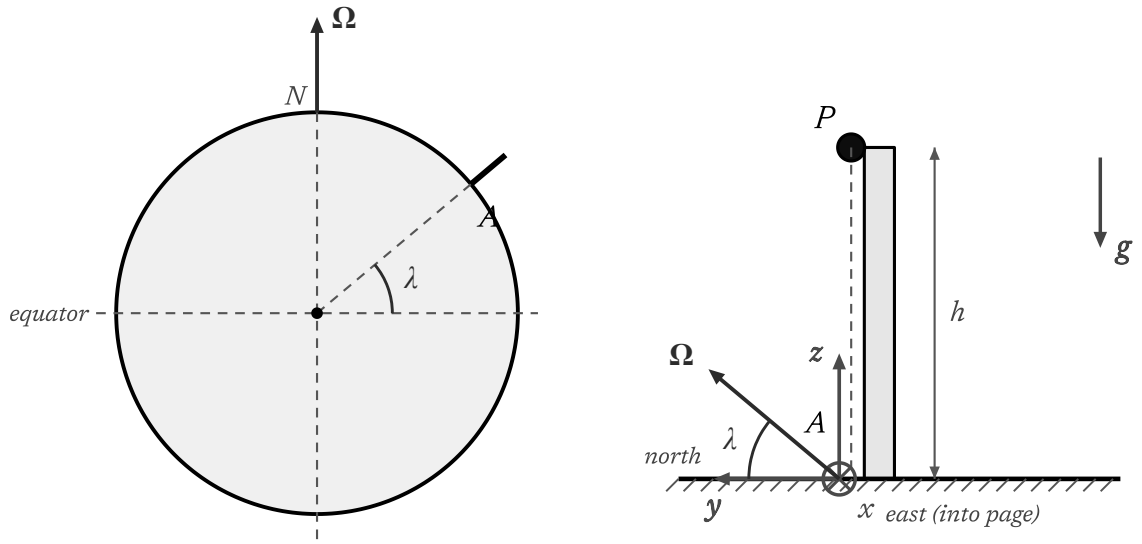


Figure 1. Free fall on the rotating Earth: point A at latitude λ, rotation $\boldsymbol{\Omega}$ along the Earth axis; local frame $S_1 = \{A; x\ \text{east},\ y\ \text{north},\ z\ \text{up}\}$ (east points into the page), $\boldsymbol{\Omega} = \Omega\cos\lambda\,\mathbf{j} + \Omega\sin\lambda\,\mathbf{k}$, particle P released from rest at height h.

Setup

- Data: m, h, g, Ω, λ. Numerical part: $h = 100$ m, $\lambda = 40^\circ$, $g = 9.81$ m s^{-2}, $\Omega = 7.292\times10^{-5}$ rad s^{-1} (2π per sidereal day of 86 164 s).
- Hypotheses: point particle, uniform effective gravity along the height of the tower, Ω constant, $\Omega t \ll 1$ during the fall (so a solution to first order in Ω is meaningful), no drag.
- S_1 is a **rotating** frame: $\boldsymbol{\omega}_{10} = \Omega\,\mathbf{k}_E$ with $\dot{\boldsymbol{\omega}}_{10} = \mathbf{0}$.

Part (a) · 1.5 points

Degrees of freedom, angular velocity of S_1 and inertia forces

How many degrees of freedom has P? Write $\boldsymbol{\omega}_{10}$ in the basis of S_1 and list the inertia forces that act on P in S_1, saying which one is absorbed into $-g\,\mathbf{k}$ and which one is kept.

Step 1. Count

P is a free particle in space: 3 degrees of freedom, the coordinates (x, y, z) of $\mathbf{r} = \overrightarrow{AP}$ in S_1. Nothing constrains it between release and impact.

Step 2. The angular velocity in the local basis

The local vertical $\mathbf{k}$ is the plumb line at A (statement), and the latitude λ is measured from this plumb line: it is the angle between $\mathbf{k}$ and the equatorial plane. The angle between $\mathbf{k}$ and $\mathbf{k}_E$ is therefore $90^\circ - \lambda$, so $\mathbf{k}_E \cdot \mathbf{k} = \sin\lambda$ by definition of λ. The polar axis lies in the meridian plane through A, which contains $\mathbf{j}$ (north) and $\mathbf{k}$, so $\mathbf{k}_E \cdot \mathbf{i} = 0$, and it leans towards the north: $\mathbf{k}_E \cdot \mathbf{j} = \cos\lambda$. Check: $\cos^2\lambda + \sin^2\lambda = 1$. The ground turns with the Earth, so

$$\boldsymbol{\omega}_{10} = \Omega\,\mathbf{k}_E = \Omega\cos\lambda\,\mathbf{j} + \Omega\sin\lambda\,\mathbf{k}, \qquad \dot{\boldsymbol{\omega}}_{10} = \mathbf{0}$$

Step 3. Change of basis

The polar axis is best described with the cylindrical basis of the Earth at A, $\{\mathbf{e}_\rho, \mathbf{e}_\varphi, \mathbf{k}_E\}$: $\mathbf{e}_\rho$ perpendicular to the polar axis and pointing away from it, in the meridian plane of A, $\mathbf{e}_\varphi = \mathbf{i}$ (east), $\mathbf{k}_E$ the polar axis. This basis turns about $\mathbf{k}_E$ with angular velocity Ω relative to S_0, like the local one. Projecting the local unit vectors ($\mathbf{j}\cdot\mathbf{e}_\rho = -\sin\lambda, \mathbf{j}\cdot\mathbf{k}_E = \cos\lambda, \mathbf{k}\cdot\mathbf{e}_\rho = \cos\lambda, \mathbf{k}\cdot\mathbf{k}_E = \sin\lambda$):

$$\begin{pmatrix}\mathbf{i}\\ \mathbf{j}\\ \mathbf{k}\end{pmatrix} = [R]\begin{pmatrix}\mathbf{e}_\rho\\ \mathbf{e}_\varphi\\ \mathbf{k}_E\end{pmatrix}, \qquad [R] = \begin{pmatrix}0 & 1 & 0\\ -\sin\lambda & 0 & \cos\lambda\\ \cos\lambda & 0 & \sin\lambda\end{pmatrix}$$

Checks: the rows have unit length (1, $\sin^2\lambda + \cos^2\lambda$, $\cos^2\lambda + \sin^2\lambda$) and are mutually orthogonal (row 1 with rows 2 and 3: 0; rows 2 and 3: $-\sin\lambda\cos\lambda + \cos\lambda\sin\lambda = 0$), so $[R][R]^T = I$; expanding along the first row, $\det[R] = -1\cdot\big((-\sin\lambda)\sin\lambda - \cos\lambda\cos\lambda\big) = +1$. The third column of $[R]$ gives back $\mathbf{k}_E = \cos\lambda\,\mathbf{j} + \sin\lambda\,\mathbf{k}$.

Step 4. Inertia forces (Notes §3.2 and §4.9)

The acceleration of P in S_0 is $\mathbf{a} = \mathbf{a}_{rel} + \mathbf{a}_{drag} + 2\boldsymbol{\omega}_{10} \times \mathbf{v}_{rel}$. The point of S_1 that coincides with P moves on a circle about the polar axis (the centre C is fixed and $\boldsymbol{\omega}_{10}$ constant), so $\mathbf{a}_{drag} = \boldsymbol{\Omega} \times \big(\boldsymbol{\Omega} \times \mathbf{r}_{CP}\big)$ with $\mathbf{r}_{CP} = \mathbf{r}_{CA} + \mathbf{r}$. In S_1, $m\mathbf{a}_{rel} = \mathbf{F} + \mathbf{F}_{cf} + \mathbf{F}_{cor}$ with

$$\mathbf{F}_{cf} = -m\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}_{CA}) - m\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}), \qquad \mathbf{F}_{cor} = -2m\boldsymbol{\Omega} \times \mathbf{v}_{rel}$$

The first term is the same at every point of the tower: added to the gravitational attraction it gives the effective weight $-mg\,\mathbf{k}$ that a plumb line at A measures (this is how $\mathbf{k}$ is defined). The second term has modulus at most $m\Omega^2 h \approx 5.3 \times 10^{-7}\, m$ m s^{-2} for $h = 100$ m: it is of order Ω^2 and is dropped. The Coriolis force is of order Ω and is the one that matters.

KEY RESULTS

- 3 degrees of freedom (x, y, z), no constraint during the fall.
- $\boldsymbol{\omega}_{10} = \Omega\cos\lambda\,\mathbf{j} + \Omega\sin\lambda\,\mathbf{k}$, constant.
- Inertia forces: centrifugal $-m\boldsymbol{\Omega} \times \big(\boldsymbol{\Omega} \times (\mathbf{r}_{CA} + \mathbf{r})\big)$ (the part with $\mathbf{r}_{CA}$ is in g; the part with $\mathbf{r}$ is $O(\Omega^2 h)$ and is neglected) and Coriolis $-2m\boldsymbol{\Omega} \times \mathbf{v}_{rel}$ (kept). No Euler force.

Part (b) · 2 points

Equations of motion in S_1

Write the vector equation of motion of P in S_1 and its three scalar components, with the initial conditions.

Step 1. Vector equation

With the results of (a), the only forces in S_1 are the effective weight and the Coriolis force:

$$m\,\ddot{\mathbf{r}} = -mg\,\mathbf{k} - 2m\,\boldsymbol{\Omega} \times \dot{\mathbf{r}}$$

Step 2. The cross product

$$\boldsymbol{\Omega} \times \dot{\mathbf{r}} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & \Omega\cos\lambda & \Omega\sin\lambda \\ \dot{x} & \dot{y} & \dot{z} \end{vmatrix} = \Omega\big(\dot{z}\cos\lambda - \dot{y}\sin\lambda\big)\mathbf{i} + \Omega\dot{x}\sin\lambda\,\mathbf{j} - \Omega\dot{x}\cos\lambda\,\mathbf{k}$$

Step 3. Components

Dividing by m and projecting on $\mathbf{i}, \mathbf{j}, \mathbf{k}$:

$$\ddot{x} = -2\Omega\big(\dot{z}\cos\lambda - \dot{y}\sin\lambda\big), \qquad \ddot{y} = -2\Omega\dot{x}\sin\lambda, \qquad \ddot{z} = -g + 2\Omega\dot{x}\cos\lambda$$

The particle starts at the top of the tower with zero velocity **relative to the ground** (in S_0 it moves east with the tower top): $\mathbf{r}(0) = h\,\mathbf{k}$, $\dot{\mathbf{r}}(0) = \mathbf{0}$.

KEY RESULTS

- $\ddot{x} = -2\Omega\big(\dot{z}\cos\lambda - \dot{y}\sin\lambda\big),\ \ddot{y} = -2\Omega\dot{x}\sin\lambda,\ \ddot{z} = -g + 2\Omega\dot{x}\cos\lambda.$
- $x(0) = y(0) = 0, z(0) = h, \dot{x}(0) = \dot{y}(0) = \dot{z}(0) = 0.$

Part (c) · 2 points

Solution to first order in Ω

Solve the equations to first order in Ω and show that the particle lands at $\Delta x = \dfrac{2\sqrt{2}}{3}\,\Omega\cos\lambda\sqrt{\dfrac{h^3}{g}}$ east of the foot of the tower. Does the time of fall change at this order?

Step 1. Zeroth order

With $\Omega = 0$: $\ddot{x}_0 = \ddot{y}_0 = 0$ and $\ddot{z}_0 = -g$, with $\dot{\mathbf{r}}_0(0) = \mathbf{0}$ and $\mathbf{r}_0(0) = h\,\mathbf{k}$. Integrating twice with the limits 0 and t (antiderivative, then Barrow's rule $[F]_0^t = F(t) - F(0)$):

$$\dot{z}_0(t) = \dot{z}_0(0) + \int_0^t (-g)\,d\tau = 0 + \big[-g\tau\big]_0^t = -gt - (-g\cdot 0) = -gt$$

$$z_0(t) = z_0(0) + \int_0^t (-g\tau)\,d\tau = h + \Big[-\tfrac{1}{2}g\tau^2\Big]_0^t = h - \tfrac{1}{2}gt^2$$

$$\dot{x}_0(t) = \dot{x}_0(0) + \int_0^t 0\,d\tau = 0, \qquad x_0(t) = x_0(0) + \int_0^t 0\,d\tau = 0$$

and in the same way $\dot{y}_0 = 0, y_0 = 0$.

Step 2. First order

Every Coriolis term carries a factor Ω, so to first order the velocity inside it is the zeroth-order one, $\dot{\mathbf{r}}_0 = -gt\,\mathbf{k}$:

$$\ddot{x}_1 = -2\Omega\big(\dot{z}_0\cos\lambda - \dot{y}_0\sin\lambda\big) = 2\Omega g t\cos\lambda, \qquad \ddot{y}_1 = -2\Omega\dot{x}_0\sin\lambda = 0, \qquad \ddot{z}_1 = 2\Omega\dot{x}_0\cos\lambda = 0$$

with zero initial conditions, so $y_1 = z_1 = 0$ and

$$\dot{x}_1 = \int_0^t 2\Omega g\cos\lambda\,\tau\,d\tau = \big[\Omega g\cos\lambda\,\tau^2\big]_0^t = \Omega g t^2\cos\lambda$$

$$x_1 = \int_0^t \Omega g\cos\lambda\,\tau^2\,d\tau = \Big[\tfrac{1}{3}\Omega g\cos\lambda\,\tau^3\Big]_0^t = \tfrac{1}{3}\Omega g t^3\cos\lambda$$

Step 3. Landing point

Since $z_1 = 0$, the height is $h - \frac{1}{2}gt^2$ up to $O(\Omega^2)$ and the particle reaches $z = 0$ at $t_f = \sqrt{2h/g}$, as without rotation. Then $t_f^3 = (2h/g)^{3/2} = 2\sqrt{2}\,h^{3/2}/g^{3/2}$ and

$$\Delta x = x_1(t_f) = \tfrac{1}{3}\Omega g\cos\lambda\cdot\frac{2\sqrt{2}\,h^{3/2}}{g^{3/2}} = \frac{2\sqrt{2}}{3}\,\Omega\cos\lambda\sqrt{\frac{h^3}{g}} > 0$$

The particle lands **east** of the foot A.

Step 4. Physical reading in S_0

At release the top of the tower is a distance $h\cos\lambda$ farther from the polar axis than the foot, so in S_0 it moves east $\Omega h\cos\lambda$ faster **than the foot**. The particle keeps that eastward speed while it falls, which is why it ends up ahead of the foot. But this lead speed $\Omega h\cos\lambda$ is measured against the foot, not against the rotating ground under the particle: relative to the ground (S_1) the particle starts at rest, and its eastward speed $\dot{x}_1 = \Omega g t^2\cos\lambda$ is zero at release and grows like t^2, as the particle descends to points of the ground that move more slowly. The crude estimate "lead speed $\Omega h\cos\lambda$ during the whole fall", $\Omega h\cos\lambda\,t_f$, assumes the full lead from the start and is therefore $3/2$ of the exact first-order value: with $h = \frac{1}{2}gt_f^2$, $\Omega h\cos\lambda\,t_f = \frac{1}{2}\Omega g\cos\lambda\,t_f^3$, against $\Delta x = \frac{1}{3}\Omega g\cos\lambda\,t_f^3$.

KEY RESULTS

- Order 0: $z_0 = h - \frac{1}{2}gt^2$, $\dot{z}_0 = -gt$, $x_0 = y_0 = 0$.
- Order 1: $\ddot{x}_1 = 2\Omega g t \cos\lambda$, $\dot{x}_1 = \Omega g t^2 \cos\lambda$, $x_1 = \frac{1}{3}\Omega g t^3 \cos\lambda$; $y_1 = z_1 = 0$.
- $t_f = \sqrt{2h/g}$ (unchanged at first order), $\Delta x = \frac{2\sqrt{2}}{3}\Omega \cos\lambda \sqrt{h^3/g}$, to the **east**.

Part (d) · 1 point

Numbers

For $h = 100$ m, $\lambda = 40°$: compute the time of fall and the eastward deflection. Where is the particle (east of the plumb line) at $t_f/2$?

Step 1. Time and deflection

$t_f = \sqrt{2 \cdot 100/9.81} = \sqrt{20.39} = 4.515$ s. With $\cos 40° = 0.7660$ and $\sqrt{h^3/g} = \sqrt{10^6/9.81} = 319.3$ s:

$$\Delta x = \frac{2\sqrt{2}}{3}\,(7.292 \times 10^{-5})(0.7660)(319.3)\ \text{m} = 1.681 \times 10^{-2}\ \text{m} = 1.681\ \text{cm}$$

A numerical integration of the exact equations of (b) (no expansion) gives the same value to four significant figures.

Step 2. Half-way

$x_1 \propto t^3$, so at $t_f/2$ the deflection is $(1/2)^3 \Delta x = 0.210$ cm: most of the deflection builds up in the second half of the fall.

KEY RESULTS

- $t_f = \sqrt{200/9.81} = 4.515$ s.
- $\Delta x = 1.681$ cm east; $x_1(t_f/2) = \Delta x/8 = 0.210$ cm.

Part (e) · 1.5 points

Is there a north-south deflection?

A student integrates the y equation once more and reports a southward deflection $y_2(t_f) = -\frac{2}{3}\Omega^2 h^2 \sin\lambda\cos\lambda/g$. Check the formula, and say whether it is a consistent result of the model of the statement.

Step 1. The second-order term

At second order the y equation receives $\dot{x}_1$: $\ddot{y}_2 = -2\Omega\dot{x}_1 \sin\lambda = -2\Omega^2 g t^2 \sin\lambda\cos\lambda$. Integrating twice from rest:

$$\dot{y}_2(t) = \int_0^t \left(-2\Omega^2 g \sin\lambda\cos\lambda\, \tau^2 \right) d\tau = \left[-\tfrac{2}{3}\Omega^2 g \sin\lambda\cos\lambda\, \tau^3 \right]_0^t = -\tfrac{2}{3}\Omega^2 g t^3 \sin\lambda\cos\lambda$$

$$y_2(t) = \int_0^t \left(-\tfrac{2}{3}\Omega^2 g \sin\lambda\cos\lambda\, \tau^3 \right) d\tau = \left[-\tfrac{1}{6}\Omega^2 g \sin\lambda\cos\lambda\, \tau^4 \right]_0^t = -\tfrac{1}{6}\Omega^2 g t^4 \sin\lambda\cos\lambda$$

(the lower limit contributes 0 in both, since $\dot{y}_2(0) = y_2(0) = 0$).

and with $t_f^4 = 4h^2/g^2$: $y_2(t_f) = -\tfrac{2}{3}\Omega^2 h^2 \sin\lambda\cos\lambda / g$. With the data: -1.78×10^{-6} m, about 10^{-4} of Δx (the ratio is $\tfrac{1}{2}\Omega t_f \sin\lambda$).

Step 2. Why it is not a consistent answer

The equation of (b) is only exact to first order in Ω: in (a) we dropped the residual centrifugal force $-m\boldsymbol{\Omega}\times(\boldsymbol{\Omega}\times\mathbf{r})$, which is $O(\Omega^2)$. For $\mathbf{r} \approx z\,\mathbf{k}$ its y component is $-m\Omega^2 z \sin\lambda\cos\lambda$; with $z = z_0(t)$ it alone displaces the particle by

$$y_{cf}(t_f) = -\Omega^2 \sin\lambda\cos\lambda \int_0^{t_f}\!\!\int_0^{\tau} z_0(s)\, ds\, d\tau = -\tfrac{5}{6}\,\frac{\Omega^2 h^2 \sin\lambda\cos\lambda}{g} = -2.22 \times 10^{-6}\ \text{m}$$

which is larger than y_2. The direction of the gravitational attraction also changes along the vertical through A by an amount of the same order. A second-order answer must collect all these terms; keeping only the Coriolis one is inconsistent. **To first order in Ω, the only deflection is the eastward one**, and $y = O(\Omega^2)$ is at the micrometre level.

KEY RESULTS

- $\ddot{y}_2 = -2\Omega\dot{x}_1 \sin\lambda = -2\Omega^2 g t^2 \sin\lambda\cos\lambda$, $y_2 = -\tfrac{1}{6}\Omega^2 g t^4 \sin\lambda\cos\lambda$; at t_f: $-\tfrac{2}{3}\Omega^2 h^2 \sin\lambda\cos\lambda / g = -1.78 \times 10^{-6}$ m.
- The formula is right but **not consistent**: it is $O(\Omega^2)$, the same order as the residual centrifugal term $-m\boldsymbol{\Omega}\times(\boldsymbol{\Omega}\times\mathbf{r})$ dropped in (a), whose own effect is $-\tfrac{5}{6}\Omega^2 h^2 \sin\lambda\cos\lambda / g = -2.22 \times 10^{-6}$ m, and as the change of direction of gravitation along the fall. To first order in Ω there is no north-south deflection.

Part (f) · 2 points

A projectile fired vertically upward

From A a projectile is fired vertically upward (along $\mathbf{k}$) with speed v_0 relative to the ground. To first order in Ω, where does it land? Compute it for $v_0 = 50\ m\,s^{-1}$ at $\lambda = 40^\circ$, and compare with the drop of (c) from the same maximum height.

Step 1. Zeroth and first order

Order 0: $z_0 = v_0 t - \tfrac12 g t^2$, $\dot z_0 = v_0 - g t$, back at $z = 0$ at $T = 2v_0/g$. Order 1, from the x equation of (b) with $\dot y_0 = 0$:

$$\ddot x_1 = -2\Omega\cos\lambda\,(v_0 - g t)$$

$$\dot x_1 = \int_0^t -2\Omega\cos\lambda\,(v_0 - g\tau)\,d\tau = -2\Omega\cos\lambda\Big[v_0\tau - \tfrac12 g\tau^2\Big]_0^t = -\Omega g\cos\lambda\,t\Big(\frac{2v_0}{g} - t\Big)$$

$$x_1 = \int_0^t -2\Omega\cos\lambda\Big(v_0\tau - \tfrac12 g\tau^2\Big)d\tau = -2\Omega\cos\lambda\Big[\tfrac12 v_0\tau^2 - \tfrac16 g\tau^3\Big]_0^t = -\Omega\cos\lambda\Big(v_0 t^2 - \tfrac13 g t^3\Big)$$

Step 2. Landing point

At $T = 2v_0/g$: $v_0 T^2 = 4v_0^3/g^2$ and $\tfrac13 g T^3 = \tfrac83 v_0^3/g^2$, so

$$\Delta x = x_1(T) = -\Omega\cos\lambda\Big(4 - \tfrac83\Big)\frac{v_0^3}{g^2} = -\frac{4}{3}\,\Omega\cos\lambda\,\frac{v_0^3}{g^2} < 0$$

The projectile lands **west** of the launch point. With $v_0 = 50$ m s^{-1}: $\Delta x = -\tfrac43(7.292\times10^{-5})(0.7660)(125\,000)/96.24$ m $= -9.674$ cm; the flight lasts $T = 10.19$ s and the apex is at $v_0^2/2g = 127.4$ m.

Step 3. Why it does not cancel

On the way up ($\dot z_0 > 0$) the Coriolis force pushes west; on the way down it pushes east. The symmetric-looking guess "the two halves cancel" confuses force with displacement: $\dot x_1 = -\Omega g \cos\lambda\, t(T-t)$ is negative during the whole flight and only returns to zero at $t = T$. The westward velocity gained on the way up is removed on the way down, but the westward **displacement** accumulated meanwhile remains.

Step 4. Comparison with the drop

Setting $v_0 = \sqrt{2gh}$ (apex at height h): $\Delta x_{up} = -\frac{4}{3}\Omega\cos\lambda (2gh)^{3/2}/g^2 = -4\cdot\frac{2\sqrt{2}}{3}\Omega\cos\lambda\sqrt{h^3/g}$, that is, -4 times the eastward deflection of (c): -6.73 cm for $h = 100$ m. The descent of the projectile alone would push it 1.68 cm east, but it starts the descent already moving west.

KEY RESULTS

- $\ddot x_1 = -2\Omega\cos\lambda\,(v_0 - gt),\ \dot x_1 = -\Omega g\cos\lambda\, t\,(T-t) \le 0,\ x_1 = -\Omega\cos\lambda\big(v_0 t^2 - \tfrac{1}{3}gt^3\big)$, with $T = 2v_0/g$.
- $\Delta x = -\frac{4}{3}\,\Omega\cos\lambda\, v_0^3/g^2$: it lands **west** of A; -9.674 cm for $v_0 = 50$ m s^{-1} ($T = 10.19$ s, apex 127.4 m).
- With $v_0 = \sqrt{2gh}$ the westward offset is 4 times the eastward deflection of the drop from h.

Interpretation and checks

- **Dimensional check:** $[\Omega g t^3] = \mathrm{s^{-1}\cdot m\,s^{-2}\cdot s^3} = \mathrm{m}$; $[\Omega\sqrt{h^3/g}] = \mathrm{s^{-1}}\sqrt{\mathrm{m^3/(m\,s^{-2})}} = \mathrm{s^{-1}\cdot m\,s} = \mathrm{m}$; $[\Omega v_0^3/g^2] = \mathrm{s^{-1}\,m^3 s^{-3}/(m^2 s^{-4})} = \mathrm{m}$.
- **Limits:** at the poles ($\lambda = 90^\circ$) $\cos\lambda = 0$: $\boldsymbol{\Omega}$ is vertical, parallel to the fall, and there is no deflection; it is largest at the equator. In the southern hemisphere ($\lambda < 0$) $\cos\lambda > 0$ still: the drop also lands **east**. The expansion requires $\Omega t_f \ll 1$ (here 3.3×10^{-4}).
- **Common mistakes in the exam:** (1) writing $\boldsymbol{\Omega} = \Omega\,\mathbf{k}$ (the local vertical) instead of $\Omega(\cos\lambda\,\mathbf{j} + \sin\lambda\,\mathbf{k})$; (2) a sign error in the Coriolis force (it is $-2m\boldsymbol{\Omega}\times\mathbf{v}_{rel}$) that sends the particle west; (3) keeping $O(\Omega^2)$ Coriolis terms while dropping the centrifugal term of the same order; (4) assuming that the up and down halves of a vertical shot cancel.

`Problem · Unit 3 · 10 points · 60 min · High difficulty / trap`

P11 — Pendulum whose string is hauled through a ring: a rheonomic constraint, angular momentum and the work of the hauling force

A bob P of mass m hangs from a light inextensible string that passes through a small, smooth ring O fixed to the ceiling. The other end of the string leaves the ring and is hauled in so that the length between O and P is

$$\ell(t) = \ell_0 - u\,t, \qquad u > 0 \text{ constant}$$

The motion is planar, in the vertical plane of the inertial frame $S_0 = \{O; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$ (x horizontal, y vertical upward, uniform gravity $-g\,\mathbf{j}$, $\mathbf{k} = \mathbf{i} \times \mathbf{j}$ out of the page). The angle θ is measured from the downward vertical, positive towards $+x$, and the string stays taut. Use the unit vectors $\mathbf{e}_r = \sin\theta\,\mathbf{i} - \cos\theta\,\mathbf{j}$ (from O to P) and $\mathbf{e}_\theta = \cos\theta\,\mathbf{i} + \sin\theta\,\mathbf{j}$.

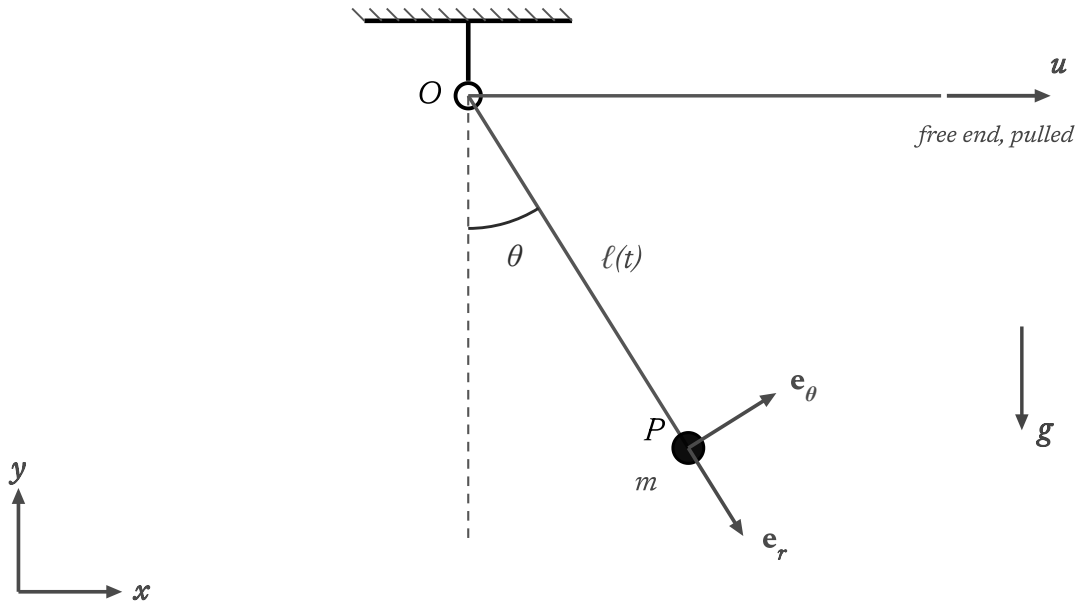


Figure 1. Pendulum hauled through a fixed smooth ring O: length $\ell(t) = \ell_0 - ut$ between O and the bob P, angle θ from the downward vertical (positive towards $+x$), unit vectors $\mathbf{e}_r$ (from O to P) and $\mathbf{e}_\theta$; the free end is pulled with constant speed u.

Setup

- Data: m, g, ℓ_0, u. Numerical parts: (d) and (e) $m = 0.5$ kg, $\ell_0 = 2$ m, $u = 0.5$ m s^{-1}, $g = 0$, initial transverse speed $\ell_0\dot{\theta}_0 = 1$ m s^{-1}; (f) $\ell_0 = 1$ m, $u = 0.01$ m s^{-1}, $g = 9.81$ m s^{-2}, $\theta(0) = 0.05$ rad, $\dot{\theta}(0) = 0$.
- Hypotheses: point particle, massless inextensible string, smooth ring (the tension T is the same on both sides), no air drag, S_0 inertial, $\ell > 0$.

Part (a) · 1 point

Degrees of freedom and the constraint

How many degrees of freedom has P? Classify the constraint imposed by the string.

Step 1. Count

In the vertical plane a free particle has 2 degrees of freedom. The taut string between O and P imposes

$$f(\mathbf{r}, t) = |\mathbf{r}| - (\ell_0 - ut) = 0$$

one equation, so 1 degree of freedom remains: θ. The length is **prescribed** by whoever pulls the string, so it is not a second coordinate (Notes §2.7).

Step 2. Classification

The constraint involves only positions and time: it is holonomic. Time appears explicitly, so it is **rheonomic**: the constraint moves, and its reaction (the tension) can do work on P, unlike the fixed string of the ordinary pendulum. A string can only pull, so the constraint is unilateral and holds while $T \geq 0$.

KEY RESULTS

- $|\overrightarrow{OP}| - \ell_0 + ut = 0$: holonomic, **rheonomic** (depends on t), and unilateral ($T \geq 0$).
- $2 - 1 = 1$ degree of freedom, coordinate $\theta(t)$; $\ell(t)$ is a given function, not an unknown.

Part (b) · 1.5 points

Kinematics in polar coordinates

Write the position, velocity and acceleration of P in $\{\mathbf{e}_r, \mathbf{e}_\theta\}$, first for a general length $\ell(t)$ and then for $\ell = \ell_0 - ut$.

Step 1. Change of basis and unit vectors

The working basis $\{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{k}\}$ turns with P; its relation to the inertial basis is

$$\begin{pmatrix}\mathbf{e}_r\\ \mathbf{e}_\theta\\ \mathbf{k}\end{pmatrix} = [R]\begin{pmatrix}\mathbf{i}\\ \mathbf{j}\\ \mathbf{k}\end{pmatrix}, \qquad [R] = \begin{pmatrix}\sin\theta & -\cos\theta & 0\\ \cos\theta & \sin\theta & 0\\ 0 & 0 & 1\end{pmatrix}$$

Checks: rows 1 and 2 have length $\sin^2\theta + \cos^2\theta = 1$ and $\sin\theta\cos\theta - \cos\theta\sin\theta = 0$, row 3 is $\mathbf{k}$, so $[R][R]^T = I$; $\det[R] = \sin\theta\cdot\sin\theta - (-\cos\theta)\cos\theta = +1$, so the basis is right-handed ($\mathbf{e}_r \times \mathbf{e}_\theta = \mathbf{k}$). By the chain rule, $\dot{\mathbf{e}}_r = \dfrac{d\mathbf{e}_r}{d\theta}\dot\theta = (\cos\theta\,\mathbf{i} + \sin\theta\,\mathbf{j})\dot\theta = \dot\theta\,\mathbf{e}_\theta$ and $\dot{\mathbf{e}}_\theta = (-\sin\theta\,\mathbf{i} + \cos\theta\,\mathbf{j})\dot\theta = -\dot\theta\,\mathbf{e}_r$.

Step 2. Position and velocity

$\mathbf{r} = \ell\,\mathbf{e}_r$. Product rule:

$$\mathbf{v} = \dot\ell\,\mathbf{e}_r + \ell\,\dot{\mathbf{e}}_r = \dot\ell\,\mathbf{e}_r + \ell\dot\theta\,\mathbf{e}_\theta$$

Step 3. Acceleration

Differentiating each product:

$$\mathbf{a} = \ddot\ell\,\mathbf{e}_r + \dot\ell\dot\theta\,\mathbf{e}_\theta + \big(\dot\ell\dot\theta + \ell\ddot\theta\big)\mathbf{e}_\theta + \ell\dot\theta\,\dot{\mathbf{e}}_\theta = \big(\ddot\ell - \ell\dot\theta^2\big)\mathbf{e}_r + \big(\ell\ddot\theta + 2\dot\ell\dot\theta\big)\mathbf{e}_\theta$$

Step 4. Prescribed length

$\ell = \ell_0 - ut$ gives $\dot\ell = -u$, $\ddot\ell = 0$:

$$\mathbf{v} = -u\,\mathbf{e}_r + \ell\dot\theta\,\mathbf{e}_\theta, \qquad \mathbf{a} = -\ell\dot\theta^2\,\mathbf{e}_r + \big(\ell\ddot\theta - 2u\dot\theta\big)\mathbf{e}_\theta$$

KEY RESULTS

- $\mathbf{v} = \dot\ell\,\mathbf{e}_r + \ell\dot\theta\,\mathbf{e}_\theta,\ \mathbf{a} = (\ddot\ell - \ell\dot\theta^2)\,\mathbf{e}_r + (\ell\ddot\theta + 2\dot\ell\dot\theta)\,\mathbf{e}_\theta$.
- With $\dot\ell = -u$, $\ddot\ell = 0$: $\mathbf{v} = -u\,\mathbf{e}_r + \ell\dot\theta\,\mathbf{e}_\theta,\ \mathbf{a} = -\ell\dot\theta^2\,\mathbf{e}_r + (\ell\ddot\theta - 2u\dot\theta)\,\mathbf{e}_\theta$.

Part (c) · 2 points

Forces and equations of motion

Write the forces on P and obtain the equation of motion free of the tension and the tension itself.

Step 1. Forces

Weight: $\mathbf{W} = -mg\,\mathbf{j}$ with $\mathbf{j}\cdot\mathbf{e}_r = -\cos\theta$ and $\mathbf{j}\cdot\mathbf{e}_\theta = \sin\theta$, so $\mathbf{W} = mg\cos\theta\,\mathbf{e}_r - mg\sin\theta\,\mathbf{e}_\theta$. Tension: the string pulls P towards the ring, $\mathbf{T} = -T\,\mathbf{e}_r$, $T \geq 0$.

Step 2. Transverse projection

$$m\big(\ell\ddot\theta + 2\dot\ell\dot\theta\big) = -mg\sin\theta \implies \ell\ddot\theta + 2\dot\ell\dot\theta + g\sin\theta = 0$$

With $\dot\ell = -u$: $(\ell_0 - ut)\ddot\theta - 2u\dot\theta + g\sin\theta = 0$. The term $-2u\dot\theta$ has the form of a **negative damping**: hauling the string in feeds the swing.

Step 3. Radial projection

$$m\big(\ddot\ell - \ell\dot\theta^2\big) = mg\cos\theta - T \implies T = m\big(g\cos\theta + \ell\dot\theta^2 - \ddot\ell\big) = m\big(g\cos\theta + \ell\dot\theta^2\big)$$

because $\ddot\ell = 0$: a constant hauling speed adds no radial inertia term. The equation of Step 2 is also what a Lagrange equation (extension, not in the course notes) with $L = \tfrac12 m(\dot\ell^2 + \ell^2\dot\theta^2) + mg\ell\cos\theta$ gives.

KEY RESULTS

- $\mathbf{W} = mg(\cos\theta\,\mathbf{e}_r - \sin\theta\,\mathbf{e}_\theta)$, $\mathbf{T} = -T\,\mathbf{e}_r$.
- $\ell\ddot\theta + 2\dot\ell\dot\theta + g\sin\theta = 0$, i.e. $(\ell_0 - ut)\ddot\theta - 2u\dot\theta + g\sin\theta = 0$.
- $T = m\big(g\cos\theta + \ell\dot\theta^2 - \ddot\ell\big) = m\big(g\cos\theta + \ell\dot\theta^2\big)$.

Part (d) · 1.5 points

Angular momentum about O and the gravity-free case

Compute $\mathbf{H}_O$ and $d\mathbf{H}_O/dt$. Is it conserved? Then remove gravity (a tether in orbit reeled in at constant speed, $g = 0$): find $T(\ell)$ and, for the data of the Setup, $\dot\theta$, the speed and T when $\ell = 0.5$ m.

Step 1. Angular momentum

$\mathbf{H}_O = \mathbf{r}\times m\mathbf{v} = \ell\,\mathbf{e}_r \times m\big(\dot\ell\,\mathbf{e}_r + \ell\dot\theta\,\mathbf{e}_\theta\big) = m\ell^2\dot\theta\,(\mathbf{e}_r\times\mathbf{e}_\theta)$, and $\mathbf{e}_r\times\mathbf{e}_\theta = \mathbf{k}$ (it is the cross product of $\mathbf{i}$ and $\mathbf{j}$ rotated by $\theta - 90^\circ$):

$$\mathbf{H}_O = m\ell^2\dot\theta\,\mathbf{k}$$

Step 2. Its rate (Notes §4.8)

The tension passes through O, so its moment is zero; the weight gives $\mathbf{r}\times\mathbf{W} = \ell\,\mathbf{e}_r \times (-mg\sin\theta\,\mathbf{e}_\theta) = -mg\ell\sin\theta\,\mathbf{k}$. Product rule:

$$\frac{d}{dt}\left(m\ell^2\dot\theta\right) = m\left(2\ell\dot\ell\dot\theta + \ell^2\ddot\theta\right) = m\ell\left(\ell\ddot\theta + 2\dot\ell\dot\theta\right) = -mg\ell\sin\theta$$

which is the equation of (c) multiplied by $m\ell$. With gravity $\mathbf{H}_O$ is not conserved.

Step 3. Gravity-free case

With $g = 0$ the right-hand side vanishes: $\ell^2\dot\theta = h$ constant. In the tension, $\dot\theta = h/\ell^2$:

$$T = m\ell\dot\theta^2 = m\ell\frac{h^2}{\ell^4} = \frac{mh^2}{\ell^3}$$

The transverse speed $\ell\dot\theta = h/\ell$ grows as the string shortens, and the tension grows like ℓ^{-3}: whoever reels the tether in must pull harder and harder.

Step 4. Numbers

$h = \ell_0\cdot(\ell_0\dot\theta_0) = 2\cdot 1 = 2\ \text{m}^2\ \text{s}^{-1}$. $\ell = 0.5$ m is reached at $t = (2-0.5)/0.5 = 3$ s. Then $\dot\theta = 2/0.25 = 8$ rad s^{-1}, $\ell\dot\theta = 4$ m s^{-1}, $|\mathbf{v}| = \sqrt{u^2 + (\ell\dot\theta)^2} = \sqrt{0.25 + 16} = 4.031$ m s^{-1} and $T = 0.5\cdot 4/0.125 = 16$ N, compared with $T = 0.5\cdot 4/8 = 0.25$ N at the start.

KEY RESULTS

- $\mathbf{H}_O = m\ell^2\dot\theta\,\mathbf{k}$; $\dfrac{d\mathbf{H}_O}{dt} = \mathbf{r}\times\mathbf{W} = -mg\ell\sin\theta\,\mathbf{k}$: **not** conserved.
- $g = 0$: $\ell^2\dot\theta = h$, $T = m\ell\dot\theta^2 = mh^2/\ell^3$, which grows without bound as $\ell \to 0$.
- Data: $h = 2\ \text{m}^2\ \text{s}^{-1}$; at $\ell = 0.5$ m ($t = 3$ s): $\dot\theta = 8$ rad s^{-1}, $\ell\dot\theta = 4$ m s^{-1}, $|\mathbf{v}| = 4.031$ m s^{-1}, $T = 16$ N (it was 0.25 N at ℓ_0).

Part (e) · 2 points

Who supplies the kinetic energy?

Show that $\dfrac{d}{dt}\big(E_{kin}+V\big)=T\,u$ *and explain why this is the power of the hauling force. For* $g=0$ *and the data of (d), compute the work done by the hauling force between* $\ell_0=2$ *m and* $\ell=0.5$ *m and check it against the change in kinetic energy.*

Step 1. Power of the tension

$\mathbf{T}\cdot\mathbf{v}=-T\,\mathbf{e}_r\cdot\big(\dot\ell\,\mathbf{e}_r+\ell\dot\theta\,\mathbf{e}_\theta\big)=-T\dot\ell=Tu$. On a fixed string ($\dot\ell=0$) the tension is orthogonal to the velocity and does no work; here the constraint moves and it does.

Step 2. Energy balance (Notes §4.4)

The work-energy theorem gives $\dfrac{dE_{kin}}{dt}=\mathbf{W}\cdot\mathbf{v}+\mathbf{T}\cdot\mathbf{v}$, and the weight derives from $V=mgy_P=-mg\ell\cos\theta$, so $\mathbf{W}\cdot\mathbf{v}=-dV/dt$. Hence

$$\frac{d}{dt}\Big(\tfrac12 m\big(\dot\ell^2+\ell^2\dot\theta^2\big)-mg\ell\cos\theta\Big)=Tu$$

The string is massless and the ring smooth, so the tension has the same value T at the free end, which moves with speed u along the pull: the hauler's power is Tu. The mechanical energy of P is **not** a first integral. Integrating the balance from 0 to t (Barrow's rule) gives instead

$$E_{kin}(t)+V(t)-W_{haul}(t)=E_{kin}(0)+V(0),\qquad W_{haul}(t)=\int_0^t T\,u\,dt'$$

that is, the energy of P grows by exactly the work done by the hauler. This is an energy balance, not a first integral in the strict sense: $W_{haul}(t)$ depends on the whole history $T(t')$, not on the state $(\theta,\dot\theta,t)$ alone.

Step 3. Work for $g=0$

With $T=mh^2/\ell^3$, change the variable to $\ell'=\ell_0-ut'$: $d\ell'=-u\,dt'$, so $dt'=-d\ell'/u$, and the limits become $t'=0\mapsto\ell'=\ell_0$, $t'=t\mapsto\ell'=\ell$:

$$W_{haul}=\int_0^t Tu\,dt'=\int_{\ell_0}^{\ell}\frac{mh^2}{\ell'^3}\,u\,\Big(-\frac{d\ell'}{u}\Big)=\int_{\ell}^{\ell_0}\frac{mh^2}{\ell'^3}\,d\ell'$$

An antiderivative of $mh^2\ell'^{-3}$ is $F(\ell')=-\dfrac{mh^2}{2\ell'^2}$ (check: $F'=-\frac{mh^2}{2}(-2)\ell'^{-3}=mh^2\ell'^{-3}$). Barrow's rule:

$$W_{haul} = F(\ell_0) - F(\ell) = -\frac{mh^2}{2\ell_0^2} - \Big(-\frac{mh^2}{2\ell^2}\Big) = \frac{mh^2}{2}\Big(\frac{1}{\ell^2} - \frac{1}{\ell_0^2}\Big)$$

With the data: $\frac{0.5\cdot 4}{2}\big(4 - 0.25\big) = 3.75$ J.

Step 4. Check

$E_{kin} = \frac{1}{2}m\big(u^2 + h^2/\ell^2\big)$: from $\frac{1}{2}\cdot 0.5\,(0.25 + 1)$ to $\frac{1}{2}\cdot 0.5\,(0.25 + 16)$, a change of $\frac{1}{2}\cdot 0.5 \cdot 15 = 3.75$ J. The radial part u^2 is the same at both ends, so all the work goes into the transverse motion.

KEY RESULTS

- $\mathbf{T}\cdot\mathbf{v} = -T\,\dot\ell = T\,u > 0$: the tension does positive work although it is "a constraint force".
- $W_{haul} = \dfrac{mh^2}{2}\Big(\dfrac{1}{\ell^2} - \dfrac{1}{\ell_0^2}\Big) = 3.75\ \text{J} = \Delta E_{kin}$.

Part (f) · 2 points

Extension: the amplitude of small swings while hauling

EXTENSION

This part goes beyond the course (adiabatic invariants and Bessel functions are not in the Notes). It is included because the result is a classic trap: neither the energy nor the angular momentum is conserved, so neither gives the amplitude.

For small angles, rewrite the equation of (c) with ℓ as the independent variable. Show that, when the hauling is slow ($u \ll \sqrt{g\ell}$), the angular amplitude grows like $\ell^{-3/4}$, and check it with the data of the Setup between $\ell_0 = 1$ m and $\ell = 0.25$ m. Grading: the 2.0 points are for the change of variable (chain rule) and the reduced equation in ℓ; the solution with Bessel functions and the $\ell^{-3/4}$ law are an extension, shown for completeness and not expected in the exam.

Step 1. Change of variable

For $\theta(t) = \Theta(\ell(t))$, the chain rule gives $\dot\theta = \Theta'\dot\ell = -u\Theta'$ and $\ddot\theta = \dfrac{d}{dt}(-u\Theta') = -u\Theta''\dot\ell = u^2\Theta''$. With $\sin\theta \approx \theta$ the equation of (c) becomes

$$\ell u^2\Theta'' + 2(-u)(-u\Theta') + g\Theta = 0 \implies \ell\,\Theta'' + 2\Theta' + \frac{g}{u^2}\Theta = 0$$

and since $\dfrac{d}{d\ell}\big(\ell^2\Theta'\big) = \ell^2\Theta'' + 2\ell\Theta'$, it is $\dfrac{1}{\ell}\big(\ell^2\Theta'\big)' + \dfrac{g}{u^2}\Theta = 0$.

Step 2. Exact solution

With $x = g\ell/u^2$ the equation is $x\,y'' + 2y' + y = 0$, whose solutions are $x^{-1/2}J_1(2\sqrt{x})$ and $x^{-1/2}Y_1(2\sqrt{x})$ (Bessel functions of order 1). So

$$\theta = \ell^{-1/2}\Big[A\,J_1\Big(\frac{2\sqrt{g\ell}}{u}\Big) + B\,Y_1\Big(\frac{2\sqrt{g\ell}}{u}\Big)\Big]$$

Step 3. Slow hauling

For large argument $J_1(z) \approx \sqrt{2/(\pi z)}\cos(z - 3\pi/4)$ (and Y_1 with a sine). With $z \propto \ell^{1/2}$ the envelope is $\ell^{-1/2}z^{-1/2} \propto \ell^{-3/4}$:

$$\theta_a \propto \ell^{-3/4}$$

The condition $z \gg 1$ is $u \ll \sqrt{g\ell}$: the length changes little during one swing. The same law follows from the adiabatic invariant E_{osc}/ω with $E_{osc} \approx \tfrac{1}{2}mg\ell\theta_a^2$ and $\omega = \sqrt{g/\ell}$: $\ell^{3/2}\theta_a^2$ stays constant.

Step 4. Numbers

At $\ell = 0.25$ m, $z = 2\sqrt{9.81 \cdot 0.25}/0.01 = 313 \gg 1$ and $u/\sqrt{g\ell} = 0.0064$. The law predicts $\theta_a = 0.05\,(0.25)^{-3/4} = 0.05 \cdot 2.828 = 0.1414$ rad: the swing angle almost triples, while the horizontal excursion $\ell\theta_a \propto \ell^{1/4}$ shrinks by the factor 0.707. A Runge-Kutta integration of $(\ell_0 - ut)\ddot\theta - 2u\dot\theta + g\sin\theta = 0$ gives 0.1401 rad at the last turning point ($\ell = 0.253$ m), where the law gives 0.1400 rad.

KEY RESULTS

- Graded: $\dot\theta = -u\,\theta'$, $\ddot\theta = u^2\theta''$ (chain rule) and the reduced equation below. Everything after it (Bessel solution, $\ell^{-3/4}$ law, numbers) is an extension, not expected in the exam.
- $\ell\,\theta'' + 2\theta' + \dfrac{g}{u^2}\,\theta = 0$, i.e. $\dfrac{1}{\ell}\dfrac{d}{d\ell}\big(\ell^2\theta'\big) + \dfrac{g}{u^2}\theta = 0$.
- $\theta = \ell^{-1/2}\big[A\,J_1(z) + B\,Y_1(z)\big]$, $z = 2\sqrt{g\ell}/u$; for $z \gg 1$, $J_1, Y_1 \sim \sqrt{2/(\pi z)}$ times a cosine, so the amplitude is $\propto \ell^{-1/2}\ell^{-1/4} = \ell^{-3/4}$.
- From 1 m to 0.25 m: $\theta_a = 0.05 \cdot 4^{3/4} = 0.1414$ rad; a numerical integration of the full equation agrees within $0.2\,\%$.

Interpretation and checks

- **Dimensional check:** every term of $\ell\ddot\theta + 2\dot\ell\dot\theta + g\sin\theta$ is $\mathrm{m\,s^{-2}}$; $[mh^2/\ell^3] = \mathrm{kg\,m^4 s^{-2}/m^3} = \mathrm{N}$; $[Tu] = \mathrm{W}$; $g\ell/u^2$ is dimensionless.
- **Limits:** $u = 0$ gives the simple pendulum with $T = m(g\cos\theta + \ell\dot\theta^2)$ and energy conservation; $u < 0$ (paying the string out) turns $-2u\dot\theta$ into a true damping and the amplitude decreases like $\ell^{-3/4}$.
- **Common mistakes in the exam:** (1) forgetting the Coriolis-like term $2\dot\ell\dot\theta$ when ℓ varies; (2) claiming that the tension does no work because "it is a constraint force" (it does, because the constraint is rheonomic); (3) writing energy conservation for P; (4) using $\ell^2\dot\theta = \text{const}$ with gravity (the weight has a moment about O); (5) believing that T stays bounded in the gravity-free case.

Problem · Unit 3 · 10 points · 60 min · Standard exam level

P12 — Block on an accelerating turntable: the friction cone, the instant of slip and the direction in which the block starts to slide

A horizontal turntable rotates about the fixed vertical axis through its centre O. It starts from rest at $t = 0$ with constant angular acceleration $\beta > 0$, so its angular rate is $\Omega(t) = \beta t$ (counter-clockwise seen from above). A small block P of mass m rests on the table at the distance r_0 from the axis; at $t = 0$ it lies on the fixed x axis. The contact is rough, with static Coulomb coefficient μ and kinetic coefficient $\mu_k \le \mu$; gravity is $-g\,\mathbf{k}$.

Use the inertial frame $S_0 = \{O; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$ ($\mathbf{k}$ vertical, upward), the turntable frame S_1 (rotating with the table), the angle φ of $\overrightarrow{OP}$ from the x axis and the unit vectors $\mathbf{e}_r = \cos\varphi\,\mathbf{i} + \sin\varphi\,\mathbf{j}$, $\mathbf{e}_\varphi = \cos\varphi\,\mathbf{j} - \sin\varphi\,\mathbf{i}$.

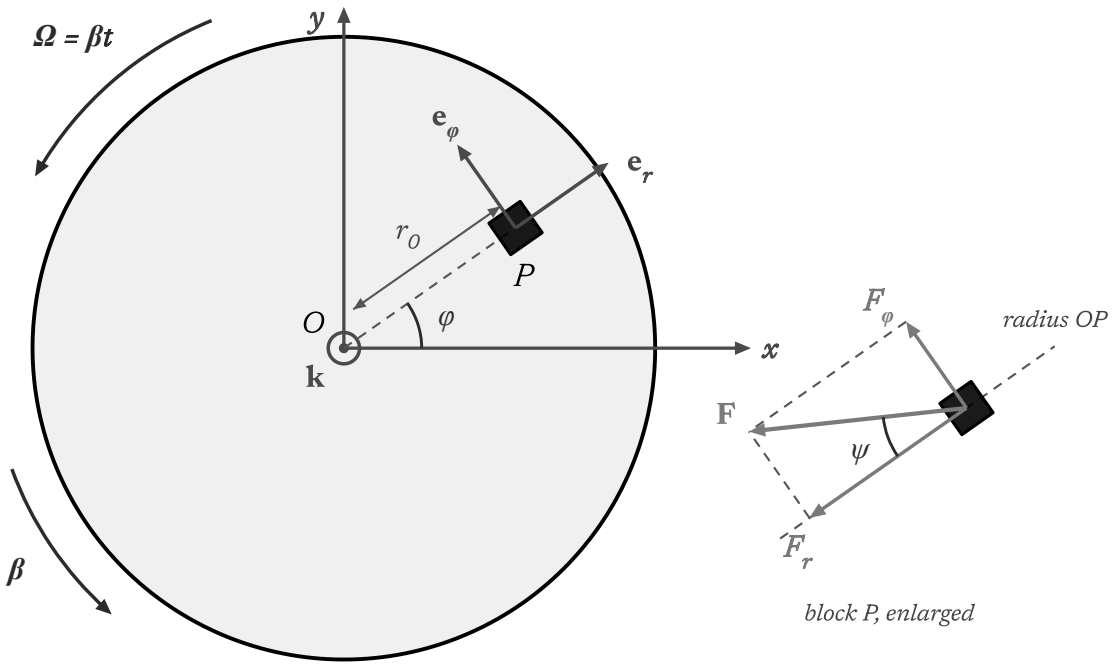


Figure 1. Turntable seen from above: fixed axes x, y and $\mathbf{k}$ (out of the page, upwards) at the axis O, rate $\Omega = \beta t$ and angular acceleration β (counter-clockwise), block P at distance r_0; while it sticks, the friction $\mathbf{F}$ has a component $F_r < 0$ towards O and a component $F_\varphi > 0$ along $\mathbf{e}_\varphi$.

Setup

- Data: $m, g, \mu, \mu_k, r_0, \beta$. Numbers: $\mu = 0.4$, $\mu_k = 0.3$, $r_0 = 0.5$ m, $\beta = 0.2$ rad s^{-2}, $g = 9.81$ m s^{-2}.
- Hypotheses: point particle (the block does not tip), rigid horizontal table, Coulomb friction, no air drag.
- In (e) and (f) the friction force on P is written $\mathbf{F} = F_r\,\mathbf{e}_r + F_\varphi\,\mathbf{e}_\varphi$ and ψ is its angle from the inward radius $-\mathbf{e}_r$, positive towards $+\mathbf{e}_\varphi$ (figure).

Part (a) · 1 point

Degrees of freedom and frames

How many degrees of freedom has P while it sticks to the table, and while it slides on it? Give the angular velocity and the angular acceleration of S_1 relative to S_0.

Step 1. Count

A free particle has 3 degrees of freedom. While it touches the table, $z = 0$ (bilateral as long as $N \geq 0$): 2 remain, the polar coordinates (r, φ). While it does not slip, it moves with the table: $r = r_0$ and φ is the angle turned by the table, $\varphi = \int_0^t \beta\tau \, d\tau = \left[\tfrac{1}{2}\beta\tau^2\right]_0^t = \tfrac{1}{2}\beta t^2$. Nothing is left free: 0 degrees of freedom, and the friction force is an **unknown reaction** that adjusts to keep the block in place, as long as it can (Notes §6.6).

Step 2. Frames

S_1 turns about $\mathbf{k}$ with $\boldsymbol{\omega}_{10} = \Omega\,\mathbf{k} = \beta t\,\mathbf{k}$; its angular acceleration $\dot{\boldsymbol{\omega}}_{10} = \beta\,\mathbf{k}$ is not zero, so in S_1 the block feels an Euler force as well as the centrifugal and Coriolis forces.

KEY RESULTS

- Sliding: 2 degrees of freedom (r, φ) on the plane $z = 0$. Sticking: 0 degrees of freedom, $r = r_0$, $\varphi = \tfrac{1}{2}\beta t^2$ prescribed.
- $\boldsymbol{\omega}_{10} = \beta t\,\mathbf{k}$, $\dot{\boldsymbol{\omega}}_{10} = \beta\,\mathbf{k}$.

Part (b) · 1.5 points

Kinematics while the block sticks

Compute the velocity and acceleration of P in S_0 in the basis $\{\mathbf{e}_r, \mathbf{e}_\varphi\}$.

Step 1. Change of basis and unit vectors

The polar basis $\{\mathbf{e}_r, \mathbf{e}_\varphi, \mathbf{k}\}$ turns with the table; its relation to the inertial basis is

$$\begin{pmatrix} \mathbf{e}_r \\ \mathbf{e}_\varphi \\ \mathbf{k} \end{pmatrix} = [R] \begin{pmatrix} \mathbf{i} \\ \mathbf{j} \\ \mathbf{k} \end{pmatrix}, \qquad [R] = \begin{pmatrix} \cos\varphi & \sin\varphi & 0 \\ -\sin\varphi & \cos\varphi & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Checks: rows 1 and 2 have length $\cos^2\varphi + \sin^2\varphi = 1$ and $-\cos\varphi\sin\varphi + \sin\varphi\cos\varphi = 0$, row 3 is $\mathbf{k}$, so $[R][R]^T = I$; $\det[R] = \cos^2\varphi + \sin^2\varphi = +1$ (right-handed, $\mathbf{e}_r \times \mathbf{e}_\varphi = \mathbf{k}$). Then $\dot{\mathbf{e}}_r = \dot\varphi\,(-\sin\varphi\,\mathbf{i} + \cos\varphi\,\mathbf{j}) = \dot\varphi\,\mathbf{e}_\varphi$ and $\dot{\mathbf{e}}_\varphi = -\dot\varphi\,\mathbf{e}_r$ (chain rule through $\varphi(t)$).

Step 2. Velocity and acceleration

$\mathbf{r} = r_0\,\mathbf{e}_r$ with r_0 constant:

$$\mathbf{v} = r_0\dot{\varphi}\,\mathbf{e}_\varphi = r_0\Omega\,\mathbf{e}_\varphi$$

$$\mathbf{a} = r_0\ddot{\varphi}\,\mathbf{e}_\varphi + r_0\dot{\varphi}\,\dot{\mathbf{e}}_\varphi = r_0\beta\,\mathbf{e}_\varphi - r_0\Omega^2\,\mathbf{e}_r$$

The general polar formula $\mathbf{a} = (\ddot{r} - r\dot{\varphi}^2)\mathbf{e}_r + (r\ddot{\varphi} + 2\dot{r}\dot{\varphi})\mathbf{e}_\varphi$ gives the same with $\dot{r} = \ddot{r} = 0$.

KEY RESULTS

- $\mathbf{v} = r_0\Omega\,\mathbf{e}_\varphi$.
- $\mathbf{a} = -r_0\Omega^2\,\mathbf{e}_r + r_0\beta\,\mathbf{e}_\varphi$: centripetal **and** tangential parts.

Part (c) · 2 points

Forces and the friction needed to stick

List the forces on P and find the normal reaction and the friction force needed for the block to move with the table. Check the result in S_1.

Step 1. Forces

Weight $-mg\,\mathbf{k}$; normal reaction $N\,\mathbf{k}$ ($N \geq 0$); friction $\mathbf{F} = F_r\,\mathbf{e}_r + F_\varphi\,\mathbf{e}_\varphi$ in the plane of contact, unknown while the block sticks.

Step 2. Newton in S_0

$m\mathbf{a} = -mg\,\mathbf{k} + N\,\mathbf{k} + \mathbf{F}$. Projecting:

$$\mathbf{k}:\ 0 = N - mg \Rightarrow N = mg; \qquad \mathbf{e}_r:\ -mr_0\Omega^2 = F_r; \qquad \mathbf{e}_\varphi:\ mr_0\beta = F_\varphi$$

Friction must pull **inward** (to bend the path) and **forward** (to speed the block up along with the table).

Step 3. Check in S_1 (Notes §4.9)

In S_1 the block is at rest, $\mathbf{v}_{rel} = \mathbf{0}$, so the Coriolis force vanishes. Centrifugal: $-m\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}) = -m\Omega^2\,\mathbf{k} \times (\mathbf{k} \times r_0\mathbf{e}_r) = -m\Omega^2\,\mathbf{k} \times r_0\mathbf{e}_\varphi = mr_0\Omega^2\,\mathbf{e}_r$. Euler: $-m\dot{\boldsymbol{\omega}} \times \mathbf{r} = -m\beta\,\mathbf{k} \times r_0\mathbf{e}_r = -mr_0\beta\,\mathbf{e}_\varphi$. Balance in the plane: $\mathbf{F} + mr_0\Omega^2\mathbf{e}_r - mr_0\beta\,\mathbf{e}_\varphi = \mathbf{0}$, the same F_r and F_φ.

KEY RESULTS

- $N = mg;\ F_r = -mr_0\Omega^2 = -mr_0\beta^2t^2;\ F_\varphi = mr_0\beta$.
- $|\mathbf{F}| = mr_0\sqrt{\Omega^4 + \beta^2}$, increasing in time.
- In S_1: $\mathbf{F} + mr_0\Omega^2\,\mathbf{e}_r - mr_0\beta\,\mathbf{e}_\varphi = \mathbf{0}$ (centrifugal and Euler forces, no Coriolis).

Part (d) · 2 points

When does the block slip?

Using the Coulomb condition, find the instant t^* at which the block starts to slide. What condition must β satisfy for the block to stick at all?

Step 1. The friction cone

Static friction holds while $|\mathbf{F}| \le \mu N$ (Notes §6.6). With (c):

$$m^2r_0^2\left(\Omega^4 + \beta^2\right) \le \mu^2m^2g^2 \iff \Omega^4 \le \left(\frac{\mu g}{r_0}\right)^2 - \beta^2$$

Step 2. Slip instant

The left side grows with t ($\Omega = \beta t$), so the inequality fails for the first time when it becomes an equality:

$$\Omega^* = \left[\left(\frac{\mu g}{r_0}\right)^2 - \beta^2\right]^{1/4}, \qquad t^* = \frac{\Omega^*}{\beta}$$

Step 3. Existence

A real $\Omega^* \ge 0$ needs $\beta \le \mu g/r_0$. In the equality case $\beta = \mu g/r_0$ the formula gives $\Omega^* = 0, t^* = 0$: the friction is already at the limit of the cone at $t = 0$ and the block slides as soon as $\Omega > 0$, so it never sticks for a finite time. Sticking therefore requires the strict condition $\beta < \mu g/r_0$. If $\beta > \mu g/r_0$, already at $t = 0$ (where $\Omega = 0$) the tangential friction $mr_0\beta$ needed exceeds μmg: the table slides under the block from the start. Dropping β^2 in the cone (the usual shortcut $r_0\Omega^2 = \mu g$) hides exactly this case.

KEY RESULTS

- $r_0^2\left(\Omega^4+\beta^2\right) \le \mu^2 g^2$; slip when $\Omega^{*4} = \left(\dfrac{\mu g}{r_0}\right)^2 - \beta^2$.
- $t^* = \dfrac{\Omega^*}{\beta} = \dfrac{1}{\beta}\left[\left(\dfrac{\mu g}{r_0}\right)^2 - \beta^2\right]^{1/4}$; requires $\beta < \mu g/r_0$ (at $\beta = \mu g/r_0$, $t^* = 0$), otherwise it slips at $t = 0$.

Part (e) · 1.5 points

Numbers and the direction of the friction at the slip instant

For the data of the Setup compute Ω^*, t^*, the number of turns before slip and the speed of the block, and the angle ψ of the friction force at t^*.

Step 1. Instant of slip

$\mu g/r_0 = 0.4 \cdot 9.81/0.5 = 7.848\ \text{s}^{-2}$, so $(\mu g/r_0)^2 = 61.591\ \text{s}^{-4}$ and $\Omega^{*4} = 61.591 - 0.04 = 61.551\ \text{s}^{-4}$. Then $\Omega^{*2} = 7.8454\ \text{s}^{-2}$, $\Omega^* = 2.801\ \text{rad s}^{-1}$ and $t^* = 2.801/0.2 = 14.00$ s. Here $\beta^2/(\mu g/r_0)^2 = 6.5 \times 10^{-4}$: the tangential part barely changes t^*.

Step 2. Turns and speed

$\varphi^* = \frac{1}{2}\beta t^{*2} = \Omega^{*2}/(2\beta) = 7.8454/0.4 = 19.61$ rad, i.e. $19.61/2\pi = 3.12$ turns. The speed is $r_0\Omega^* = 1.400\ \text{m s}^{-1}$, along $\mathbf{e}_\varphi$.

Step 3. Direction of the friction

At t^*: $|F_r| = mr_0\Omega^{*2} = 3.923\,m$ N and $F_\varphi = mr_0\beta = 0.1\,m$ N, so $\tan\psi = \beta/\Omega^{*2} = 0.2/7.8454 = 0.0255$ and $\psi = 1.460^\circ$. Check: $\cos\psi = r_0\Omega^{*2}/(\mu g)$, and $|\mathbf{F}| = \mu m g = 3.924\,m$ N. The friction points to the axis, tilted 1.46° **forward**.

KEY RESULTS

- $\mu g/r_0 = 7.848\ \text{s}^{-2}$, $\Omega^{*4} = 61.551\ \text{s}^{-4}$, $\Omega^* = 2.801\ \text{rad s}^{-1}$, $t^* = 14.00$ s.
- $\varphi^* = \Omega^{*2}/2\beta = 19.61$ rad $= 3.12$ turns; $|\mathbf{v}| = r_0\Omega^* = 1.400\ \text{m s}^{-1}$.
- $\tan\psi = \beta/\Omega^{*2} = 0.0255$, $\psi = 1.460^\circ$, inward and slightly forward.

Part (f) · 2 points

How does the block start to slide?

Just after t^* the friction is kinetic, of modulus $\mu_k mg$. Find the initial acceleration of P relative to the table, its modulus and its direction. Does the block leave along the radius? Describe its motion in S_0 at that instant.

Step 1. Inertia forces at t^*

In S_1, with $\mathbf{v}_{rel} = \mathbf{0}$, the inertia forces are those of (c): $\mathbf{F}_{in} = mr_0\Omega^{*2}\,\mathbf{e}_r - mr_0\beta\,\mathbf{e}_\varphi$. By (c) and (d), $\mathbf{F}_{in} = -\mathbf{F}_{needed}$ and $|\mathbf{F}_{in}| = \mu m g$. It points outward and **backward** (against the rotation), at the angle ψ from $+\mathbf{e}_r$.

Step 2. Direction of sliding

Once static friction fails, the block starts to move relative to the table along the unit vector $\hat{\mathbf{u}}$ of its relative acceleration, and kinetic friction opposes that motion: $\mathbf{F}_k = -\mu_k m g\,\hat{\mathbf{u}}$. In S_1, $m\mathbf{a}_{rel} = \mathbf{F}_{in} - \mu_k m g\,\hat{\mathbf{u}}$ must be parallel to $\hat{\mathbf{u}}$, which is possible only if $\hat{\mathbf{u}}$ is along $\mathbf{F}_{in}$:

$$\mathbf{a}_{rel}(t^{*+}) = (\mu - \mu_k)\, g\, \frac{\mathbf{F}_{in}}{|\mathbf{F}_{in}|}, \qquad |\mathbf{a}_{rel}| = (0.4 - 0.3)\cdot 9.81 = 0.981 \text{ m s}^{-2}$$

Components: $0.981\cos\psi = 0.9807$ m s^{-2} outward and $-0.981\sin\psi = -0.0250$ m s^{-2} along $\mathbf{e}_\varphi$: the block **lags behind** the table as it drifts out. It does not start along the radius.

Step 3. Cross-check in S_0

In S_0, $\mathbf{a} = \mathbf{F}_k/m = \mu_k g\,\hat{\mathbf{F}}_{needed}$ (inward and forward, 2.943 m s^{-2}), while the point of the table under the block has $\mathbf{a}_{drag} = \mu g\,\hat{\mathbf{F}}_{needed}$ (3.924 m s^{-2}, part (c)). With $\mathbf{v}_{rel} = \mathbf{0}$ the Coriolis term vanishes and $\mathbf{a}_{rel} = \mathbf{a} - \mathbf{a}_{drag} = (\mu_k - \mu) g\,\hat{\mathbf{F}}_{needed}$: the same result. In S_0 the block moves at 1.400 m s^{-1} **tangentially**; it is not "flung out along the radius". It simply cannot be pulled inward hard enough to stay on its circle, so its path opens out.

Step 4. Equal coefficients

If $\mu_k = \mu$, $\mathbf{a}_{rel}(t^{*+}) = \mathbf{0}$: the sliding starts smoothly, driven by the growth of Ω after t^*, but in the same direction.

KEY RESULTS

- In S_1: centrifugal + Euler $= -\mathbf{F}_{needed}$, of modulus $\mu m g$, pointing **outward and backward** (at ψ behind the outward radius).
- $\mathbf{a}_{rel}(t^{*+}) = (\mu - \mu_k)\, g$ in that direction: 0.981 m s^{-2}, components 0.9807 m s^{-2} along $\mathbf{e}_r$ and -0.0250 m s^{-2} along $\mathbf{e}_\varphi$.
- In S_0 the velocity is tangential (1.400 m s^{-1}, no radial launch) and the acceleration $\mu_k g = 2.943$ m s^{-2} is smaller than the $\mu g = 3.924$ m s^{-2} needed to follow the table.

Interpretation and checks

- **Dimensional check:** $\mu g/r_0$ and β are s^{-2}, so $\Omega^{*4} = (\mu g/r_0)^2 - \beta^2$ is s^{-4} and $t^* = \Omega^*/\beta$ is s; $r_0\Omega^2$, $r_0\beta$, μg are m s^{-2}.
- **Limits:** $\beta \to 0$ (very slow spin-up) gives $r_0\Omega^{*2} = \mu g$, the classic result for uniform rotation, with $\psi \to 0$ and sliding along the outward radius in S_1; $\beta \to \mu g/r_0$ gives $t^* \to 0$ and $\psi \to 90^\circ$: the block slides straight backward.
- **Common mistakes in the exam:** (1) writing $r_0\Omega^2 \le \mu g$ and forgetting the tangential friction $m r_0 \beta$; (2) forgetting the Euler force $-m\dot{\boldsymbol{\omega}} \times \mathbf{r}$ in S_1; (3) putting $|\mathbf{F}| = \mu N$ before slip (static friction is only bounded by μN); (4) claiming the block flies off radially in S_0.

Problem · Unit 2 · 10 points · 60 min · Standard exam level

P13 — Capsule descent under quadratic drag with two parachute stages: terminal speed, quadratures and the work of drag

*In a drop test, a capsule of mass m is released **from rest** from a hovering helicopter at the point O and falls vertically. Its drogue parachute is open from the release: the air exerts on the capsule the drag $-c_1 v^2$ along the motion, where $v = \dot{y} \geq 0$ is the speed and y the distance fallen, measured **downward** from O. When the speed reaches v_1, the main parachute opens instantaneously and the drag becomes $-c_2 v^2$, with $c_2 \gg c_1$. Gravity g is uniform and points down; buoyancy and the variation of air density with height are neglected.*

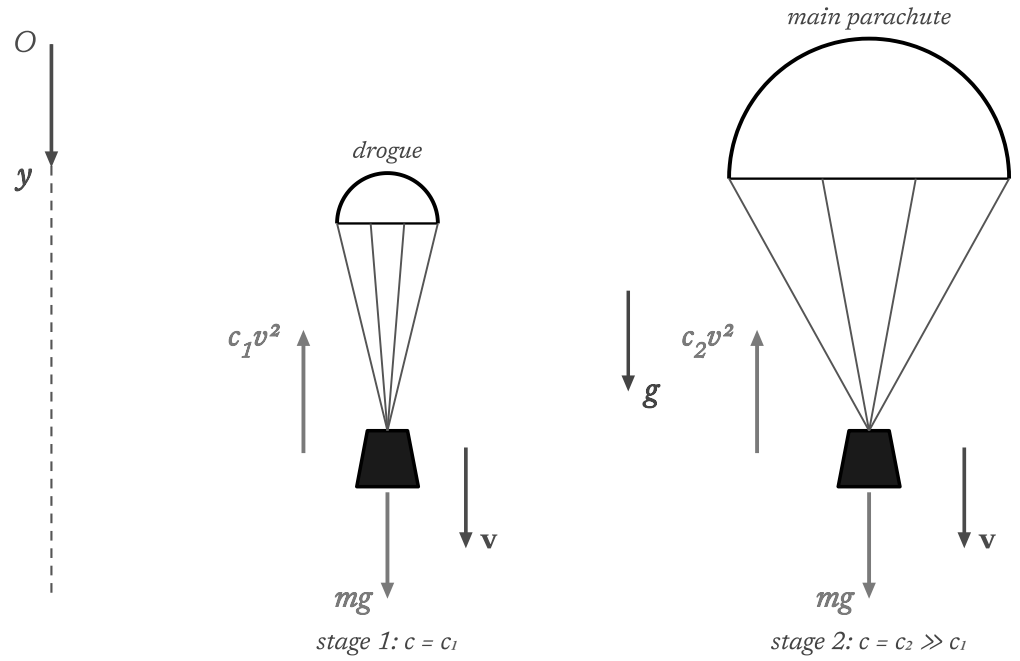


Figure 1. Two-stage descent: coordinate y downward from the release point O, velocity $v = \dot{y}$; forces on the capsule: weight mg (down) and drag cv^2 (up), with $c = c_1$ under the drogue (stage 1) and $c = c_2 \gg c_1$ under the main parachute (stage 2).

Setup

- Data: $m = 1000$ kg, $g = 9.81$ m s^{-2}. The parachutes are given by their terminal speeds: $v_{t1} = 50$ m s^{-1} (drogue) and $v_{t2} = 8$ m s^{-1} (main). The main parachute opens at $v_1 = 40$ m s^{-1}.
- Hypotheses: point particle, vertical motion (released with no horizontal velocity, no wind), constant drag coefficients within each stage, instantaneous opening.
- The time of stage 2 is counted from the opening, and Δy is the distance fallen since the opening.

Part (a) · 1 point

Degrees of freedom, forces and equation of motion

How many degrees of freedom has the capsule? Write the forces and the equation of motion for v.

Step 1. Count

With no horizontal force and no horizontal initial velocity the motion is along the vertical: 1 degree of freedom, the coordinate y, with unit vector $\mathbf{e}_y$ pointing **down**.

Step 2. Forces and Newton

Weight $mg\,\mathbf{e}_y$ (positive: it points down). Drag $-c|\mathbf{v}|\mathbf{v} = -cv^2\,\mathbf{e}_y$ for $v \geq 0$ (upward). Newton (Notes §4.3), with $\mathbf{a} = \dot{v}\,\mathbf{e}_y$:

$$m\dot{v} = mg - cv^2$$

KEY RESULTS

- 1 degree of freedom, $y(t)$, positive downward.
- $m\dot{v} = mg - cv^2$ (for $v \geq 0$).

Part (b) · 2 points

Terminal speed and $v(t)$ in stage 1

Define the terminal speed v_t and integrate the equation for the drogue stage, from rest at $t = 0$, to get $v(t)$ and $y(t)$.

Step 1. Terminal speed

The acceleration vanishes when $mg = cv^2$: $v_t = \sqrt{mg/c}$. With it the equation reads $\dot{v} = g\big(1 - v^2/v_t^2\big)$. For the drogue, $c_1 = mg/v_{t1}^2 = 9810/2500 = 3.924\ \text{kg m}^{-1}$.

Step 2. Separation of variables

For $0 \leq v < v_t$:

$$\frac{dv}{1 - v^2/v_t^2} = g\,dt$$

The antiderivative of the left side is $F(v) = v_t\,\operatorname{artanh}(v/v_t)$. Check with the chain rule: $\dfrac{dF}{dv} = v_t \cdot \dfrac{1}{1-(v/v_t)^2}\cdot\dfrac{1}{v_t} = \dfrac{1}{1 - v^2/v_t^2}$.

Step 3. Limits (Barrow's rule)

Integrating from $(t=0, v=0)$ to (t, v):

$$\left[v_t \operatorname{artanh}\frac{v'}{v_t}\right]_0^{v} = \left[g\,t'\right]_0^{t} \implies v_t \operatorname{artanh}\frac{v}{v_t} - 0 = gt \implies v = v_t \tanh\frac{gt}{v_t}$$

Since $\tanh < 1$, v approaches v_t but never reaches it in finite time.

Step 4. Position

$y = \int_0^t v\,dt'$. With $w = \cosh(gt'/v_t)$, $dw = \dfrac{g}{v_t}\sinh(gt'/v_t)\,dt'$, so $\tanh(gt'/v_t)\,dt' = \dfrac{v_t}{g}\dfrac{dw}{w}$ and

$$y = v_t \cdot \frac{v_t}{g}\left[\ln w\right]_{w=1}^{w=\cosh(gt/v_t)} = \frac{v_t^2}{g}\ln\cosh\frac{gt}{v_t}$$

KEY RESULTS

- $v_t = \sqrt{mg/c}$, i.e. $c = mg/v_t^2$; $c_1 = 3.924\ \mathrm{kg\,m^{-1}}$.
- $\displaystyle\int \frac{dv}{1 - v^2/v_t^2} = v_t \operatorname{artanh}\frac{v}{v_t}$ for $v < v_t$.
- $v = v_t \tanh\dfrac{gt}{v_t}$, $y = \dfrac{v_t^2}{g}\ln\cosh\dfrac{gt}{v_t}$.

Part (c) · 1.5 points

Speed as a function of the distance fallen

Obtain $v(y)$ for stage 1 directly from the equation of motion, without using t.

Step 1. Chain rule

With $v = v(y(t))$: $\dot v = \dfrac{dv}{dy}\dot y = v\dfrac{dv}{dy}$, so $v\dfrac{dv}{dy} = g\left(1 - v^2/v_t^2\right)$ and, separating, $\dfrac{v\,dv}{1 - v^2/v_t^2} = g\,dy$.

Step 2. Substitution

$w = 1 - v^2/v_t^2$, $dw = -\dfrac{2v}{v_t^2}dv$, so $v\,dv = -\dfrac{v_t^2}{2}dw$. Limits: $v = 0 \mapsto w = 1$, $v \mapsto w = 1 - v^2/v_t^2$:

$$-\frac{v_t^2}{2}\int_1^{1-v^2/v_t^2}\frac{dw}{w} = -\frac{v_t^2}{2}\Big[\ln w\Big]_1^{1-v^2/v_t^2} = -\frac{v_t^2}{2}\ln\left(1 - \frac{v^2}{v_t^2}\right) = g\,y$$

Step 3. Result

$$v^2 = v_t^2\left(1 - e^{-2gy/v_t^2}\right)$$

Cross-check: inserting $y(t)$ of (b), $e^{-2gy/v_t^2} = \cosh^{-2}(gt/v_t)$ and $1 - \cosh^{-2} = \tanh^2$, which is $v(t)^2/v_t^2$ of (b).

KEY RESULTS

- $v\dfrac{dv}{dy} = g\left(1 - \dfrac{v^2}{v_t^2}\right)$.
- $v^2 = v_t^2\left(1 - e^{-2gy/v_t^2}\right)$.

Part (d) · 1.5 points

Numbers for the drogue stage

If the main parachute never opened, how long and how far would the capsule fall before reaching $95\ \%$ *of* v_{t1}*? When and where does it actually reach* $v_1 = 40\ m\ s^{-1}$*?*

Step 1. $95\ \%$ **of the terminal speed**

$v_{t1}/g = 50/9.81 = 5.097$ s and $\operatorname{artanh} 0.95 = \tfrac{1}{2}\ln(1.95/0.05) = 1.8318$, so $t_{95} = 9.336$ s. From (c), $y = -\dfrac{v_t^2}{2g}\ln\left(1 - v^2/v_t^2\right)$ with $v_t^2/2g = 127.4$ m and $\ln(1/0.0975) = 2.3279$: $y_{95} = 296.6$ m.

Step 2. Opening of the main parachute

$\operatorname{artanh} 0.8 = \tfrac{1}{2}\ln 9 = 1.0986$, so $t_1 = 5.097 \cdot 1.0986 = 5.600$ s, and $y_1 = 127.4 \cdot \ln(1/0.36) = 127.4 \cdot 1.0217 = 130.2$ m. A numerical integration of the equation of motion gives the same values.

KEY RESULTS

- $t_{95} = \dfrac{v_{t1}}{g}\operatorname{artanh} 0.95 = 9.336$ s; $y_{95} = \dfrac{v_{t1}^2}{2g}\ln\dfrac{1}{1 - 0.95^2} = 296.6$ m.
- $v_1 = 0.8\,v_{t1}$ is reached at $t_1 = 5.600$ s, $y_1 = 130.2$ m.

Part (e) · 2 points

Stage 2: the main parachute

At the opening $v_1 > v_{t2}$. Find the acceleration just after the opening, solve for $v(t)$ in stage 2, and compute the time and the distance needed to come down to $1.05\,v_{t2}$. Why can the formula of (b) not be used?

Step 1. Acceleration at the opening

$c_2 = mg/v_{t2}^2 = 9810/64 = 153.3$ kg m^{-1} ($c_2/c_1 = 39.06$). Just after the opening $v = 40$ m s^{-1}, so $\dot v = g\big(1 - (40/8)^2\big) = -24g = -235.4$ m s^{-2}: the drag $c_2 v_1^2 = 245.3$ kN is 25 times the weight.

Step 2. Why not tanh

The formula $v = v_t \tanh(\dots)$ is bounded by v_t, so it can never start at $v_1 = 5v_{t2}$; equivalently the constant of integration would be $\operatorname{artanh} 5$, which is not real. For $v > v_t$ the antiderivative of $1/(1 - v^2/v_t^2)$ is $v_t \operatorname{arcoth}(v/v_t)$: $\dfrac{d}{dv}\big[v_t \operatorname{arcoth}(v/v_t)\big] = \dfrac{1}{1 - v^2/v_t^2}$, the same derivative as artanh on the other branch.

Step 3. Integration

From $(t = 0, v = v_1)$ to (t, v):

$$\Big[v_{t2} \operatorname{arcoth}\frac{v'}{v_{t2}}\Big]_{v_1}^{v} = \big[g\,t'\big]_0^t \;\Longrightarrow\; v = v_{t2} \coth\Big(\frac{gt}{v_{t2}} + \operatorname{arcoth}\frac{v_1}{v_{t2}}\Big)$$

$\coth > 1$ and decreases, so v decreases monotonically to v_{t2} from above.

Step 4. Time to $1.05\,v_{t2}$

$t = \dfrac{v_{t2}}{g}\Big(\operatorname{arcoth} 1.05 - \operatorname{arcoth} 5\Big)$, with $\operatorname{arcoth} 1.05 = \tfrac12 \ln 41 = 1.8568$ and $\operatorname{arcoth} 5 = \tfrac12 \ln 1.5 = 0.2027$: $t = 0.8155 \cdot 1.6541 = 1.349$ s.

Step 5. Distance

As in (c), $\dfrac{v\,dv}{1 - v^2/v_{t2}^2} = g\,dy$, now integrated from $(\Delta y = 0,\ v = v_1)$ to $(\Delta y,\ v)$. Substitute $w = 1 - v'^2/v_{t2}^2$, $dw = -\dfrac{2v'}{v_{t2}^2}dv'$, so $v'\,dv' = -\dfrac{v_{t2}^2}{2}dw$; the limits become $w_1 = 1 - v_1^2/v_{t2}^2 = -24$ and $w = 1 - v^2/v_{t2}^2$. In stage 2 $v' > v_{t2}$, so $w < 0$ on the whole interval and the antiderivative of $1/w$ is $\ln|w|$ (not $\ln w$):

$$g\,\Delta y = -\frac{v_{t2}^2}{2}\int_{w_1}^{w}\frac{dw'}{w'} = -\frac{v_{t2}^2}{2}\Big[\ln|w'|\Big]_{w_1}^{w} = -\frac{v_{t2}^2}{2}\big(\ln|w| - \ln|w_1|\big) = \frac{v_{t2}^2}{2}\ln\frac{|w_1|}{|w|}$$

With $|w_1| = (v_1^2 - v_{t2}^2)/v_{t2}^2$ and $|w| = (v^2 - v_{t2}^2)/v_{t2}^2$ the factors v_{t2}^2 cancel in the quotient; solving for Δy and, inversely, for v^2:

$$v^2 = v_{t2}^2 + \left(v_1^2 - v_{t2}^2\right) e^{-2g\Delta y / v_{t2}^2}, \qquad \Delta y = \frac{v_{t2}^2}{2g} \ln \frac{v_1^2 - v_{t2}^2}{v^2 - v_{t2}^2}$$

With $v = 8.4$ m s^{-1}: $\Delta y = 3.262 \cdot \ln(1536/6.56) = 3.262 \cdot 5.456 = 17.80$ m.

KEY RESULTS

- $c_2 = 153.3$ kg m^{-1}; $\dot v(0^+) = g\left(1 - v_1^2/v_{t2}^2\right) = -24\,g = -235.4$ m s^{-2} (a **deceleration**; drag 245.3 kN against a weight of 9.81 kN).
- $v = v_{t2} \coth\left(\dfrac{gt}{v_{t2}} + \operatorname{arcoth}\dfrac{v_1}{v_{t2}}\right)$, decreasing towards v_{t2} from above.
- To 8.4 m s^{-1}: $t = 1.349$ s, $\Delta y = 17.80$ m.

Part (f) · 2 points

Work of the drag and energy balance

Is the mechanical energy conserved? Compute the work done by the drag (i) in stage 1, from the release to the opening, and (ii) in stage 2, from the opening to $v = 8.4$ m s^{-1}, from the energy balance; then check (ii) by integrating the drag force along the path.

Step 1. Energy is not conserved

The power of the drag is $(-cv^2\,\mathbf{e}_y)\cdot(v\,\mathbf{e}_y) = -cv^3 < 0$ (Notes §4.4): it removes mechanical energy as long as the capsule moves. Only the weight is conservative ($V = -mgy$ with y downward), so $\dfrac{d}{dt}\left(\tfrac12 mv^2 - mgy\right) = -cv^3$. At the opening the dissipated power is $c_2 v_1^3 = 153.3 \cdot 64\,000$ W $= 9.81$ MW.

Step 2. Stage 1

From $v = 0, y = 0$ to $v_1 = 40$ m s^{-1}, $y_1 = 130.2$ m: $\Delta E_{kin} = \tfrac12 \cdot 1000 \cdot 1600 = 800.0$ kJ and $W_{weight} = mgy_1 = 1277.1$ kJ, so $W_{drag,1} = 800.0 - 1277.1 = -477.1$ kJ.

Step 3. Stage 2 from the energy balance

$\Delta E_{kin} = \tfrac12 \cdot 1000\,(70.56 - 1600) = -764.7$ kJ and $W_{weight} = 9810 \cdot 17.80 = 174.6$ kJ, so $W_{drag,2} = -764.7 - 174.6 = -939.3$ kJ: in under 18 m the main parachute absorbs almost twice what the drogue absorbed in 130 m.

Step 4. Direct integral

$W_{drag,2} = -\int_0^{\Delta y} c_2 v^2\, ds$ with $v^2 = v_{t2}^2 + (v_1^2 - v_{t2}^2)e^{-2gs/v_{t2}^2}$ and $c_2 v_{t2}^2 = mg$:

$$W_{drag,2} = -mg\,\Delta y - c_2\left(v_1^2 - v_{t2}^2\right)\Big[-\frac{v_{t2}^2}{2g} e^{-2gs/v_{t2}^2}\Big]_0^{\Delta y} = -mg\,\Delta y - \frac{m}{2}\left(v_1^2 - v_{t2}^2\right)\left(1 - e^{-2g\Delta y/v_{t2}^2}\right)$$

and by (e), $\left(v_1^2 - v_{t2}^2\right)\left(1 - e^{-2g\Delta y / v_{t2}^2}\right) = v_1^2 - v^2$, so $W_{drag,2} = -\left(mg\Delta y + \tfrac{1}{2}m(v_1^2 - v^2)\right) = -939.3$ kJ, the value of Step 3.

KEY RESULTS

- No: the drag power $-cv^3 < 0$ dissipates energy (9.81 MW just after the opening).
- (i) $W_{drag,1} = \tfrac{1}{2}mv_1^2 - mgy_1 = 800.0 - 1277.1 = -477.1$ kJ.
- (ii) $W_{drag,2} = \tfrac{1}{2}m(8.4^2 - 40^2) - mg\Delta y = -764.7 - 174.6 = -939.3$ kJ; the direct integral gives $-\left(mg\Delta y + \tfrac{1}{2}m(v_1^2 - v^2)\right)$, the same.

Interpretation and checks

- **Dimensional check:** $[c] = \text{N}/(\text{m s}^{-1})^2 = \text{kg m}^{-1}$, so $\sqrt{mg/c}$ is m s^{-1}; gt/v_t and gy/v_t^2 are dimensionless (arguments of tanh and exp); $[cv^3] = \text{W}$.
- **Limits:** for $gt \ll v_t$, $\tanh x \approx x$ gives $v \approx gt$ and $y \approx \tfrac{1}{2}gt^2$ (free fall); for $t \to \infty$ both tanh and coth tend to 1 and $v \to v_t$, from below in stage 1 and from above in stage 2; $y \approx v_t t - \tfrac{v_t^2}{g}\ln 2$ for large t in stage 1.
- **Common mistakes in the exam:** (1) writing the drag as $+cv^2$ or $-cv$ (wrong sign or wrong law); (2) using $v = v_t\tanh(\dots)$ after the main parachute opens, which would need artanh of a number larger than 1; (3) skipping the chain rule $\dot v = v\,dv/dy$; (4) assuming the terminal speed is reached in a finite time; (5) writing energy conservation with drag acting.

Problem · Unit 1 · 10 points · 60 min · Standard exam level

P14 — A spiral trajectory flown at constant speed: polar and intrinsic kinematics, a non-central force and zero power

*A drone P of mass m flies in a **horizontal** plane (its weight is balanced by the vertical lift, which is not studied). In this plane take the inertial frame $S_0 = \{O; \mathbf{i}, \mathbf{j}\}$ and the polar coordinates (r, φ) of P about O, with φ measured from $\mathbf{i}$ counter-clockwise and the polar basis $\mathbf{e}_r = \cos\varphi\,\mathbf{i} + \sin\varphi\,\mathbf{j}$, $\mathbf{e}_\varphi = -\sin\varphi\,\mathbf{i} + \cos\varphi\,\mathbf{j}$. The autopilot keeps P on the logarithmic spiral*

$$r = r_0\, e^{b\varphi} \qquad (r_0 > 0,\ b > 0)$$

*with a **constant speed** $|\mathbf{v}| = v_0$ and φ increasing. At $t = 0$ the drone is at $\varphi = 0$, $r = r_0$. The resultant of the horizontal forces on the drone (thrust and drag) is called $\mathbf{F}$. Let $\mathbf{e}_t$, $\mathbf{e}_n$ be the unit tangent and the principal normal of the trajectory, ρ its radius of curvature and α the angle between $\mathbf{e}_r$ and $\mathbf{e}_t$.*

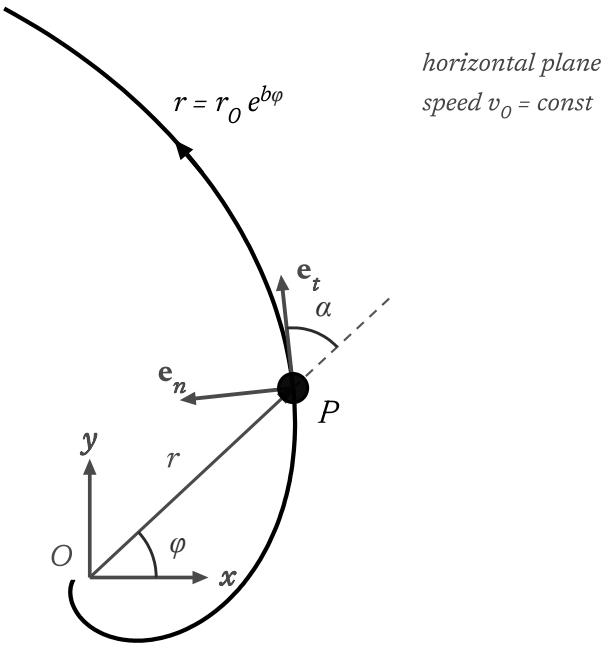


Figure 1. Drone P on the spiral $r = r_0 e^{b\varphi}$ flown with constant speed v_0 (φ increasing): polar coordinates (r, φ) about O, unit tangent $\mathbf{e}_t$ and principal normal $\mathbf{e}_n$, constant angle α between $\mathbf{e}_r$ and $\mathbf{e}_t$ (drawn for $b = 3/4$).

Setup

- Data: r_0, b, v_0, m. In the numerical parts: $r_0 = 20$ m, $b = 3/4$, $v_0 = 10$ m/s, $m = 2$ kg, and the drone flies from $r_1 = r_0 = 20$ m to $r_2 = 50$ m.
- Hypotheses: point particle, planar motion in the horizontal plane, S_0 inertial; the trajectory and the speed are imposed by the autopilot (the force $\mathbf{F}$ is whatever is needed to keep them).
- Useful constant: $k \equiv \sqrt{1+b^2}$; for $b = 3/4$, $k = \sqrt{1+9/16} = 5/4$.

Part (a) · 2 points

Velocity in polar coordinates and the constant angle

How many parameters fix the position of P on the trajectory? Write $\mathbf{v}$ in the polar basis and obtain $\dot r$ and $\dot\varphi$ as functions of r from the two conditions of the statement. Show that the angle α between the velocity and the radius is constant. Give the numbers for the data of the Setup.

Step 1. Parameters

A free point of the plane has two coordinates (r, φ). The trajectory $r = r_0 e^{b\varphi}$ is one relation between them, so one parameter (for instance φ, or r, or the arc length s) fixes the position on the curve. The speed law $|\mathbf{v}| = v_0$ then fixes how that parameter changes with time: nothing is left free, which is why this is a pure kinematics problem and $\mathbf{F}$ is an output.

Step 2. Velocity in the polar basis

The polar basis turns with φ; its relation to the inertial basis is

$$\begin{pmatrix}\mathbf{e}_r & \mathbf{e}_\varphi\end{pmatrix} = [R]\begin{pmatrix}\mathbf{i} & \mathbf{j}\end{pmatrix}, \qquad [R] = \begin{pmatrix}\cos\varphi & \sin\varphi \\ -\sin\varphi & \cos\varphi\end{pmatrix}$$

with $[R][R]^T = \begin{pmatrix}\cos^2\varphi+\sin^2\varphi & -\cos\varphi\sin\varphi+\sin\varphi\cos\varphi \\ -\sin\varphi\cos\varphi+\cos\varphi\sin\varphi & \sin^2\varphi+\cos^2\varphi\end{pmatrix} = I$ and $\det[R] = \cos^2\varphi+\sin^2\varphi = +1$ (same orientation as $\{\mathbf{i},\mathbf{j}\}$). Differentiating the rows with the chain rule gives $\dot{\mathbf{e}}_r = \dot\varphi\,\mathbf{e}_\varphi$ and $\dot{\mathbf{e}}_\varphi = -\dot\varphi\,\mathbf{e}_r$.

With $\dot{\mathbf{e}}_r = \dot\varphi\,\mathbf{e}_\varphi$ and $\dot{\mathbf{e}}_\varphi = -\dot\varphi\,\mathbf{e}_r$ (Notes §2.6), differentiating $\mathbf{r} = r\,\mathbf{e}_r$ with the product rule:

$$\mathbf{v} = \dot r\,\mathbf{e}_r + r\,\dot{\mathbf{e}}_r = \dot r\,\mathbf{e}_r + r\dot\varphi\,\mathbf{e}_\varphi$$

Step 3. The trajectory relates $\dot r$ and $\dot\varphi$

By the chain rule $\frac{d}{dt}r(\varphi(t)) = \frac{dr}{d\varphi}\dot\varphi$, with $\frac{dr}{d\varphi} = b\,r_0 e^{b\varphi} = b\,r$:

$$\dot r = b\,r\,\dot\varphi$$

Step 4. The speed fixes $\dot\varphi$

$$|\mathbf{v}|^2 = \dot r^2 + r^2\dot\varphi^2 = b^2r^2\dot\varphi^2 + r^2\dot\varphi^2 = (1+b^2)\,r^2\dot\varphi^2 = v_0^2$$

With $\dot\varphi > 0$ (the statement):

$$\dot\varphi = \frac{v_0}{r\sqrt{1+b^2}}, \qquad \dot r = b\,r\dot\varphi = \frac{b\,v_0}{\sqrt{1+b^2}}$$

The radial velocity is **constant** although $\dot\varphi$ is not: $\dot\varphi$ decreases as $1/r$.

Step 5. The angle with the radius

$\mathbf{v} = v_0\,\mathbf{e}_t$, so $\cos\alpha = \mathbf{e}_r\cdot\mathbf{e}_t = \dot r/v_0$ and $\sin\alpha = \mathbf{e}_\varphi\cdot\mathbf{e}_t = r\dot\varphi/v_0$:

$$\tan\alpha = \frac{r\dot\varphi}{\dot r} = \frac{r\dot\varphi}{b\,r\dot\varphi} = \frac{1}{b}$$

independent of the point: the logarithmic spiral cuts every radius at the same angle (it is the *equiangular* spiral).

Step 6. Numbers

$k = 5/4$: $\dot r = \frac{(3/4)\cdot 10}{5/4} = 6$ m/s, $r\dot\varphi = \frac{v_0}{k} = 8$ m/s (check: $6^2+8^2 = 10^2$), $\tan\alpha = 4/3$, $\alpha = 53.13^\circ$.

KEY RESULTS

- One parameter (φ, or r, or s); with the speed imposed, the motion is fully determined.
- $\dot r = b\,r\dot\varphi$, $\dot\varphi = \dfrac{v_0}{r\sqrt{1+b^2}}$, $\dot r = \dfrac{b\,v_0}{\sqrt{1+b^2}}$ **constant**.
- $\tan\alpha = \dfrac{r\dot\varphi}{\dot r} = \dfrac{1}{b}$; numbers: $\dot r = 6$ m/s, $r\dot\varphi = 8$ m/s, $\alpha = \arctan(4/3) = 53.13^\circ$.

Part (b) · 1.5 points

Time law: $r(t)$, $\varphi(t)$ and the flight time

Integrate the results of (a) to obtain $r(t)$ and $\varphi(t)$, and check that they lie on the spiral. Compute the time needed to go from $r_1 = 20$ m to $r_2 = 50$ m and the angle turned.

Step 1. Radial law

$\dot r = c$ with $c = b v_0/k$ constant. Integrating from 0 to t with Barrow's rule, $r(t) - r(0) = \int_0^t c\,d\tau = c\,t$:

$$r(t) = r_0 + \frac{b\,v_0}{\sqrt{1+b^2}}\,t$$

Step 2. Angular law (quadrature)

$$\varphi(t) = \varphi(0) + \int_0^t \frac{v_0}{k\,(r_0 + c\tau)}\,d\tau$$

Substitution $u = r_0 + c\tau$, $du = c\,d\tau$; the limits $\tau = 0 \mapsto u = r_0$ and $\tau = t \mapsto u = r(t)$:

$$\varphi(t) = \frac{v_0}{k\,c}\int_{r_0}^{r(t)} \frac{du}{u} = \frac{v_0}{k\,c}\Big[\ln u\Big]_{r_0}^{r(t)} = \frac{v_0}{k\,c}\big(\ln r(t) - \ln r_0\big) = \frac{1}{b}\ln\frac{r(t)}{r_0}$$

since $\frac{v_0}{kc} = \frac{v_0}{k}\cdot\frac{k}{b v_0} = \frac{1}{b}$. Inverting, $r(t) = r_0 e^{b\varphi(t)}$: the time law lies on the spiral, as it must.

Step 3. Flight time and angle turned

$r_2 - r_1 = c\,t_{12}$:

$$t_{12} = \frac{(r_2 - r_1)\sqrt{1+b^2}}{b\,v_0} = \frac{30\cdot(5/4)}{(3/4)\cdot 10} = 5\ \text{s}$$

$$\Delta\varphi = \frac{1}{b}\ln\frac{r_2}{r_1} = \frac{4}{3}\ln 2.5 = 1.2217\ \text{rad} = 70.00^\circ$$

KEY RESULTS

- $r(t) = r_0 + \dfrac{b\,v_0}{\sqrt{1+b^2}}\,t,\ \varphi(t) = \dfrac{1}{b}\ln\dfrac{r(t)}{r_0}.$
- $t_{12} = \dfrac{(r_2 - r_1)\sqrt{1+b^2}}{b\,v_0} = 5\ \text{s};\ \Delta\varphi = \dfrac{\ln(5/2)}{3/4} = 1.2217\ \text{rad} = 70.00^\circ.$

Part (c) · 1.5 points

Arc length and the intrinsic basis

Obtain the arc length $s(r)$ measured from r_0, and check it against $v_0\,t_{12}$. Write $\mathbf{e}_t$ and the principal normal $\mathbf{e}_n$ in the polar basis.

Step 1. Arc length

From $\varphi(r) = \frac{1}{b}\ln\frac{r}{r_0}$, $\frac{d\varphi}{dr} = \frac{1}{br}$, so $r\,d\varphi = \frac{dr}{b}$ and

$$ds = \sqrt{dr^2 + r^2 d\varphi^2} = \sqrt{1 + \frac{1}{b^2}}\,dr = \frac{\sqrt{1+b^2}}{b}\,dr$$

Integrating from r_0 to r (Barrow):

$$s(r) = \int_{r_0}^{r} \frac{k}{b}\,dr' = \frac{k}{b}\Big[r'\Big]_{r_0}^{r} = \frac{\sqrt{1+b^2}}{b}\,(r - r_0)$$

From 20 m to 50 m: $s = \frac{5/4}{3/4}\cdot 30 = 50$ m, equal to $v_0 t_{12} = 10\cdot 5 = 50$ m, as it must at constant speed.

Step 2. Unit tangent

$\mathbf{e}_t = \mathbf{v}/v_0$ with the components of (a):

$$\mathbf{e}_t = \frac{\dot r}{v_0}\,\mathbf{e}_r + \frac{r\dot\varphi}{v_0}\,\mathbf{e}_\varphi = \frac{b\,\mathbf{e}_r + \mathbf{e}_\varphi}{\sqrt{1+b^2}} = \cos\alpha\,\mathbf{e}_r + \sin\alpha\,\mathbf{e}_\varphi$$

Step 3. Principal normal

The components of $\mathbf{e}_t$ are constant, so only the basis vectors change. With the product rule and $\dot{\mathbf{e}}_r = \dot\varphi\,\mathbf{e}_\varphi$, $\dot{\mathbf{e}}_\varphi = -\dot\varphi\,\mathbf{e}_r$:

$$\dot{\mathbf{e}}_t = \frac{b\,\dot\varphi\,\mathbf{e}_\varphi - \dot\varphi\,\mathbf{e}_r}{k} = \dot\varphi\,\frac{-\mathbf{e}_r + b\,\mathbf{e}_\varphi}{k}$$

By the chain rule $\frac{d\mathbf{e}_t}{ds} = \frac{\dot{\mathbf{e}}_t}{\dot s}$ with $\dot s = v_0$, and the principal normal is the unit vector of $d\mathbf{e}_t/ds$ (Notes §2.4). The vector $(-\mathbf{e}_r + b\,\mathbf{e}_\varphi)/k$ has modulus $\sqrt{1+b^2}/k = 1$, so

$$\mathbf{e}_n = \frac{-\mathbf{e}_r + b\,\mathbf{e}_\varphi}{\sqrt{1+b^2}}, \qquad \left|\frac{d\mathbf{e}_t}{ds}\right| = \frac{\dot\varphi}{v_0}$$

Check: $\mathbf{e}_t \cdot \mathbf{e}_n = (-b+b)/k^2 = 0$, and $\mathbf{e}_n$ is $\mathbf{e}_t$ turned $+90^\circ$ (to the left of the motion, towards the concave side and partly towards O).

KEY RESULTS

- $s(r) = \dfrac{\sqrt{1+b^2}}{b}\,(r - r_0)$; from 20 m to 50 m: $s = 50\ \text{m} = v_0 t_{12}$.
- $\mathbf{e}_t = \dfrac{b\,\mathbf{e}_r + \mathbf{e}_\varphi}{\sqrt{1+b^2}}$, $\mathbf{e}_n = \dfrac{-\mathbf{e}_r + b\,\mathbf{e}_\varphi}{\sqrt{1+b^2}}$ (towards the concave side, to the left of the motion).

Part (d) · 2 points

Acceleration, radius of curvature

Compute the acceleration in polar components. Obtain the radius of curvature ρ and show that $\mathbf{a} = \dfrac{v_0^2}{\rho}\,\mathbf{e}_n$. Numbers at $r = 20$ m and at $r = 50$ m.

Step 1. Polar acceleration

Differentiating $\mathbf{v} = \dot r\,\mathbf{e}_r + r\dot\varphi\,\mathbf{e}_\varphi$ with the product rule (Notes §2.6):

$$\mathbf{a} = (\ddot r - r\dot\varphi^2)\,\mathbf{e}_r + (r\ddot\varphi + 2\dot r\dot\varphi)\,\mathbf{e}_\varphi$$

Step 2. The derivatives needed

$\ddot r = 0$ because $\dot r$ is constant. From $\dot\varphi = \frac{v_0}{k}\, r^{-1}$, with the chain rule $\frac{d}{dt} r^{-1} = -r^{-2}\dot r$:

$$\ddot\varphi = -\frac{v_0}{k}\,\frac{\dot r}{r^2} = -\frac{\dot r\,\dot\varphi}{r}$$

Step 3. Components

$$a_r = 0 - r\dot\varphi^2 = -r\,\frac{v_0^2}{r^2 k^2} = -\frac{v_0^2}{r(1+b^2)}$$

$$a_\varphi = r\Big(-\frac{\dot r\dot\varphi}{r}\Big) + 2\dot r\dot\varphi = \dot r\dot\varphi = \frac{b v_0}{k}\cdot\frac{v_0}{rk} = \frac{b\,v_0^2}{r(1+b^2)}$$

Step 4. Intrinsic form and radius of curvature

The curvature is $1/\rho = |d\mathbf{e}_t/ds| = \dot\varphi/v_0 = \frac{1}{rk}$ (part (c)), so $\rho = r\sqrt{1+b^2}$. In the intrinsic basis (Notes §2.5) $\mathbf{a} = \dot v\,\mathbf{e}_t + \frac{v^2}{\rho}\,\mathbf{e}_n$ with $\dot v = 0$:

$$\mathbf{a} = \frac{v_0^2}{r\sqrt{1+b^2}}\,\mathbf{e}_n = \frac{v_0^2}{r\sqrt{1+b^2}}\cdot\frac{-\mathbf{e}_r + b\,\mathbf{e}_\varphi}{\sqrt{1+b^2}} = -\frac{v_0^2}{r(1+b^2)}\,\mathbf{e}_r + \frac{b\,v_0^2}{r(1+b^2)}\,\mathbf{e}_\varphi$$

which coincides with Step 3: the two routes agree. Check: $\sqrt{a_r^2 + a_\varphi^2} = \frac{v_0^2\sqrt{1+b^2}}{r(1+b^2)} = \frac{v_0^2}{\rho}$.

Step 5. Numbers

At $r = 20$ m: $\rho = 20\cdot\frac{5}{4} = 25$ m, $|\mathbf{a}| = \frac{100}{25} = 4$ m/s^2, $a_r = -\frac{100}{20\cdot 25/16} = -3.2$ m/s^2, $a_\varphi = \frac{3}{4}\cdot 3.2 = 2.4$ m/s^2 (check: $3.2^2 + 2.4^2 = 16$). At $r = 50$ m: $\rho = 62.5$ m and $|\mathbf{a}| = 1.6$ m/s^2: the spiral opens up and the turn becomes gentler.

KEY RESULTS

- $a_r = -\dfrac{v_0^2}{r(1+b^2)},\ a_\varphi = \dfrac{b\,v_0^2}{r(1+b^2)},\ \rho = r\sqrt{1+b^2},\ |\mathbf{a}| = \dfrac{v_0^2}{r\sqrt{1+b^2}}.$
- At $r = 20$ m: $\rho = 25$ m, $a_r = -3.2$ m/s^2, $a_\varphi = 2.4$ m/s^2, $|\mathbf{a}| = 4$ m/s^2; at $r = 50$ m: $|\mathbf{a}| = 1.6$ m/s^2.

Part (e) · 1.5 points

Force, torque and angular momentum about O

Find the horizontal force $\mathbf{F}$ needed at $r = 20$ m. Is it a central force about O? Compute the torque $\mathbf{M}_O$ and the angular momentum $\mathbf{H}_O(t)$, and verify the angular momentum theorem between r_1 and r_2.

Step 1. The force

Newton's second law in the inertial frame, $\mathbf{F} = m\mathbf{a}$, with $m = 2$ kg and the numbers of (d):

$$\mathbf{F} = 2\,(-3.2\,\mathbf{e}_r + 2.4\,\mathbf{e}_\varphi) = -6.4\,\mathbf{e}_r + 4.8\,\mathbf{e}_\varphi \ \text{N}, \qquad |\mathbf{F}| = 8 \ \text{N}$$

In general $\mathbf{F} = \frac{m v_0^2}{\rho}\mathbf{e}_n$. Since $\mathbf{e}_n \cdot (-\mathbf{e}_r) = 1/k$, the angle γ between $\mathbf{F}$ and $-\mathbf{e}_r$ has $\cos\gamma = 1/\sqrt{1+b^2}$, i.e. $\tan\gamma = b$: $\gamma = \arctan(3/4) = 36.87^\circ$. The force is **not** central: $F_\varphi \neq 0$ at every point.

Step 2. Torque

$\mathbf{M}_O = \mathbf{r} \times \mathbf{F} = r\,\mathbf{e}_r \times (F_r\mathbf{e}_r + F_\varphi \mathbf{e}_\varphi) = r\,F_\varphi\,\mathbf{k}$ (since $\mathbf{e}_r \times \mathbf{e}_\varphi = \mathbf{k}$):

$$\mathbf{M}_O = m\,r\,a_\varphi\,\mathbf{k} = m\,r\,\frac{b\,v_0^2}{r(1+b^2)}\,\mathbf{k} = \frac{m\,b\,v_0^2}{1+b^2}\,\mathbf{k} = \frac{2\cdot 0.75\cdot 100}{1.5625}\,\mathbf{k} = 96\,\mathbf{k}\ \text{N m}$$

constant along the whole flight, although $\mathbf{F}$ changes.

Step 3. Angular momentum

$\mathbf{r}\times\mathbf{v} = r\,\mathbf{e}_r \times (\dot r\,\mathbf{e}_r + r\dot\varphi\,\mathbf{e}_\varphi) = r^2\dot\varphi\,\mathbf{k}$, so with $\dot\varphi$ of (a):

$$\mathbf{H}_O = m\,r^2\dot\varphi\,\mathbf{k} = \frac{m\,r(t)\,v_0}{\sqrt{1+b^2}}\,\mathbf{k}$$

At $r = 20$ m: $H_O = \frac{2\cdot 20\cdot 10}{1.25} = 320$ kg m^2/s; at $r = 50$ m: 800 kg m^2/s.

Step 4. Angular momentum theorem

About the fixed point O, $\dot{\mathbf{H}}_O = \mathbf{M}_O$ (Notes §4.8). By the chain rule $\frac{d}{dt} H_O = \frac{m v_0}{k} \dot{r} = \frac{m v_0}{k} \cdot \frac{b v_0}{k} = \frac{m\, b\, v_0^2}{1+b^2} = M_O$. Integrating from 0 to t_{12} (Barrow): $H_O(t_{12}) - H_O(0) = \int_0^{t_{12}} M_O \, dt = 96 \cdot 5 = 480$ kg m^2/s, and indeed $800 - 320 = 480$. The angular momentum is **not** conserved, and the areal velocity $\frac{1}{2} r^2 \dot{\varphi}$ grows: Kepler's second law does not hold for this force.

KEY RESULTS

- At $r = 20$ m: $F_r = -6.4$ N, $F_\varphi = 4.8$ N, $|\mathbf{F}| = 8$ N; it makes $\arctan b = 36.87^\circ$ with $-\mathbf{e}_r$: **not central**.
- $\mathbf{M}_O = m\, r\, a_\varphi\, \mathbf{k} = \dfrac{m\, b\, v_0^2}{1+b^2}\, \mathbf{k} = 96$ N m (constant); $\mathbf{H}_O = \dfrac{m\, r(t)\, v_0}{\sqrt{1+b^2}}\, \mathbf{k}$: 320 kg m^2/s at 20 m, 800 kg m^2/s at 50 m; $\Delta H_O = 96 \cdot 5 = 480$ kg m^2/s.

Part (f) · 1.5 points

Power, kinetic energy and the trap

Compute the power of $\mathbf{F}$ and the work it does between r_1 and r_2, and check the result with the kinetic energy theorem. A student argues: "the speed is constant, so the drone is in equilibrium and the net force is zero; and since $\mathbf{F}$ does no work, it must be a central (or conservative) force". Discuss.

Step 1. Power and work

The power of $\mathbf{F}$ is $P = \mathbf{F} \cdot \mathbf{v}$ (Notes §4.5). With $\mathbf{F} = \frac{m v_0^2}{\rho} \mathbf{e}_n$ and $\mathbf{v} = v_0 \mathbf{e}_t$:

$$P = \frac{m\, v_0^3}{\rho} \, (\mathbf{e}_n \cdot \mathbf{e}_t) = 0$$

In polar components the same result is $m(a_r \dot{r} + a_\varphi r \dot{\varphi}) = m \frac{v_0^2}{r k^2} \left(- \frac{b v_0}{k} + b \, \frac{v_0}{k} \right) = 0$. The work between r_1 and r_2 is $W_{12} = \int_0^{t_{12}} P \, dt = 0$.

Step 2. Kinetic energy theorem

$\frac{d}{dt} \left(\frac{1}{2} m \, \mathbf{v} \cdot \mathbf{v} \right) = m \, \mathbf{a} \cdot \mathbf{v} = P = 0$, consistent with $|\mathbf{v}| = v_0$: $E_{kin} = \frac{1}{2} \cdot 2 \cdot 10^2 = 100$ J at every instant.

Step 3. The trap

- "Constant speed means zero force" confuses the speed $|\mathbf{v}|$ with the velocity vector. The direction of $\mathbf{v}$ turns with angular rate $\dot{\varphi}$ (since α is constant, $\mathbf{e}_t$ turns exactly as $\mathbf{e}_r$), so $\mathbf{a} = \frac{v_0^2}{\rho} \mathbf{e}_n \neq \mathbf{0}$: at 20 m the autopilot must supply 8 N.
- "No work, so central" is false: zero power only means $\mathbf{F} \perp \mathbf{v}$ at each instant. Here $\mathbf{F}$ is perpendicular to $\mathbf{v}$ but makes 36.87° with the radius, so it has a moment about O and $\mathbf{H}_O$ grows by 480 kg m^2/s.
- A force perpendicular to the velocity is not conservative in general either: being workless along **this** path says nothing about the work along other paths. What matters for the energy budget is that it does no work along the actual motion.

KEY RESULTS

- $P = \mathbf{F}\cdot\mathbf{v} = \frac{mv_0^2}{\rho}\,v_0\,(\mathbf{e}_n\cdot\mathbf{e}_t) = 0$; $W_{12} = 0$; $E_{kin} = \frac{1}{2}mv_0^2 = 100$ J at every instant.
- Constant **speed** is not constant **velocity**: $|\mathbf{F}| = mv_0^2/\rho \neq 0$. Zero power only says $\mathbf{F}\perp\mathbf{v}$; the force is neither zero nor central ($\mathbf{M}_O \neq \mathbf{0}$).

Interpretation and checks

- **Dimensional check:** $\dot r = bv_0/k$ and $r\dot\varphi$ are m/s; $\rho = rk$ is m; $a = v_0^2/\rho$ is $\mathrm{m^2 s^{-2}/m}$ = $\mathrm{m\ s^{-2}}$; $M_O = mbv_0^2/k^2$ is $\mathrm{kg\ m^2 s^{-2}}$ = N m; $H_O = mrv_0/k$ is $\mathrm{kg\ m^2/s}$; $t_{12} = (r_2 - r_1)k/(bv_0)$ is s.
- **Limiting cases:** $b \to 0$ gives $r = r_0$, $\dot r = 0$, $\alpha = 90^\circ$, $\rho = r_0$, $a_\varphi = 0$: uniform circular motion with a central force $-mv_0^2/r_0\,\mathbf{e}_r$ and $\mathbf{H}_O$ constant. $b \to \infty$ gives $\alpha \to 0$: an almost radial straight flight with $\rho \to \infty$ and $\mathbf{a} \to \mathbf{0}$.
- **Flying inwards:** with φ decreasing, r decreases linearly and reaches O after the finite time $r_0k/(bv_0) = 10/3$ s and the finite length $r_0k/b = 100/3$ m, while $\varphi = \frac{1}{b}\ln(r/r_0) \to -\infty$: infinitely many turns in a finite time, with $\dot\varphi$ and $|\mathbf{a}|$ growing without bound. A real autopilot cannot follow the spiral to the end.
- **Common mistakes in the exam:** (1) setting $a_r = 0$ because $\dot r$ is constant (the centripetal term $-r\dot\varphi^2$ remains); (2) taking $\dot\varphi$ constant and dropping $r\ddot\varphi$ in a_φ; (3) writing $\rho = r$ as for a circle; (4) concluding from $P = 0$ that $\mathbf{F} = \mathbf{0}$ or that $\mathbf{H}_O$ is conserved.

Problem · Unit 5 · 10 points · 60 min · High difficulty / trap

P15 — Relative orbital motion near a target on a circular orbit: Clohessy-Wiltshire equations, drift and a counter-intuitive rendezvous

*A target T moves on a circular orbit of radius R_T around the Earth, whose centre O is the origin of the inertial frame S_0 and whose gravitational parameter is μ (the Earth's gravity on a mass m at $\mathbf{r}$ is $-m\mu\,\mathbf{r}/|\mathbf{r}|^3$). The target's angular velocity is constant, $n\,\mathbf{k}$, with the **mean motion** $n = \sqrt{\mu/R_T^3}$ ($\mathbf{k}$ normal to the orbital plane). A chaser P of mass m, subject only to the Earth's gravity, moves in the same plane close to T.*

Use the frame $S_1 = \{T; \mathbf{e}_x, \mathbf{e}_y, \mathbf{k}\}$ attached to the target: $\mathbf{e}_x$ radial, pointing away from O; $\mathbf{e}_y = \mathbf{k} \times \mathbf{e}_x$ along the target's velocity (along-track); S_1 rotates with $\boldsymbol{\omega}_{10} = n\,\mathbf{k}$. The relative position of the chaser is $\boldsymbol{\rho} = x\,\mathbf{e}_x + y\,\mathbf{e}_y$ with $|\boldsymbol{\rho}| \ll R_T$. Initial conditions at $t = 0$: x_0, y_0, $\dot{x}_0$, $\dot{y}_0$ (relative to S_1).

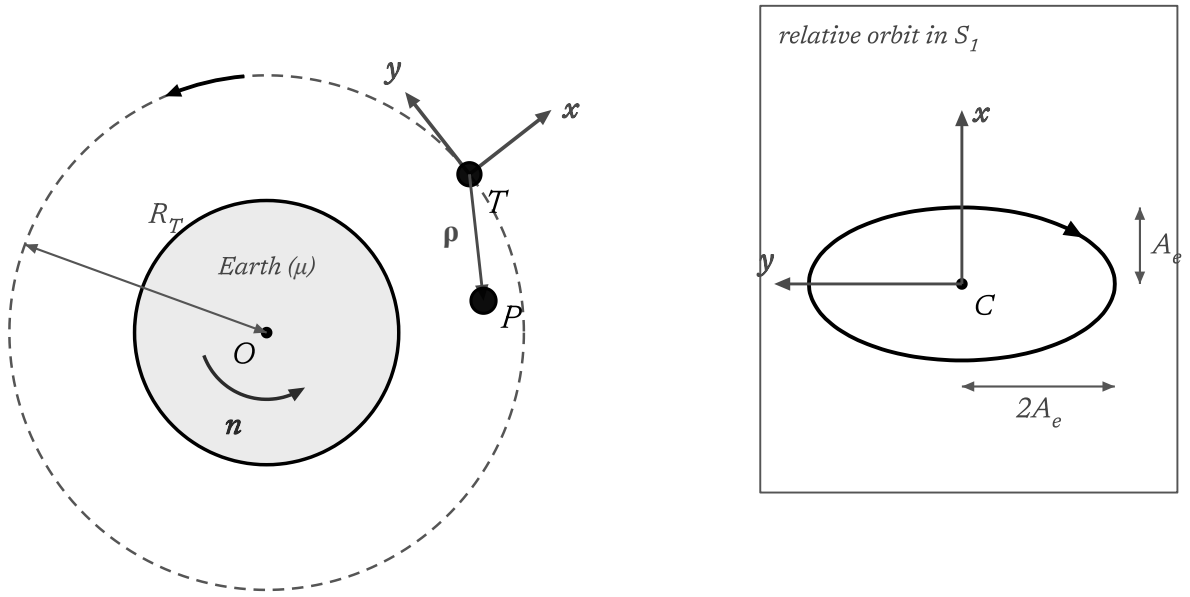


Figure 1. Target T on a circular orbit of radius R_T with mean motion n; rotating frame $S_1 = \{T; \mathbf{e}_x, \mathbf{e}_y\}$ (x radial outwards, y along-track); chaser P at $\boldsymbol{\rho} = x\,\mathbf{e}_x + y\,\mathbf{e}_y$ (distance exaggerated). Inset: bounded relative orbit ($\dot{y}_0 = -2nx_0$), a $2{:}1$ ellipse about its centre C, run in the sense of the arrow.

Setup

- Data: μ, R_T, $n = \sqrt{\mu/R_T^3}$, the initial conditions. Numerical part (f): $\mu = 3.986 \times 10^{14}$ m^3/s^2, $R_T = 6778$ km (low Earth orbit), chaser 1 km behind the target ($x_0 = 0$, $y_0 = -1000$ m).
- Hypotheses: point masses, planar motion in the orbital plane, S_0 inertial, no drag, no other bodies; the target's orbit is not affected by the chaser; terms of order $|\boldsymbol{\rho}|^2/R_T^2$ are neglected in the gravity difference.
- The frame S_1 has a uniformly rotating basis ($\dot{\boldsymbol{\omega}}_{10} = \mathbf{0}$) and an accelerated origin ($\mathbf{a}_T = -n^2 R_T\,\mathbf{e}_x$).

Part (a) · 1.5 points

Degrees of freedom, frame and relative kinematics

How many degrees of freedom has P in this planar problem? Write the absolute acceleration of P in the basis of S_1 as the sum of the acceleration of T, the relative acceleration, the Coriolis and the centripetal terms, and list the inertia forces that act on P in S_1.

Step 1. Count

P is free in the orbital plane: 2 degrees of freedom, the coordinates (x, y) in S_1. There are no constraints and no reactions; the only force is gravity.

Step 2. Motion of the frame

T moves on a circle of radius R_T with constant angular velocity $n\mathbf{k}$, so its acceleration is purely centripetal: $\mathbf{a}_T = -n^2 R_T\,\mathbf{e}_x$. The basis of S_1 turns with $\boldsymbol{\omega}_{10} = n\mathbf{k}$, and $\dot{\boldsymbol{\omega}}_{10} = \mathbf{0}$.

Change of basis: take the inertial basis $\{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$ of S_0 with $\mathbf{i}$ along $\overrightarrow{OT}$ at $t = 0$. Then $\mathbf{e}_x = \cos nt\,\mathbf{i} + \sin nt\,\mathbf{j}$, $\mathbf{e}_y = \mathbf{k} \times \mathbf{e}_x = -\sin nt\,\mathbf{i} + \cos nt\,\mathbf{j}$, and

$$\begin{pmatrix} \mathbf{e}_x \\ \mathbf{e}_y \\ \mathbf{k} \end{pmatrix} = [R] \begin{pmatrix} \mathbf{i} \\ \mathbf{j} \\ \mathbf{k} \end{pmatrix}, \qquad [R] = \begin{pmatrix} \cos nt & \sin nt & 0 \\ -\sin nt & \cos nt & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

The rows have length $\cos^2 nt + \sin^2 nt = 1$, rows 1 and 2 are orthogonal ($-\cos nt \sin nt + \sin nt \cos nt = 0$) and row 3 is $\mathbf{k}$, so $[R][R]^T = I$; $\det[R] = \cos^2 nt + \sin^2 nt = +1$. Differentiating, $\dot{\mathbf{e}}_x = n\,\mathbf{e}_y = n\mathbf{k} \times \mathbf{e}_x$ and $\dot{\mathbf{e}}_y = -n\,\mathbf{e}_x = n\mathbf{k} \times \mathbf{e}_y$ (Poisson), which confirms $\boldsymbol{\omega}_{10} = n\,\mathbf{k}$.

Step 3. Relative kinematics

In S_1: $\mathbf{v}_{rel} = \dot{x}\,\mathbf{e}_x + \dot{y}\,\mathbf{e}_y$, $\mathbf{a}_{rel} = \ddot{x}\,\mathbf{e}_x + \ddot{y}\,\mathbf{e}_y$. With $\mathbf{k} \times \mathbf{e}_x = \mathbf{e}_y$, $\mathbf{k} \times \mathbf{e}_y = -\mathbf{e}_x$:

$$2\boldsymbol{\omega} \times \mathbf{v}_{rel} = 2n\big(\dot{x}\,\mathbf{e}_y - \dot{y}\,\mathbf{e}_x\big), \qquad \boldsymbol{\omega} \times \boldsymbol{\rho} = n\big(x\,\mathbf{e}_y - y\,\mathbf{e}_x\big), \qquad \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \boldsymbol{\rho}) = -n^2\big(x\,\mathbf{e}_x + y\,\mathbf{e}_y\big)$$

The composition of accelerations (Notes §3.2), with the Euler term zero, gives

$$\mathbf{a}_P = \big(-n^2 R_T + \ddot{x} - 2n\dot{y} - n^2 x \big)\,\mathbf{e}_x + \big(\ddot{y} + 2n\dot{x} - n^2 y \big)\,\mathbf{e}_y$$

Step 4. Inertia forces

Written as $m\mathbf{a}_{rel} = \mathbf{F} + \mathbf{F}_{in}$ (Notes §4.9), the inertia force is $\mathbf{F}_{in} = -m\mathbf{a}_T - 2m\boldsymbol{\omega} \times \mathbf{v}_{rel} - m\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \boldsymbol{\rho})$: the dragging force of the origin $mn^2 R_T\,\mathbf{e}_x$, the centrifugal force $mn^2\boldsymbol{\rho}$ (about T) and the Coriolis force $-2mn\,\mathbf{k} \times \mathbf{v}_{rel}$. The Coriolis force is perpendicular to $\mathbf{v}_{rel}$ and does no work in S_1.

KEY RESULTS

- 2 degrees of freedom, (x, y); $\boldsymbol{\omega}_{10} = n\mathbf{k}$ constant, $\mathbf{a}_T = -n^2 R_T\,\mathbf{e}_x$.
- $\mathbf{a}_P = \big(-n^2 R_T + \ddot{x} - 2n\dot{y} - n^2 x \big)\mathbf{e}_x + \big(\ddot{y} + 2n\dot{x} - n^2 y \big)\mathbf{e}_y$.
- Inertia forces in S_1: $-m\mathbf{a}_T = mn^2 R_T\,\mathbf{e}_x$, centrifugal $mn^2\boldsymbol{\rho}$, Coriolis $-2mn\mathbf{k} \times \mathbf{v}_{rel}$; no Euler force.

Part (b) · 2 points

Linearised gravity and the equations of motion

Expand the gravity on P to first order in $|\boldsymbol{\rho}|/R_T$ and obtain the two equations of relative motion. Explain why the terms in R_T disappear.

Step 1. The exact gravity

$\mathbf{r}_P = \overrightarrow{OT} + \boldsymbol{\rho} = (R_T + x)\,\mathbf{e}_x + y\,\mathbf{e}_y$ and $|\mathbf{r}_P|^2 = R_T^2\big(1 + \frac{2x}{R_T} + \frac{x^2+y^2}{R_T^2}\big)$.

Step 2. First-order expansion

Keeping only first-order terms, $u = 2x/R_T$ and $(1+u)^{-3/2} = 1 - \frac{3}{2}u + O(u^2)$:

$$\frac{1}{|\mathbf{r}_P|^3} \approx \frac{1}{R_T^3}\Big(1 - \frac{3x}{R_T}\Big)$$

$$-\frac{\mu\,\mathbf{r}_P}{|\mathbf{r}_P|^3} \approx -\frac{\mu}{R_T^3}\Big(1 - \frac{3x}{R_T}\Big)\big[(R_T + x)\,\mathbf{e}_x + y\,\mathbf{e}_y\big] \approx -\frac{\mu}{R_T^2}\,\mathbf{e}_x + \frac{\mu}{R_T^3}\big(2x\,\mathbf{e}_x - y\,\mathbf{e}_y\big)$$

(the product $x \cdot 3x/R_T$ and $y \cdot 3x/R_T$ are second order and dropped). The circular orbit of T gives $n^2 R_T = \mu/R_T^2$ (Notes §8.3), i.e. $\mu/R_T^3 = n^2$.

Step 3. Newton in S_0, projected on the basis of S_1

$m\mathbf{a}_P = -m\mu\mathbf{r}_P/|\mathbf{r}_P|^3$ with $\mathbf{a}_P$ of (a):

$$\mathbf{e}_x: \quad -n^2R_T + \ddot{x} - 2n\dot{y} - n^2x = -n^2R_T + 2n^2x$$

$$\mathbf{e}_y: \quad \ddot{y} + 2n\dot{x} - n^2y = -n^2y$$

The terms $-n^2R_T$ cancel: the gravity at T is exactly what keeps T (and S_1) on its orbit, so only the **difference** of gravity between P and T (the tidal term) drives the relative motion. Likewise the centrifugal term $-n^2y$ cancels the along-track gravity component. Hence

$$\ddot{x} - 2n\dot{y} - 3n^2x = 0, \qquad \ddot{y} + 2n\dot{x} = 0$$

KEY RESULTS

- $-\mu\,\mathbf{r}_P/|\mathbf{r}_P|^3 \approx -n^2R_T\,\mathbf{e}_x + n^2\big(2x\,\mathbf{e}_x - y\,\mathbf{e}_y\big)$ (using $\mu = n^2R_T^3$).
- $\ddot{x} - 2n\dot{y} - 3n^2x = 0, \qquad \ddot{y} + 2n\dot{x} = 0.$

Part (c) · 2 points

Closed-form solution

Solve the equations of (b) for the initial conditions $x_0, y_0, \dot{x}_0, \dot{y}_0$.

Step 1. First integral

$\ddot{y} + 2n\dot{x} = \frac{d}{dt}\big(\dot{y} + 2nx\big) = 0$. Integrating from 0 to t (Barrow): $\big[\dot{y} + 2nx\big]_0^t = 0$, so

$$\dot{y} + 2nx = C, \qquad C = \dot{y}_0 + 2nx_0$$

Step 2. Radial equation

Substitute $\dot{y} = C - 2nx$ in the radial equation: $\ddot{x} - 2n(C - 2nx) - 3n^2 x = 0$, i.e.

$$\ddot{x} + n^2 x = 2nC$$

A particular solution is the constant $x_p = 2C/n$; the homogeneous solution is $A\cos nt + B\sin nt$. The initial conditions give $x(0) = 2C/n + A = x_0$ and $\dot{x}(0) = nB = \dot{x}_0$:

$$A = x_0 - \frac{2C}{n} = -3x_0 - \frac{2\dot{y}_0}{n}, \qquad B = \frac{\dot{x}_0}{n}$$

$$x(t) = 4x_0 + \frac{2\dot{y}_0}{n} - \left(3x_0 + \frac{2\dot{y}_0}{n}\right)\cos nt + \frac{\dot{x}_0}{n}\sin nt$$

Step 3. Along-track position

From Step 1, $\dot{y} = C - 2nx = -3C - 2nA\cos nt - 2nB\sin nt$. Integrating from 0 to t with the antiderivatives $\int \cos n\tau\, d\tau = \frac{\sin n\tau}{n}$ and $\int \sin n\tau\, d\tau = -\frac{\cos n\tau}{n}$ (chain rule: $\frac{d}{d\tau}\sin n\tau = n\cos n\tau$) and Barrow's rule:

$$y(t) - y_0 = -3Ct - 2nA\left[\frac{\sin n\tau}{n}\right]_0^t - 2nB\left[-\frac{\cos n\tau}{n}\right]_0^t = -3Ct - 2A\sin nt + 2B(\cos nt - 1)$$

With C, A, B:

$$y(t) = y_0 - (6nx_0 + 3\dot{y}_0)\,t + \left(6x_0 + \frac{4\dot{y}_0}{n}\right)\sin nt + \frac{2\dot{x}_0}{n}(\cos nt - 1)$$

Check: $x(0) = x_0$, $y(0) = y_0$, $\dot{x}(0) = \dot{x}_0$, $\dot{y}(0) = -6nx_0 - 3\dot{y}_0 + 6nx_0 + 4\dot{y}_0 = \dot{y}_0$.

KEY RESULTS

- First integral $\dot{y} + 2nx = C \equiv \dot{y}_0 + 2nx_0$; then $\ddot{x} + n^2 x = 2nC$.
- $x(t) = 4x_0 + \dfrac{2\dot{y}_0}{n} - \left(3x_0 + \dfrac{2\dot{y}_0}{n}\right)\cos nt + \dfrac{\dot{x}_0}{n}\sin nt$.
- $y(t) = y_0 - (6nx_0 + 3\dot{y}_0)\,t + \left(6x_0 + \dfrac{4\dot{y}_0}{n}\right)\sin nt + \dfrac{2\dot{x}_0}{n}(\cos nt - 1)$.

Part (d) · 1.5 points

Secular drift: the trap of the lower orbit

*Find the along-track drift per orbit and the condition for no drift. A chaser is placed 1 km **below** the target on its **own circular orbit** (radius $R_T - 1$ km): does it keep its position below T? Compare with a chaser released 1 km below T at rest in S_1.*

Step 1. Drift per orbit

The periodic terms of $y(t)$ repeat after one orbital period $2\pi/n$, and $x(2\pi/n) = x_0$; only the secular term changes:

$$\Delta y_{orbit} = y\Big(\frac{2\pi}{n}\Big) - y_0 = -(6nx_0 + 3\dot{y}_0)\,\frac{2\pi}{n} = -12\pi x_0 - \frac{6\pi\,\dot{y}_0}{n}$$

It vanishes iff $6nx_0 + 3\dot{y}_0 = 0$:

$$\dot{y}_0 = -2n\,x_0 \quad (C = 0)$$

Step 2. Chaser on its own circular orbit

The circular orbit of radius $R_T + x_0$ has angular rate $n' = \sqrt{\mu/(R_T + x_0)^3} = n\,(1 + x_0/R_T)^{-3/2} \approx n\big(1 - \frac{3x_0}{2R_T}\big)$. Its velocity in S_1 is the absolute velocity minus the transport velocity $n\mathbf{k} \times \mathbf{r}$: $\dot{y}_0 = (R_T + x_0)(n' - n) \approx R_T \cdot n \cdot \big(-\frac{3x_0}{2R_T}\big) = -\frac{3}{2}nx_0$ (first order). Then $C = \frac{1}{2}nx_0$, $A = -3x_0 + 3x_0 = 0$, so $x(t) = x_0$ for all t and

$$y(t) = y_0 - \tfrac{3}{2}n\,x_0\,t, \qquad \Delta y_{orbit} = -3\pi x_0$$

For $x_0 = -1000$ m: $\Delta y_{orbit} = +3000\pi = +9425$ m per orbit. **The trap:** "same circular motion, so it stays below" is wrong. The lower orbit has a shorter period (Kepler), so the chaser moves ahead of the target at $\frac{3}{2}n|x_0|$. The no-drift value $-2nx_0$ is **not** the circular-orbit value $-\frac{3}{2}nx_0$: the drift-free relative motion is the ellipse of (e), not a fixed point.

Step 3. Chaser at rest in S_1

With $\dot{y}_0 = 0$ (and $\dot{x}_0 = 0$): $\Delta y_{orbit} = -12\pi x_0 = +12\,000\pi$ m $= +37.70$ km per orbit, while x oscillates between x_0 and $7x_0$ ($x(t) = 4x_0 - 3x_0 \cos nt$, i.e. down to 7 km below T): a point "at rest" below the target is moving slower than the local circular speed and falls into an eccentric orbit.

KEY RESULTS

- $\Delta y_{orbit} = -\dfrac{2\pi}{n}\left(6nx_0 + 3\dot{y}_0\right) = -12\pi x_0 - \dfrac{6\pi\dot{y}_0}{n}$; no drift iff $\dot{y}_0 = -2nx_0$ (i.e. $C = 0$).
- Own circular orbit: $\dot{y}_0 \approx -\frac{3}{2}nx_0$, $x \equiv x_0$, $\Delta y_{orbit} = -3\pi x_0 = +9.42$ km per orbit for $x_0 = -1$ km: it **moves ahead** of T.
- At rest in S_1 ($\dot{y}_0 = 0$): $\Delta y_{orbit} = -12\pi x_0 = +37.70$ km per orbit.

Part (e) · 1.5 points

Bounded relative orbit

For $\dot{y}_0 = -2nx_0$, show that the relative trajectory is an ellipse centred on the along-track axis, with the along-track semi-axis twice the radial one. Find its centre, its period and its sense of motion.

Step 1. Substitution

With $\dot{y}_0 = -2nx_0$: $4x_0 + 2\dot{y}_0/n = 0$, $3x_0 + 2\dot{y}_0/n = -x_0$, $6nx_0 + 3\dot{y}_0 = 0$ and $6x_0 + 4\dot{y}_0/n = -2x_0$. Then (c) gives

$$x = x_0\cos nt + \frac{\dot{x}_0}{n}\sin nt, \qquad y = y_0 - \frac{2\dot{x}_0}{n} - 2x_0\sin nt + \frac{2\dot{x}_0}{n}\cos nt$$

Step 2. The ellipse

With $A_e\cos\delta = x_0$, $A_e\sin\delta = \dot{x}_0/n$ and the addition formulas, $x = A_e\cos(nt - \delta)$ and $y - y_c = -2\left(x_0\sin nt - \frac{\dot{x}_0}{n}\cos nt\right) = -2A_e\sin(nt - \delta)$. Using $\cos^2 + \sin^2 = 1$:

$$\frac{x^2}{A_e^2} + \frac{(y - y_c)^2}{4A_e^2} = 1$$

an ellipse centred at $(0, y_c)$, with semi-axes A_e (radial) and $2A_e$ (along-track), run once per orbital period $2\pi/n$.

Step 3. Sense of motion

The $\mathbf{k}$-component of $\left(x\,\mathbf{e}_x + (y-y_c)\,\mathbf{e}_y\right) \times \left(\dot{x}\,\mathbf{e}_x + \dot{y}\,\mathbf{e}_y\right)$ is $x\dot{y} - (y-y_c)\dot{x} = A_e\cos\theta\,(-2A_e n\cos\theta) - (-2A_e\sin\theta)(-A_e n\sin\theta) = -2nA_e^2 < 0$ (with $\theta = nt - \delta$): the ellipse is run in the sense of $-\mathbf{k}$. Above the target ($x > 0$) the chaser moves backwards ($\dot{y} < 0$), below it moves forwards: higher means slower.

KEY RESULTS

- $x = x_0\cos nt + \dfrac{\dot{x}_0}{n}\sin nt,\ y = y_c - 2x_0\sin nt + \dfrac{2\dot{x}_0}{n}\cos nt,\ y_c = y_0 - \dfrac{2\dot{x}_0}{n}.$
- $\dfrac{x^2}{A_e^2} + \dfrac{(y-y_c)^2}{4A_e^2} = 1$, $A_e = \sqrt{x_0^2 + \dot{x}_0^2/n^2}$; period $2\pi/n$; run in the sense of $-\mathbf{k}$ (opposite to the orbital motion).

Part (f) · 1.5 points

A low Earth orbit case: two manoeuvres

*Data: $\mu = 3.986\times10^{14}$ m^3/s^2, $R_T = 6778$ km; the chaser is 1 km behind the target, $x_0 = 0$, $y_0 = -1000$ m, at rest in S_1, and receives an impulse at $t = 0$. (i) The impulse is radial, $\dot{x}_0 = +0.5$ m/s. Describe the motion and give the position after half an orbit and after one orbit. (ii) Which **along-track** impulse $\dot{y}_0$ (with $\dot{x}_0 = 0$) brings the chaser onto the target after exactly one orbit? Which way does it point, how far does the chaser go from the orbit, and with what relative velocity does it arrive?*

Step 1. Mean motion

$$n = \sqrt{\frac{3.986\times10^{14}}{(6.778\times10^{6})^3}} = 1.1314\times10^{-3}\ \mathrm{s}^{-1}, \qquad \frac{2\pi}{n} = 5553\ \mathrm{s} = 92.6\ \mathrm{min}$$

Step 2. Radial impulse

$x_0 = 0$, $\dot{y}_0 = 0$: $C = \dot{y}_0 + 2nx_0 = 0$, so the motion is the bounded ellipse of (e), with $A_e = \dot{x}_0/n = 0.5/1.1314\times10^{-3} = 441.9$ m:

$$x = 441.9\,\sin nt\ \mathrm{m}, \qquad y = -1000 + 883.9\,(\cos nt - 1)\ \mathrm{m}$$

At half an orbit ($nt = \pi$): $x = 0$, $y = -1000 - 4\cdot441.9 = -2768$ m (the chaser rose and fell behind); centre of the ellipse $y_c = y_0 - 2\dot{x}_0/n = -1884$ m. After one orbit it is back at $(0, -1000)$ m: a radial impulse alone gains nothing.

Step 3. Along-track impulse for a one-orbit rendezvous

With $x_0 = \dot{x}_0 = 0$, (c) gives $x = \frac{2\dot{y}_0}{n}(1 - \cos nt)$ and $y = y_0 - 3\dot{y}_0 t + \frac{4\dot{y}_0}{n}\sin nt$. At $t = 2\pi/n$, $\sin 2\pi = 0$, $\cos 2\pi = 1$: $x = 0$ and

$$0 = y_0 - 3\dot{y}_0\,\frac{2\pi}{n} \implies \dot{y}_0 = \frac{y_0\, n}{6\pi} = \frac{-1000}{3 \cdot 5553} = -0.0600 \text{ m/s}$$

The trap: to catch a target 1 km **ahead**, the chaser must thrust **backwards**. The backward impulse lowers its orbit (x reaches $\frac{4\dot{y}_0}{n} = -212.2$ m at half orbit), and the lower orbit is faster, so it gains $3|\dot{y}_0| \cdot 2\pi/n = 1000$ m in one period. A forward impulse would raise the chaser and make it fall behind.

Step 4. Arrival

At $t = 2\pi/n$: $\dot{x} = 2\dot{y}_0 \sin nt = 0$ and $\dot{y} = -3\dot{y}_0 + 4\dot{y}_0 \cos nt = \dot{y}_0$. The chaser arrives at T with $\mathbf{v}_{rel} = -0.0600\,\mathbf{e}_y$ m/s, so a second impulse of $+0.0600$ m/s along $\mathbf{e}_y$ stops it at the target (total 0.120 m/s). A numerical integration of the full nonlinear two-body equations from the same initial state misses the target by about 3 m (2.8 m) after one orbit, consistent with neglecting terms of order $|\boldsymbol{\rho}|^2/R_T$.

KEY RESULTS

- $n = 1.1314 \times 10^{-3}$ rad/s, period $2\pi/n = 5553$ s $= 92.6$ min.
- (i) $C = 0$: bounded ellipse with $A_e = \dot{x}_0/n = 441.9$ m, centre $y_c = -1884$ m; at half orbit $(x, y) = (0, -2768\text{ m})$; after one orbit back at $(0, -1000\text{ m})$: no gain.
- (ii) $\dot{y}_0 = \dfrac{y_0\, n}{6\pi} = \dfrac{y_0}{3 \cdot 2\pi/n} = -0.0600$ m/s (**backwards**); x reaches $4\dot{y}_0/n = -212.2$ m at half orbit; arrival with $\mathbf{v}_{rel} = \dot{y}_0\,\mathbf{e}_y$, so a braking impulse of $+0.0600$ m/s is needed. A nonlinear two-body integration misses the target by about 3 m (2.8 m).

Interpretation and checks

- **Dimensional check:** n is s^{-1} (μ/R^3 is $\text{m}^3\text{s}^{-2}/\text{m}^3$); every term of $\ddot{x} - 2n\dot{y} - 3n^2x$ is m s^{-2}; $\dot{y}_0/n$ is m, so $x(t)$ and $y(t)$ are m; $\Delta y_{orbit} = -12\pi x_0 - 6\pi\dot{y}_0/n$ is m.
- **Limiting cases:** $\boldsymbol{\rho} = \mathbf{0}$ with zero relative velocity stays at the origin (the target itself). $n \to 0$ (no gravity, non-rotating frame): expanding $\cos nt \approx 1 - \frac{1}{2}n^2t^2$ and $\sin nt \approx nt - \frac{1}{6}n^3t^3$ in (c) gives $x \to x_0 + \dot{x}_0 t$, $y \to y_0 + \dot{y}_0 t$, uniform straight motion, as it must.
- **Physical meaning of C:** $C = \dot{y} + 2nx$ is constant; it measures the difference of orbital period between chaser and target (to first order the chaser's semi-major axis differs from R_T by $2C/n$). Equal periods ($C = 0$) means no drift; that is why the radial impulse of (f)(i) gives a closed loop.
- **Common mistakes in the exam:** (1) forgetting the acceleration of the origin $\mathbf{a}_T$ or the centrifugal term, which leaves an R_T term that does not cancel; (2) expanding $|\mathbf{r}_P|^{-3}$ with the wrong exponent ($1 - 2x/R_T$ instead of $1 - 3x/R_T$); (3) assuming that a chaser below the target on its own circular orbit, or at rest in S_1, keeps its relative position; (4) thrusting towards the target to catch it.

Problem · Unit 3 · 10 points · 60 min · Standard exam level

P16 — Particle on a free wedge: a moving constraint, the reaction, momentum and energy, and the limit of a massless wedge

*A wedge of mass M rests on a smooth horizontal table. Its upper face is a smooth plane inclined at the angle α ($0 < \alpha < \pi/2$) to the horizontal, descending towards $-x$; the top edge is A and the corner of the wedge under A is B. A particle P of mass m is placed on the incline. The inertial frame is $S_0 = \{O; \mathbf{i}, \mathbf{j}\}$ (x horizontal, y vertical upwards, gravity $-g\,\mathbf{j}$). The wedge translates without rotating (it does not tip), and its position is the abscissa X of B. The position of P on the incline is the distance $s = |AP|$ measured from A **down the slope**, along the unit vector $\mathbf{u} = -\cos\alpha\,\mathbf{i} - \sin\alpha\,\mathbf{j}$; $\mathbf{n} = -\sin\alpha\,\mathbf{i} + \cos\alpha\,\mathbf{j}$ is the unit normal of the incline pointing out of the wedge. Both bodies are released from rest at $t = 0$ with $s = 0$.*

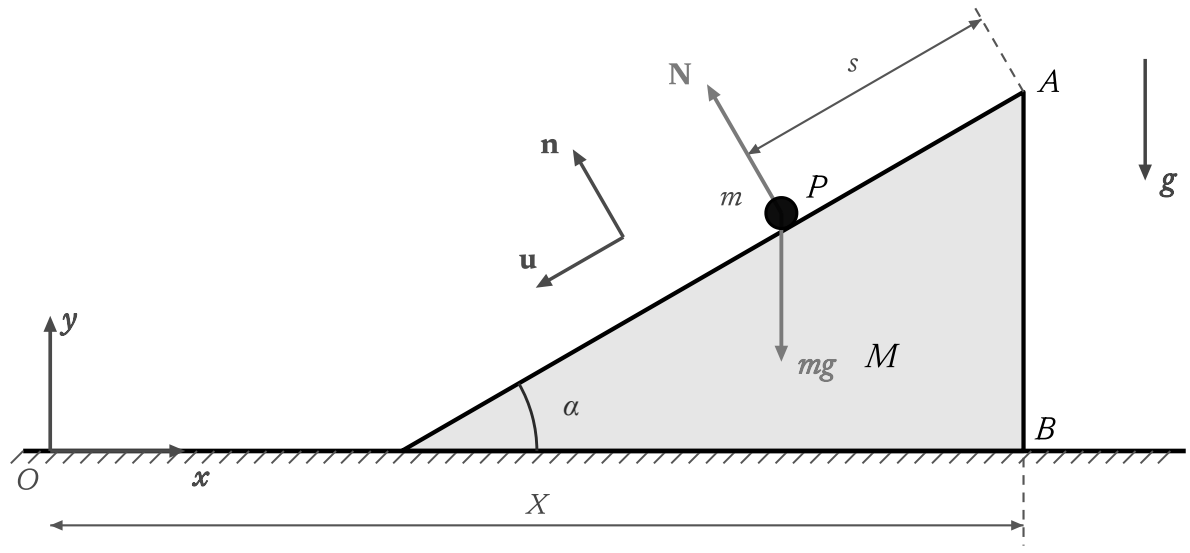


Figure 1. Free wedge: the wedge (mass M) slides on the smooth table, its position is X (abscissa of the corner B); P (mass m) slides on the smooth incline of angle α, at the distance s from the top edge A measured down the slope (unit vector $\mathbf{u}$); $\mathbf{N} = N\mathbf{n}$ is the normal reaction on P (outward normal $\mathbf{n}$).

Setup

- Data: m, M, α, g. Numerical parts: $\alpha = 30^\circ$, $M = 2m$, and the particle has slid a distance L along the incline.
- Hypotheses: point particle, rigid wedge translating on the table, all contacts smooth (no friction), S_0 inertial; the contact of P with the incline is unilateral ($N \geq 0$, checked in (d)).
- Frames: S_0 and the frame $S_1 = \{A; \mathbf{n}, \mathbf{u}\}$ attached to the wedge, ordered so that it is right-handed: $\mathbf{n} \times \mathbf{u} = \big[(-\sin\alpha)(-\sin\alpha) - \cos\alpha\,(-\cos\alpha)\big]\mathbf{k} = \mathbf{k}$ (the order $\{\mathbf{u}, \mathbf{n}\}$ would give $\mathbf{u} \times \mathbf{n} = -\mathbf{k}$, a left-handed basis). S_1 **translates** with acceleration $\ddot{X}\,\mathbf{i}$ and does not rotate.

- Basis change: $\mathbf{i} = -\sin\alpha\,\mathbf{n} - \cos\alpha\,\mathbf{u},\ \mathbf{j} = \cos\alpha\,\mathbf{n} - \sin\alpha\,\mathbf{u}$ (from $\mathbf{i}\cdot\mathbf{n} = -\sin\alpha, \mathbf{i}\cdot\mathbf{u} = -\cos\alpha$, $\mathbf{j}\cdot\mathbf{n} = \cos\alpha, \mathbf{j}\cdot\mathbf{u} = -\sin\alpha$).

In matrix form:

$$\begin{pmatrix}\mathbf{i}\\ \mathbf{j}\end{pmatrix} = [R]\begin{pmatrix}\mathbf{n}\\ \mathbf{u}\end{pmatrix}, \qquad [R] = \begin{pmatrix}-\sin\alpha & -\cos\alpha\\ \cos\alpha & -\sin\alpha\end{pmatrix}$$

with $[R][R]^T = \begin{pmatrix}\sin^2\alpha + \cos^2\alpha & -\sin\alpha\cos\alpha + \cos\alpha\sin\alpha\\ -\sin\alpha\cos\alpha + \cos\alpha\sin\alpha & \cos^2\alpha + \sin^2\alpha\end{pmatrix} = I$ and $\det[R] = \sin^2\alpha + \cos^2\alpha = +1$.

Part (a) · 1 point

Degrees of freedom and constraints

How many degrees of freedom has the system particle + wedge? Classify the constraint that keeps P on the incline, as seen in S_0 and in S_1.

Step 1. Count

The wedge is a rigid body in the plane (3 coordinates); the table fixes its height and, since it does not tip, its orientation: 1 coordinate X remains. The particle has 2 coordinates in the plane; staying on the incline removes one, leaving s. In total **2 degrees of freedom**, (X, s). Unlike a fixed incline, $X(t)$ is an unknown: the motion of the support depends on the motion of the particle.

Step 2. Classification

The condition "P on the incline" reads $\overrightarrow{AP}\cdot\mathbf{n} = 0$, i.e. $-(x_P - X)\sin\alpha + (y_P - h)\cos\alpha = 0$ with $\overrightarrow{OA} = X\,\mathbf{i} + h\,\mathbf{j}$, a relation between the positions of P and of the wedge: it is **holonomic** and, written in the system coordinates (X, s) (or (X, x_P, y_P)), it contains no explicit time, so it is **scleronomic**. It would be rheonomic for P alone only if the motion $X(t)$ of the wedge were prescribed from outside; here X is an unknown, so the time enters only through the coordinates. In S_1 the plane is fixed. It is **unilateral**: P can leave the plane upwards, so it holds only while $N \geq 0$. Smooth: the reaction is $\mathbf{N} = N\,\mathbf{n}$, with no component along $\mathbf{u}$ (Notes §6.4).

KEY RESULTS

- 2 degrees of freedom, (X, s).
- P on the incline: holonomic, unilateral, smooth; scleronomic as a relation between the coordinates (X, s) of the system (no explicit t); it would be rheonomic for P alone only if $X(t)$ were a prescribed function of time, which it is not here.

Part (b) · 1.5 points

Kinematics of P

Write the position, velocity and acceleration of P relative to S_1 and in S_0, in the basis $\{\mathbf{n}, \mathbf{u}\}$.

Step 1. Relative motion

$\mathbf{u}$ is fixed in S_1, so $\mathbf{r}_{rel} = \overrightarrow{AP} = s\,\mathbf{u}$, $\mathbf{v}_{rel} = \dot{s}\,\mathbf{u}$, $\mathbf{a}_{rel} = \ddot{s}\,\mathbf{u}$. The relative velocity is tangent to the incline.

Step 2. Absolute motion

$\overrightarrow{OA} = X\,\mathbf{i} + h\,\mathbf{j}$ with the height h of the wedge constant, so $\mathbf{v}_A = \dot{X}\,\mathbf{i}$, $\mathbf{a}_A = \ddot{X}\,\mathbf{i}$. Since $\boldsymbol{\omega}_{10} = \mathbf{0}$ (Notes §3.2):

$$\mathbf{v}_P = \dot{X}\,\mathbf{i} + \dot{s}\,\mathbf{u}, \qquad \mathbf{a}_P = \ddot{X}\,\mathbf{i} + \ddot{s}\,\mathbf{u}$$

Substituting $\mathbf{i} = -\cos\alpha\,\mathbf{u} - \sin\alpha\,\mathbf{n}$:

$$\mathbf{a}_P = (\ddot{s} - \ddot{X}\cos\alpha)\,\mathbf{u} - \ddot{X}\sin\alpha\,\mathbf{n}$$

and in Cartesian components $\mathbf{v}_P = (\dot{X} - \dot{s}\cos\alpha)\,\mathbf{i} - \dot{s}\sin\alpha\,\mathbf{j}$.

KEY RESULTS

- $\mathbf{r}_{rel} = s\,\mathbf{u}$, $\mathbf{v}_{rel} = \dot{s}\,\mathbf{u}$, $\mathbf{a}_{rel} = \ddot{s}\,\mathbf{u}$.
- $\mathbf{a}_P = \ddot{X}\,\mathbf{i} + \ddot{s}\,\mathbf{u} = (\ddot{s} - \ddot{X}\cos\alpha)\,\mathbf{u} - \ddot{X}\sin\alpha\,\mathbf{n}$.

Part (c) · 2.5 points

Equations of motion and the reaction

Write the forces on P in S_1, including the inertia force, and on the wedge. Obtain $\ddot{X}$, $\ddot{s}$ and N, and the reaction of the table on the wedge.

Step 1. Forces on P in S_1

Weight $\mathbf{W} = -mg\,\mathbf{j} = mg\sin\alpha\,\mathbf{u} - mg\cos\alpha\,\mathbf{n}$; reaction $\mathbf{N} = N\,\mathbf{n}$; inertia force of the translating frame (Notes §4.9) $\mathbf{F}_{in} = -m\,\mathbf{a}_A = -m\ddot{X}\,\mathbf{i} = m\ddot{X}\cos\alpha\,\mathbf{u} + m\ddot{X}\sin\alpha\,\mathbf{n}$. There is no Coriolis or centrifugal force ($\boldsymbol{\omega}_{10} = \mathbf{0}$).

Step 2. Projections for P

$m\,\ddot{s}\,\mathbf{u} = \mathbf{W} + \mathbf{N} + \mathbf{F}_{in}$:

$$\mathbf{u}: \quad m\ddot{s} = mg\sin\alpha + m\ddot{X}\cos\alpha \implies \ddot{s} = g\sin\alpha + \ddot{X}\cos\alpha$$

$$\mathbf{n}: \quad 0 = -mg\cos\alpha + N + m\ddot{X}\sin\alpha \implies N = m\big(g\cos\alpha - \ddot{X}\sin\alpha\big)$$

The $\mathbf{u}$ equation is free of the reaction (dynamical decoupling, Notes §6.5), but it still contains the unknown $\ddot{X}$.

Step 3. The wedge

Forces on the wedge: its weight $-Mg\,\mathbf{j}$, the table reaction $R\,\mathbf{j}$ (smooth table, vertical) and $-\mathbf{N} = -N\mathbf{n} = N\sin\alpha\,\mathbf{i} - N\cos\alpha\,\mathbf{j}$. Newton for the translating wedge, $M\ddot{X}\,\mathbf{i} = -Mg\,\mathbf{j} + R\,\mathbf{j} - N\mathbf{n}$:

$$\mathbf{i}: \quad M\ddot{X} = N\sin\alpha, \qquad \mathbf{j}: \quad 0 = R - Mg - N\cos\alpha$$

Step 4. Solve

Substituting N of Step 2 in the wedge equation: $M\ddot{X} = m\sin\alpha\,(g\cos\alpha - \ddot{X}\sin\alpha)$, so $\ddot{X}\,(M + m\sin^2\alpha) = m\,g\sin\alpha\cos\alpha$:

$$\ddot{X} = \frac{m\,g\sin\alpha\cos\alpha}{M + m\sin^2\alpha}$$

Then, with $\sin^2\alpha + \cos^2\alpha = 1$:

$$\ddot{s} = g\sin\alpha + \frac{m\,g\sin\alpha\cos^2\alpha}{M + m\sin^2\alpha} = \frac{g\sin\alpha\,(M + m\sin^2\alpha + m\cos^2\alpha)}{M + m\sin^2\alpha} = \frac{(M+m)\,g\sin\alpha}{M + m\sin^2\alpha}$$

$$N = mg\cos\alpha - \frac{m^2 g\sin^2\alpha\cos\alpha}{M + m\sin^2\alpha} = \frac{M\,m\,g\cos\alpha}{M + m\sin^2\alpha}$$

and $R = Mg + N\cos\alpha$. All three accelerations are **constant**. Check with the system: the vertical external forces are $R - (M+m)g = N\cos\alpha - mg$, which equals $m\,\mathbf{a}_P\cdot\mathbf{j} = -m\ddot{s}\sin\alpha$ (the wedge has no vertical acceleration). An independent check with the Lagrange equations in (X, s) gives the same $\ddot{X}$ and $\ddot{s}$.

Step 5. Numbers ($\alpha = 30^\circ$, $M = 2m$)

$M + m\sin^2\alpha = 2m + m/4 = \frac{9}{4}m$: $\ddot{X} = \dfrac{mg\cdot\frac{1}{2}\cdot\frac{\sqrt{3}}{2}}{\frac{9}{4}m} = \dfrac{\sqrt{3}}{9}g = 0.1925\,g$, $\ddot{s} = \dfrac{3m\,g\cdot\frac{1}{2}}{\frac{9}{4}m} = \dfrac{2}{3}g$, $N = \dfrac{2m\cdot m\,g\frac{\sqrt{3}}{2}}{\frac{9}{4}m} = \dfrac{4\sqrt{3}}{9}mg = 0.7698\,mg$ (smaller than $mg\cos 30^\circ = 0.866\,mg$ on a fixed wedge).

KEY RESULTS

- $\ddot{s} = g\sin\alpha + \ddot{X}\cos\alpha$, $N = m(g\cos\alpha - \ddot{X}\sin\alpha)$, $M\ddot{X} = N\sin\alpha$.
- $\ddot{X} = \dfrac{m\,g\sin\alpha\cos\alpha}{M + m\sin^2\alpha}$, $\ddot{s} = \dfrac{(M+m)\,g\sin\alpha}{M + m\sin^2\alpha}$, $N = \dfrac{M\,m\,g\cos\alpha}{M + m\sin^2\alpha}$.
- Table reaction $R = Mg + N\cos\alpha$.

Part (d) · 1.5 points

Limits, the sign of N and the massless wedge

Check that $N > 0$ for every $M > 0$. Study the limits $M \to \infty$ and $M \to 0$. In the second one a student says: "the particle slides down a plane, so it accelerates along the incline with $g\sin\alpha$ times some factor and presses on it". Discuss.

Step 1. Sign of the reaction

$N = \dfrac{Mmg\cos\alpha}{M + m\sin^2\alpha}$: numerator and denominator are positive for $M > 0$ and $0 < \alpha < \pi/2$, so $N > 0$ throughout the motion (the accelerations are constant, so N is too). The unilateral constraint is never violated, and $R = Mg + N\cos\alpha > Mg > 0$: the wedge stays on the table.

Step 2. Heavy wedge

Dividing numerator and denominator by M and letting $M \to \infty$ ($m/M \to 0$): $\ddot{X} \to 0$, $\ddot{s} \to g\sin\alpha$, $N \to mg\cos\alpha$, the results of a fixed incline.

Step 3. Massless wedge (the trap)

Setting $M = 0$: $N \to 0$, $\ddot{X} \to \dfrac{mg\sin\alpha\cos\alpha}{m\sin^2\alpha} = g\cot\alpha$, $\ddot{s} \to \dfrac{mg\sin\alpha}{m\sin^2\alpha} = \dfrac{g}{\sin\alpha}$. The absolute acceleration is

$$\mathbf{a}_P = g\cot\alpha\,\mathbf{i} + \frac{g}{\sin\alpha}\big(-\cos\alpha\,\mathbf{i} - \sin\alpha\,\mathbf{j}\big) = -g\,\mathbf{j}$$

The particle **falls freely**; the incline exerts no force on it. A massless wedge cannot push back ($M\ddot{X} = N\sin\alpha$ with $M = 0$ forces $N = 0$), so it flies away horizontally with $g\cot\alpha$, exactly the motion that keeps the surface under the falling particle. The first half of the student's statement is harmless: $\ddot{s} = g\sin\alpha \cdot \dfrac{M+m}{M+m\sin^2\alpha}$ is indeed $g\sin\alpha$ times a factor, and this factor is ≥ 1 for every M (because $M + m \geq M + m\sin^2\alpha$), growing from 1 ($M \to \infty$) to $1/\sin^2\alpha$ ($M \to 0$), which gives $\ddot{s} \to g/\sin\alpha$. The error is the second half, "the particle presses on the wedge": $N \to 0$, the incline exerts no force and the particle is in free fall, merely accompanied by the surface.

KEY RESULTS

- $N > 0$ for $M > 0$ and $0 < \alpha < \pi/2$: the particle never leaves the incline; $R > Mg > 0$.
- $M \to \infty$: $\ddot{X} \to 0$, $\ddot{s} \to g\sin\alpha$, $N \to mg\cos\alpha$ (fixed incline).
- $M \to 0$: $N \to 0$, $\ddot{X} \to g\cot\alpha$, $\ddot{s} \to g/\sin\alpha$ and $\mathbf{a}_P \to -g\,\mathbf{j}$: **free fall**; the massless wedge is pushed away just fast enough to stay in contact without force.
- The factor $\ddot{s}/(g\sin\alpha) = (M+m)/(M+m\sin^2\alpha) \geq 1$ grows to $1/\sin^2\alpha$ as $M \to 0$; the student's error is "presses on the wedge", since $N \to 0$.

Part (e) · 2 points

First integrals and the speeds after sliding a distance L

Show that the horizontal linear momentum of the system and its mechanical energy are conserved, explaining why the reaction $\mathbf{N}$ does no net work although it works on each body. For $\alpha = 30°$, $M = 2m$, find $\dot{s}$, $\dot{X}$ and $|\mathbf{v}_P|$ when P has slid a distance L, and the work of $\mathbf{N}$ on P and on the wedge.

Step 1. Horizontal momentum

The external forces on the system $\{P, \text{wedge}\}$ are $-mg\,\mathbf{j}$, $-Mg\,\mathbf{j}$ and $R\,\mathbf{j}$: no horizontal component. $\mathbf{N}$ is internal (action-reaction pair). The linear momentum theorem gives $\frac{d}{dt}p_x = 0$, with $p_x = M\dot{X} + m\,\mathbf{v}_P\cdot\mathbf{i} = M\dot{X} + m(\dot{X} - \dot{s}\cos\alpha)$. From rest, $p_x = 0$:

$$\dot{X} = \frac{m\cos\alpha}{M+m}\,\dot{s}$$

Consequently the centre of mass has $\ddot{x}_G = 0$ and moves only vertically.

Step 2. Energy

Powers: the weights are conservative; $R\,\mathbf{j}\cdot\dot{X}\,\mathbf{i} = 0$; the pair $\mathbf{N}$, $-\mathbf{N}$ has total power

$$\mathbf{N}\cdot\mathbf{v}_P + (-\mathbf{N})\cdot\mathbf{v}_A = \mathbf{N}\cdot(\mathbf{v}_P - \mathbf{v}_A) = N\,\mathbf{n}\cdot\dot{s}\,\mathbf{u} = 0$$

because the relative velocity is tangent to the smooth incline (Notes §6.4). So the mechanical energy of the system is constant (Notes §4.5). With the zero of potential energy at $s = 0$, the height of P changes by $-s\sin\alpha$:

$$\tfrac{1}{2}M\dot{X}^2 + \tfrac{1}{2}m\big[(\dot{X} - \dot{s}\cos\alpha)^2 + \dot{s}^2\sin^2\alpha\big] - mg\,s\sin\alpha = 0$$

Step 3. Kinetic energy for $\alpha = 30^\circ$, $M = 2m$

$\dot{X} = \frac{m\cdot\frac{\sqrt{3}}{2}}{3m}\,\dot{s} = \frac{\sqrt{3}}{6}\,\dot{s}$. For P: $\dot{X} - \dot{s}\cos\alpha = \big(\frac{\sqrt{3}}{6} - \frac{3\sqrt{3}}{6}\big)\dot{s} = -\frac{\sqrt{3}}{3}\dot{s}$, whose square is $\frac{1}{3}\dot{s}^2$, and $\dot{s}^2\sin^2\alpha = \frac{1}{4}\dot{s}^2$, so $|\mathbf{v}_P|^2 = \frac{7}{12}\dot{s}^2$ and $\frac{1}{2}m|\mathbf{v}_P|^2 = \frac{7}{24}m\dot{s}^2$. For the wedge: $\dot{X}^2 = \frac{3}{36}\dot{s}^2 = \frac{1}{12}\dot{s}^2$ and $\frac{1}{2}(2m)\frac{1}{12}\dot{s}^2 = \frac{2}{24}m\dot{s}^2$. Total:

$$E_{kin} = \frac{2+7}{24}\,m\dot{s}^2 = \frac{3}{8}\,m\dot{s}^2$$

Step 4. Speeds after sliding L

With $s = L$ and $mgL\sin 30^\circ = \frac{1}{2}mgL$:

$$\tfrac{3}{8}m\dot{s}^2 = \tfrac{1}{2}mgL \implies \dot{s}^2 = \tfrac{4}{3}gL, \quad \dot{s} = 1.1547\sqrt{gL}, \quad \dot{X} = \tfrac{\sqrt{3}}{6}\cdot\tfrac{2}{\sqrt{3}}\sqrt{gL} = \tfrac{1}{3}\sqrt{gL}$$

$$|\mathbf{v}_P|^2 = \tfrac{7}{12}\cdot\tfrac{4}{3}gL = \tfrac{7}{9}gL, \qquad |\mathbf{v}_P| = \tfrac{\sqrt{7}}{3}\sqrt{gL} = 0.8819\sqrt{gL}$$

Check with (c): constant $\ddot{s} = \frac{2}{3}g$ from rest gives $\dot{s}^2 = 2\ddot{s}L = \frac{4}{3}gL$. The time to slide L is $\sqrt{2L/\ddot{s}} = \sqrt{3L/g}$ and the wedge has moved $\Delta X = \frac{\sqrt{3}}{6}L = 0.2887\,L$ towards $+x$.

Step 5. Work of the reaction on each body

Kinetic energy theorem for P alone: $\frac{1}{2}m|\mathbf{v}_P|^2 = W_{weight} + W_{N\to P}$ with $W_{weight} = \frac{1}{2}mgL$: $W_{N\to P} = \frac{7}{18}mgL - \frac{9}{18}mgL = -\frac{1}{9}mgL$. For the wedge: $W_{-N\to wedge} = \frac{1}{2}(2m)\dot{X}^2 = \frac{1}{9}mgL$. The reaction **does** work on each body (the incline moves), but the two works cancel: energy is transferred from P to the wedge.

KEY RESULTS

- $p_x = M\dot{X} + m(\dot{X} - \dot{s}\cos\alpha) = 0 \Rightarrow \dot{X} = \dfrac{m\cos\alpha}{M+m}\,\dot{s}$; $\frac{1}{2}M\dot{X}^2 + \frac{1}{2}m|\mathbf{v}_P|^2 - mgs\sin\alpha = 0$.
- $\alpha = 30^\circ$, $M = 2m$: $E_{kin} = \frac{3}{8}m\dot{s}^2$, $\dot{s}^2 = \frac{4}{3}gL$ ($\dot{s} = 1.1547\sqrt{gL}$), $\dot{X} = \frac{1}{3}\sqrt{gL}$, $|\mathbf{v}_P| = \frac{\sqrt{7}}{3}\sqrt{gL} = 0.8819\sqrt{gL}$.
- Work of $\mathbf{N}$ on P: $-\frac{1}{9}mgL$; on the wedge: $+\frac{1}{9}mgL$; total 0.

Part (f) · 1.5 points

The absolute path of P

Show that P moves along a straight line in S_0 and find the angle β of this line with the horizontal. Compare it with α for $\alpha = 30^\circ$, $M = 2m$.

Step 1. A straight path

$\mathbf{a}_P = \ddot{X}\,\mathbf{i} + \ddot{s}\,\mathbf{u}$ is a constant vector (part (c)). Integrating twice from rest (Barrow): $\mathbf{v}_P = \mathbf{a}_P\,t$ and $\mathbf{r}_P - \mathbf{r}_P(0) = \frac{1}{2}\mathbf{a}_P\,t^2$, so P moves along the fixed straight line through its initial position with direction $\mathbf{a}_P$.

Step 2. Its slope

In Cartesian components $\mathbf{a}_P = (\ddot{X} - \ddot{s}\cos\alpha)\,\mathbf{i} - \ddot{s}\sin\alpha\,\mathbf{j}$; the line descends towards $-x$ with

$$\tan\beta = \frac{\ddot{s}\sin\alpha}{\ddot{s}\cos\alpha - \ddot{X}} = \frac{\sin\alpha}{\cos\alpha - \dfrac{\ddot{X}}{\ddot{s}}}$$

From (c), $\dfrac{\ddot{X}}{\ddot{s}} = \dfrac{m\cos\alpha}{M+m}$ (the same ratio as $\dot{X}/\dot{s}$ in (e)), so $\cos\alpha - \dfrac{m\cos\alpha}{M+m} = \dfrac{M\cos\alpha}{M+m}$ and

$$\tan\beta = \frac{(M+m)\sin\alpha}{M\cos\alpha} = \left(1 + \frac{m}{M}\right)\tan\alpha$$

Step 3. Numbers

$\tan\beta = \frac{3}{2}\cdot\frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{2} = 0.8660$, $\beta = 40.89^\circ$. The particle descends **more steeply** than the incline, because the incline recedes under it. Check with (e): $|v_{Py}|/|v_{Px}| = \sqrt{gL/3}\big/\frac{2}{3}\sqrt{gL} = \frac{\sqrt{3}}{2}$.

KEY RESULTS

- $\tan\beta = \dfrac{\ddot{s}\sin\alpha}{\ddot{s}\cos\alpha - \ddot{X}} = \left(1+\dfrac{m}{M}\right)\tan\alpha.$
- $\alpha = 30^\circ$, $M = 2m$: $\tan\beta = \frac{\sqrt{3}}{2}$, $\beta = 40.89^\circ > \alpha$.

Interpretation and checks

- **Dimensional check:** $\ddot{X}$ and $\ddot{s}$ are g times a ratio of masses times trigonometric factors (m s^{-2}); $N = Mmg\cos\alpha/(M + m\sin^2\alpha)$ is kg·m s^{-2} = N; $\dot{s}^2 = \frac{4}{3}gL$ is m^{2}s^{-2}; the works are mgL times numbers (J).
- **Limiting cases:** $M \to \infty$ recovers the fixed incline ($\beta \to \alpha$); $M \to 0$ gives free fall ($\beta \to 90^\circ$, $\tan\beta \to \infty$); $\alpha \to 0$ gives $\ddot{X} = \ddot{s} = 0$, $N = mg$ (nothing moves); $\alpha \to \pi/2$ gives $N \to 0$, $\ddot{X} \to 0$, $\ddot{s} \to g$ (the particle falls along the vertical face).
- **Common mistakes in the exam:** (1) using $N = mg\cos\alpha$ as on a fixed incline; (2) forgetting the inertia force $-m\ddot{X}\,\mathbf{i}$ in S_1, or giving it the wrong sign; (3) writing energy conservation for P alone (the reaction works on P: $-\frac{1}{9}mgL$ here); (4) assuming the particle's absolute path is parallel to the incline.